

Total	100	-	100
Final	50	-	50
Sessional	50	-	50
	Theory	Practical	Total

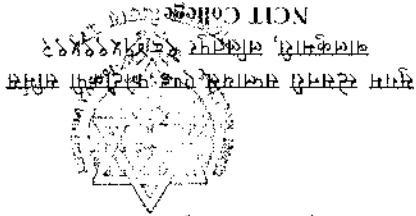
Course Objective:

After completing this course, students will be able to

- understand and describe the basic concept of economics, engineering economics, cost accounting and time value of money,
- assist in the valuation of engineering projects in the public and private sector to take investment decisions,
- analyze the project risk and understand the concept of ecological limit and economic development,
- calculate depreciation, taxation and its application in analysis and identify different financing options and general accounting procedures.

Course Contents:

1. Basics of Engineering Economics (3 hrs)
 - 1.1. Definition of Economics, Demand, the Law of Demand, Law of Diminishing Utility, Marginal Utility, Supply, Law of Supply and Demand
 - 1.2. Engineering Economics, Principles of Engineering Economy and its application
2. Cost Concept and Fundamentals of Cost Accounting (3 hrs)
 - 2.1. Cost Terminology: Manufacturing Cost and Non-Manufacturing Cost
 - 2.2. Cost for Business Decision: Differential Cost and Revenue; Opportunity Cost, Sunk Cost and Marginal Cost
3. Time Value of Money (4 hrs)
 - 3.1. Interest, Simple Interest, Compound Interest, Nominal Rate of Interest, Effective Rate of Interest
 - 3.2. Economic Equivalence: Present Worth, Future Worth and Annual Worth
 - 3.3. Development of Formulas for Equivalence Calculation
4. Basic Methods of Engineering Economic Studies (7 hrs)
 - 4.1. Minimum Attractive Rate of Return - MARR
 - 4.2. Payback Period Method - Simple and Discounted
 - 4.3. Equivalent Worth Methods; Present Worth Method, Future Worth Method and Annual Worth Method
 - 4.4. Rate of Return Methods: Internal Rate of Return (IRR) Method and External/Modified Rate of Return (ERR/MIRR) Method
 - 4.5. Benefit Cost Ratio Method
5. Comparative Analysis of Alternatives (6 hrs)
 - 5.1. Comparing Mutually Exclusive Alternatives having Same useful life by Payback Period Method, Equivalent Worth Method; Rate of Return Methods and Benefit Cost Ratio Method
 - 5.2. Comparing Mutually Exclusive Alternatives having different useful lives by Repeatability Assumption, Co-terminated Assumption, Capitalized Worth Method
 - 5.3. Comparing Mutually Exclusive, Contingent and Independent Projects in Combination.





1. E. Paul De Garmo, William G. Sullivan and James A. Bontadelli. *Engineering Economy*. MC Milian Publishing Company.
2. James L. Riggs, David D. Bedworth and Sabah U. Randhawa. *Engineering Economics*. Tata McGraw Hill Education Private Limited.
3. N.N. Borish and S. Kaplan. *Economic Analysis for Engineering and Managerial Decision Making*. MC Gran Hill Publishing Company.
4. Adhikari, D. *Principles of Engineering Economic Analysis*. Nepal: Global Publication.
5. Sen Gupta, Ramprasad. *Ecological Limits and Economic Development*. Oxford University Press.

References:

1. Chan S. Park. *Contemporary Engineering Economics*. PHI Learning Private Limited.

Text Book:

Two assignments and 1 case study.

Tutorials:

10. **Basic Accounting Procedure** (6 hrs)
 - 10.1. Accounting Terminologies; Asset and liabilities: Fundamental equation of accounting
 - 10.2. Financial statements: The Balance Sheet, Income Statement and Cashflow Statements
 - 10.3. Using Ratios to make Decisions: Debt Ratio, Current Ratio, Quick Ratio – Acid Test Ratio, Inventory Turnover Ratio, Total Asset Turnover, Profit Margin on Sales, Return on Total Assets, Price Earnings Ratio and Book Value per Share
9. **Enterprise Financing and Capital Investment** (4 hrs)
 - 9.1. Method of Financing: Equity Financing, Debt Financing and Capital Structure
 - 9.2. Cost of Capital: Cost of Equity, Cost of Debt and calculating cost of capital
 - 9.3. Project Funding Mechanism: Government budget, Public Private Partnership and Private Investment
 - 9.4. FIRR, EIRR and Return on Equity
8. **Depreciation and Corporate Income Taxes** (5 hrs)
 - 8.1. Depreciation and its causes, Asset Depreciation and Accounting Depreciation
 - 8.2. Basic Methods of Depreciation: Straight line method, Declining Balance Method, Sinking Fund Method, Sum of the Year Digit Method, Unit of Production Method, Modified Accelerated Cost Recovery System (MACRS)
 - 8.3. Introduction to Corporate Income Tax, Taxation Law, Depreciation Rates, Personal Tax, Corporate Tax, VAT
 - 8.4. After Tax Cash flow Estimate, General Procedure for Making After Tax Economic Analysis
7. **Ecological Limits and Economic Development** (3 hrs)
 - 7.1. Economic Theory and Ecological Limit
 - 7.2. Concept of sustainable development
 - 7.3. Ecological Footprint and
 - 7.4. Overcoming Ecological Limits
6. **Risk Analysis** (4 hrs)
 - 6.1. Origin/Sources of Project Risks.
 - 6.2. Methods of Describing Project Risks: Sensitivity Analysis, Break-even Analysis, Scenario Analysis

BASICS OF ENGINEERING ECONOMICS

CHAPTER - 1

1.1. Definition of Economics, Demand, Law of Demand, Law of Diminishing Utility, Marginal Utility, Supply, Law of supply, Law of supply and Demand.

Definition of Economics :- Economics is the science that deals with the production and consumption of goods and services and the distribution and rendering of these for human welfare.

According to Alfred Marshall (British economists), "Economics is on the one side, a study of wealth and on the other more important side is a part of the study of man."

According to Lionel Robbins (British economist), "Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."

Some Economics Terminology

1. Annuity: - A amount of money payable to a beneficiary at regular intervals for a prescribed period of time like annually, monthly, weekly, etc. out of a fund reserved for that purpose is known as annuity.

2. Asset: -

Capital owned by a company.
An economic resource of entity (including money

resources, physical researches and intangible resources.
 Real asset: Land, building, plant, machinery, etc.
 Financial asset: - cash, shares, securities, etc.

2. Break-even point (BEP): -
 A point that indicates graphical representation of the relation between total revenue and total cost for that level of production and sales at which revenue is just sufficient to cover all expenses. The percentage of capacity operation of a manufacturing plant at which income will just cover expenses.
 A point where the organization is neither gain nor loss state.

4. Capital: -
 The man-made factors that contribute to the production of goods and services as well as the non-human ingredients (factors) that may be utilized to economic advantage that includes cash, buildings, raw materials, semi-finished & finished products, etc.

5. Sunk cost: -
 It is a cost that an entity has incurred and which can no longer recover by any means.

6. Salvage value: -
 The value of capital asset at the end of economic life and known as market or scrap or residual value.

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Demand:-

The various quantities of an item that a buyer is willing to buy at alternative prices, other things being equal.

Desire with ability to pay and willingness to pay.

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The Law of Demand

"Other things being equal demand varies inversely with price."

Everybody is known that demand depends upon price. Law of demand states the relationship between the quantity demanded and price of the commodity.

Although there are many other factors (income, taste, weather, population size etc.) to influence on demand, price is the most important determinant of demand in the short-run as well as long-run. Law of demand explains the inverse relationship between price and quantity demanded. It is defined as "All other things remaining the same, the quantity demanded of a commodity increase when its price decrease and vice-versa."

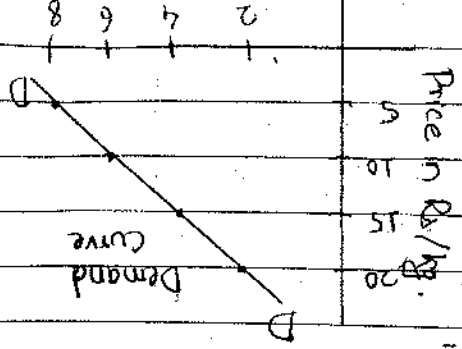
This law assumes income, fashion, price of related goods, population size, taste and preferences, etc. are remained constant.

This can be illustrated as:-

Price in Rs/kg	Quantity demanded in kg
25	2
20	4
15	6
10	8
5	10

Table 1:

Quantity Demanded.



In above Table, as price increases from Rs. 5/kg to Rs. 25/kg, quantity demanded decreases from 10kg to 2kg gradually.

Demand curve is a graphical presentation of demand schedule. When the data given in the above demand schedule plots in a graphical form, then we can derive a demand curve.

In above fig, DD is a demand curve and slope of the demand curve is negative i.e. downward sloping from left to right. Demand curve has explained the meaning of law of demand i.e. quantity demanded is increased when price decreased.

Factors influencing to demand.

1. Price of commodity.
2. Income of consumer.
3. Price of related goods.
4. Weather.
5. Custom and fashion.
6. Size of population.
7. Future expectations, etc.

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Law of Diminishing Marginal Utility:-

It states that in the process of fulfilling human wants, when a consumer consumes more and more units of a commodity, the utility derived from each successive unit of the commodity goes on decreasing. Hence, the relationship between quantity consumed and utility derived from each successive unit is called the law of diminishing marginal utility.

According to Marshall,

"The additional benefit which a person derives from an increase of his stock of a thing diminishes with the growth of the stock that he already has".

This law is based and explained under certain

conditions or assumption such as, utility is cardinally measurable, consumer's taste and preference must be unchanged, suitable and similar units of commodity, etc.

Let us consider the following table:-

Units of apple	Total utility	Marginal utility
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

Suppose a consumer is consuming units of apple. He/she

starts eating apple one after another. The first apple

gives him/her great pleasure/satisfaction/utility. Right that

time, he starts second one and he compares it with

first. Second apple gives him less satisfaction than first

apple and the satisfaction of the third apple will be

lesser and so on. In this way, consumer goes on

consuming till it drop down to zero. If there further

force on consumption, the satisfaction will be negative

which leads to decrease in total utility. According

to above table, the utility derived from the first

unit of apple is 10 and it is decreasing on

successive unit like 8, 6, 4 and so on. At sixth unit,

consumer is getting zero marginal utility and total utility

is 30. At this point, the satisfaction derived is

maximum, so it is the point of maximum satisfaction to the consumer. He does not want to consume another i.e. seventh unit gives him negative utility (-2) instead of positive satisfaction which leads to decrease his total utility by 28.

Supply:- The various quantities of an item that a seller is willing to sell at alternative prices, other things being equal.

Desire to sell with ability to sell and willingness to sell.

Law of supply. "Other things being equal, supply varies positively with price."

The relationship between price and quantity supplied is known as law of supply. Supply varies directly and positively with price. It can be defined as "Other things remaining the same, as the price of a commodity rises its supply is extended and as price falls, its supply is contracted." Suppliers are always profit motivated, therefore they expect prices will go up and they will be able to make more profit. Hence, when price rises, there increase in production and supply will extend i.e. higher price will lead to high return and profit.

For example:-

Suppose a commodity like tomato has different prices in a market within a week as shown in the following table.

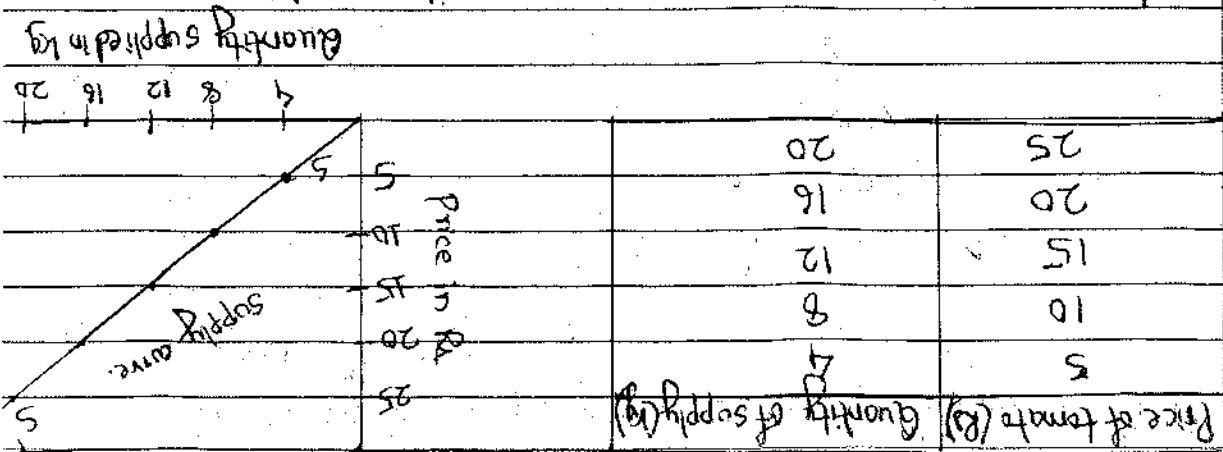
An interesting aspect of the economy is that the demand and supply of a product are interdependent and they are sensitive with respect to the price of

Law of supply and demand:-

1. Price of commodity
2. Price of factors of production
3. Price of related goods
4. Production technology
5. Change in money income
6. New inventions
7. Taxes and subsidies
8. Development of infrastructures
9. State of natural resources, etc.

In above table, Price is rising and quantity supplied is also rising. The graph is shown above.

Supply curve is a graphical representation of supply schedule. When the data given in the above example plots in the graphical form, we can derive a supply curve SS. Supply curve always slopes upward to the right as shown in figure.



that product. The inter-relationships between them is given below.

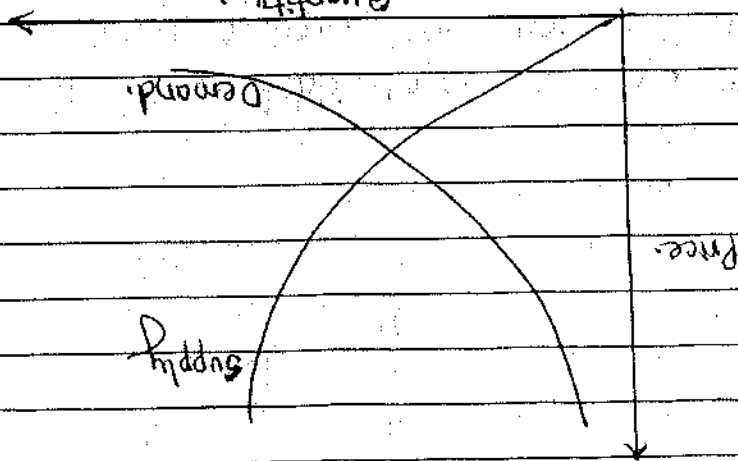


Fig. Demand & Supply curve.

From above fig., it is clear that when there is a decrease in the price of a product, the demand for the product increases and its supply decreases. Also, the product is more in demand and hence the demand of the product increases. At the same time, lowering of the price of the product makes the producers restrain from releasing more quantities of the product in the market. Hence, the supply of the product is decreased. The point of the intersection of the supply curve and the demand curve is known as the equilibrium point. At the price corresponding to this point, the quantity of supply is equal to the quantity of demand. Hence, this point is equilibrium point.

1.2. Engineering Economics, Principles of Engineering Economy and its application:

Engineering economics deals with the methods that enable one to take economic decisions towards minimizing cost, and/or maximizing benefits to business organizations. Engineering economics is the application of economic techniques for the evaluation of design and engineering alternatives. The role of engineering economics is to:

- Access the appropriateness of a given project.
 - Estimate its value.
 - Justify it from an engineering point of view.
- According to A.M. Wellington,
- "Engineering is the art of doing well with one dollar which any bungler can do with two after a fashion."

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Role of an engineer's in Decision Making.

The role of engineers in the decision making is that evaluate the condition, feasibility and appropriateness of an alternatives of a given project, estimate its value and justify it from an engineering stand point and finally discover the best alternative for implementation.

- Understand the problem and define the objectives.
- Collect relevant information.
- Define the feasible alternative solution and make realistic estimates.
- Identify the criteria for decision making using one or more attribute.
- Evaluate each alternative, using sensitivity analysis to enhance the evaluation.
- Select the best alternative.
- Implement the solution and monitor the result.

Principles of Engineering Economics:-

There are several principles which are as follows:-

1. Develop the alternatives:-
The feasible alternatives need to be identified and then defined for subsequent analysis. A decision involves making a choice among two or more alternatives. Developing and defining the feasible alternatives for evaluation is important because of the resulting impact on the quality of decision.

2. Focus on the differences:-
The expected differences in future outcomes among the alternatives are relevant to their comparison and should be considered in the decision making. If all prospective outcomes of the feasible alternatives are exactly same, we would be indifferent among the alternatives and could make a decision using random selection.

3. Use a consistent viewpoint (Economic viewpoint):-
To determine future outcomes of the feasible alternative the viewpoint should be consistently developed and defined. It is important that the viewpoint for the particular decision is defined and then used consistently in the description, analysis and comparison of the alternatives.

4. Use a common unit of measure:-
Common unit of measurement should follow to enumerate prospective outcomes of the feasible alternatives which will make easier the analysis and comparison for

The application of engineering economy are diverse and found in most areas of and organizations. They are typically classified as:-

Applications:-

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7. Revisit your decision (self-evaluation):-
 Improper decision results from an adaptive process, due to this the projected outcomes of the selected alternatives and actual results achieved should be subsequently compared. Organizational discipline is needed to ensure that the post-evaluations of implemented decisions are ruthlessly accomplished and the results used to future decision making.

6. Make uncertainty explicit:-
 Uncertainty is inherent in estimating magnitude and impact of future outcomes of the feasible alternatives and should be recognized in their analysis and comparison. Thus, dealing with uncertainty is an important aspect of engineering economic analysis.

5. Consider all relevant criteria (Social & economical aspect)
 An engineering economic analysis, the primary criterion relates to provide maximum monetary return to the owners. But other objectives like social, cultural, environmental justice, etc. should become the basis for additional criteria in the decision-making process.

economic consequences, a monetary unit is the common measure. It will make uniformly in measurement, analysis, result interpretation and comparison between the alternatives

1. Equipment and process selection.

2. Equipment replacement.

3. New product and production expansion.

4. Cost reduction.

5. Capital Budget allocation.

6. New project evaluation, etc.

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Cash Flow:-

Cash flow is the statement which shows inflows and outflows of cash and cash equivalents during life of project i.e. actual rupees coming into a firm or actual rupees going out from firm in different time periods. It is the basis for the evaluation of different alternatives.

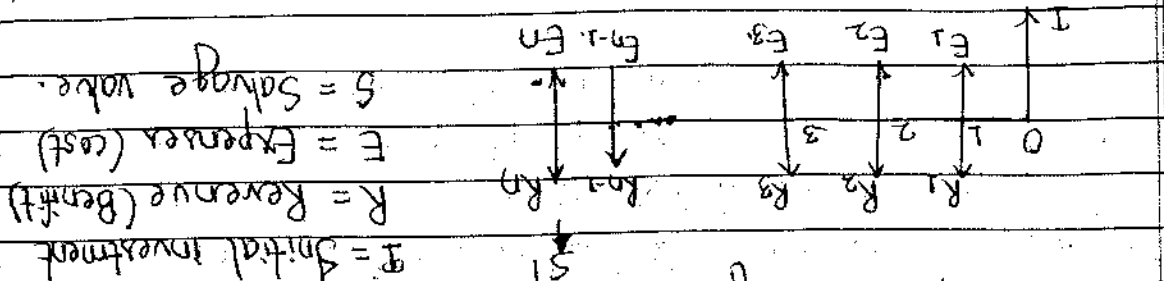
Cash flow diagram (CFD):-

The graphical representation of different cash flow streams is known as cash flow diagram. It should show three things:-

1. A time interval divided into an appropriate number of equal periods.

2. All cash out flows in each period.

3. All cash inflows for each period.



How to draw cash flow diagram?

1. Horizontal line is time scale with progression of time moving from left to right. (Normally starts from 0 to

2. Inflows and outflows of money is expressed by upward and downward arrows respectively. Here, inflows of cash represent annual benefit, salvage value, etc. and outflows represent initial capital investment, annual cost, etc.
3. It is better to show two or more cash flows occurring in the same year individually to make clear connection in between problem statement and cash flows in the diagram.

Cost Concept and Fundamentals of Cost Accounting

CHAPTER - 2

2.1 Cost Terminology: Manufacturing Cost and Non-Manufacturing Cost:-

A measure of what must be given up in order to obtain something.

The value of the inputs needed to produce any goods or service, that has to be measured in some units usually money.

The amount of money paid for a thing is called cost or cost is the amount of expenditure incurred on a given thing.

In engineering economics, the term cost is used in various ways because cost is classified according to the needs of management. To understand the cost in a manufacturing company, we must have better knowledge about manufacturing and non-manufacturing cost.

Manufacturing cost:-
In the process of converting raw materials into finished goods, a manufacturer needs various nature of costs that refers manufacturing cost. It is divided into three headings:-

a. Direct raw material costs:-
It refers all material that can be easily traced and used in the final products like: wood in furniture, steel in bridge, cloth in a shirt, paper in a books, etc. The total cost having on such direct raw materials refers direct raw material cost.

On business decisions, each alternative (project) has certain costs and revenue that must be compared with other available alternatives (projects).

Differential Cost and Revenue: -
Costs: Revenue, Opportunity cost, Sunk cost and Marginal

Cost for Business Decision: Differential Cost and

Administrative cost/expenses: -
It includes executive compensation, general accounting, public relations, secretarial support associated with general management of an organization.

Marketing & Selling cost/expenses: -
It includes secure customer order, supply of finish product into hands of customer, advertising, shipping, sales commission, travel, etc.

Non-manufacturing costs: - It includes:

Manufacturing overheads: -
It includes all costs of manufacturing process except those of direct materials and direct labour costs like costs of indirect materials (thread in a shirt), cost of indirect labour (overtime pay), cost of maintenance and repair of equipment, etc.

Direct labour cost: -
Like direct material costs, direct labour costs refers the costs of direct manpower that go into the production process like salary of managers, machine operators, full time staff, etc.

A difference in cost between any two alternatives is known as a differential cost (incremental cost).

Similarly, a difference in revenue between any two alternatives is known as differential revenue.

An investor must compare cost and revenue within a project and also between many projects and make decision accordingly.

If Revenue > Cost $\Rightarrow$ Select project
If Revenue $\leq$ Cost $\Rightarrow$ Reject project.

Differential (incremental) cost is the difference in total cost that results from selecting one alternative (project) instead of another. If several alternatives are possible, we select the one which has the minimum cost and maximum savings from the base (existing method). Problems of this type are often called trade-off problems, because cost of one project is traded off cost for another project.

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Opportunity cost:-

It is the best rejected opportunity to earn return by using available resources. In other words, it is the best opportunity to generate income that is forgone by not using the limited resources.

It is more useful in business decision.
If the asset/resource has been abandoned, then the opportunity cost will be zero.
It is hidden or expected cost but important for decision making process.

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Sunk cost:-

All the costs that already has been incurred in past.

Actions which cannot be recovered refers such costs.

Sunk costs are not relevant to decision making process because they cannot be changed regardless of what decision is made now or in the future. Cost of engineering design, wages, registration, license, depreciation, etc. are its examples.

Marginal cost:-

The change in total cost due to increase in output by single unit. The net cost involved in producing one more unit of product is called marginal cost.

$$\text{Marginal cost (MC)} = \frac{\Delta TC}{\Delta Q} \text{ or } MC = \Delta TC$$

Q. Following are the costs for the production of 100 badminton rackets;

Labor rate: Rs. 40/hr.
Leather: 50 meter @ Rs. 200/meter.
Gut: 300 meter @ Rs. 50/meter.
Graphic: 100 kg @ Rs. 200/kg.

Total labor hours needed: 200 hours.

Total annual factory overhead: Rs. 50,00,000

Total annual direct labor hours: 2,50,000 hours

Break down the costs into components of prime costs and overhead costs and find out the manufacturing cost of each racket.

Solution:-

Direct labor cost = Rs. 40 × 200 hours = Rs. 8000

Direct leather cost = Rs. 200 × 50 meter = Rs. 10,000

$$\text{Gut cost} = \text{Rs. } 50 \times 300 \text{ meter} = \text{Rs. } 15,000$$

$$\text{Graphite cost} = \text{Rs. } 200 \times 100 \text{ kg} = \text{Rs. } 20,000$$

$$\therefore \text{Total prime cost} = \text{Rs. } 53,000$$

For the overhead cost,

$$\text{Factory overhead per hour cost} = \frac{\text{Total factory annual overhead cost}}{\text{Total annual direct labour cost}}$$

$$= \text{Rs. } \frac{50,00,000}{2,50,000} = \text{Rs. } 20/\text{hr.}$$

$$\therefore \text{Total overhead cost} = \text{Factory overhead per hour cost} \times \text{Total labour hour}$$

$$= \text{Rs. } 20 \times 200 \text{ hours}$$

$$= \text{Rs. } 4,000$$

$$\therefore \text{Total cost} = \text{Total prime cost} + \text{Total overhead cost}$$

$$= \text{Rs. } 53,000 + \text{Rs. } 4,000$$

$$= \text{Rs. } 57,000$$

Now,

$$\text{Manufacturing cost of each racket} = \frac{\text{Total cost}}{\text{Total production unit}}$$

$$= \text{Rs. } \frac{57,000}{100} = \text{Rs. } 570 \text{ per racket.}$$

CHAPTER-3 TIME VALUE OF MONEY

3.1 Interest, Simple Interest, Compound Interest, Nominal Rate of Interest, Effective Rate of Interest.

Time value of money is a fundamental concept in comparison of the economic performance between alternatives. Money has a time value, the value of a given sum of money depends on when the money is received.

Time value of money is defined as the time dependent value of money because of changes in purchasing power (inflation or deflation) and real earning capacity of money over time. Since money has the ability to earn interest, its value increases with the time. Hence, it is the relationship between interest and time.

The value of a money expressed in terms of the amount of goods or services that one unit of money can buy is called purchasing power. As time left, purchasing power of money is reduced.

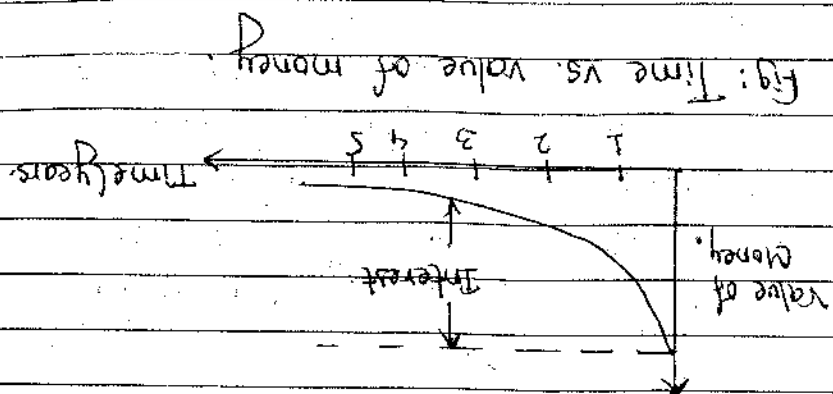


Fig: Time vs. value of money.

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Interest:-

Interest is the price of money services or it is the

the fee that is charged for the use of money. Interest depends upon the total amount of money and the length of time over which it is borrowed. Normally it is calculated as an annual rate and interest payments are made by borrowers to lenders. There are two methods of interest calculation.

(i) Simple interest

(ii) Compound interest

(1)

Simple Interest:-

When the total amount of interest earned is directly proportional to the initial principal amount, then the interest is said to be simple. Simple interest is not used in commercial practice at this modern time. For a deposit (P) at simple interest rate (r) for periods (N), the total earned interest amount (I) would be,

$$I = P \times r \times N$$

The total amount available at the end of N periods, i.e. total future amount (F) would be,

$$F = P + I$$

(ii)

Compound Interest:-

Whenever the interest charge for any interest period (year) is based on the principal loan amount plus any accumulated interest charges up to the beginning of that period, the interest is said to be compound. It is mostly used in practice.

$$F = P(1+i)^N$$

where, i = Effective interest rate per interest period.

N = Number of compounding periods.

P = Present sum of money.

F = Future sum of money.

Nominal Interest Rate:-

If the financial institution uses a unit of time less than one year in length such as a month, a quarter, etc., the institution usually quotes the interest rate on the annual basis. In this situation, the basic annual rate of interest is known as the nominal interest rate and it is represented by 'r'. Hence, nominal interest rate may be stated for any time less than one year such as, 6 months (semi annual), quarter, month, week, etc.

For example, if the interest rate is 12% per

year and compound semi-annually (or if the interest

rate is 6% per interest period and the interest period

is six months), in this situation, nominal interest is 12%

(i.e. $r = 12$). But the actual rate of interest would be

greater than 12% because of the compounding twice

during a year.

Effective Interest Rate:-

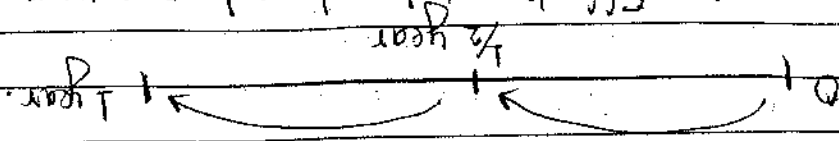
The actual rate of interest earned during one year is known as effective rate and it is also expressed on the annual basis unless specifically stated otherwise;

effective interest rate is represented by 'i'.

$$i.e. i = (1 + \frac{r}{m})^m - 1$$

where, m = compounding period per year.

Then, the interest rate per single compounding period
 $m = \text{No. of corresponding periods in year.}$
 $r = \text{Nominal interest rate per year.}$
 where, $i = \text{Effective interest rate per year.}$



We know, $F = P(1+i)^N$

Semiannually, i.e. Let us suppose that, the interest rate is compounded

Interest Rate:-
 Relationship between Effective (i) and Nominal (r)

use of continuous compounding.
 $r\%$ is used to denote the nominal rate and the

$$F = P \{ F/P, r\%, N \}$$

$$\text{or, } F = P e^{rN}$$

$$F = P(1+i)^N$$

We have,

$$\text{or, } e^r = 1+i$$

$$\text{or, } i = e^r - 1$$

$$\left[(1 + r/m)^m - 1 \right]$$

$$\text{So, } i = \lim_{m \rightarrow \infty}$$

by taking limit of m (No. of compounding periods in year) to infinity.
 compounded continuously. Therefore, it can be achieved
 flows occurs at discrete interval but it is
 compounding period infinitesimally small. Here, cash
 Continuous compounding can be thought of as making

Continuous Compounding:-

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$$\begin{aligned} l_{\text{eff}} &= (1 + l_{\text{monthly}})^{12} - 1 \\ l_{\text{quarterly}} &= (1 + l_{\text{eff}})^{1/4} - 1 \\ l_{\text{semi-annual}} &= (1 + l_{\text{quarterly}})^2 - 1 \\ l_{\text{eff}} &= (1 + l_{\text{quarterly}})^6 - 1 \end{aligned}$$

Similarly

$$l_{\text{monthly}} = (1 + l_{\text{eff}})^{1/m} - 1$$

$$\text{or, } 1 = (1 + l_{\text{eff}})^{1/m} - 1$$

$$\text{or, } 1 + 1 = (1 + l_{\text{eff}})^{1/m}$$

$$\text{or, } (1 + 1)^m = 1 + l_{\text{eff}}$$

How to find monthly when l_{eff} is given?

$$l_{\text{eff}} = (1 + l_{\text{monthly}})^m - 1$$

$$\text{or, } 1 + l_{\text{eff}} = (1 + l_{\text{monthly}})^m$$

$$P(1 + l_{\text{monthly}})^m = P(1 + l_{\text{eff}})$$

We can also write,

$$l = \frac{m}{r}$$

would be

8.2 Economic Equivalence: Present Worth, Future Worth and Annual Worth:

The process of comparing two different cash amounts at different points in time is called economic equivalence.

Different sums of money at different times are equal in economic value. Calculations for determining the economic effects of one or more cash flows are based on the concept of economic equivalence. Economic equivalence refers to the fact that a cash flow whether a single payment or a series of payments can be converted to an equivalent cash flow at any point in time.

For example, we could find the equivalent future value of F or a present amount P at interest rate i at period n , or we could determine the equivalent present value of P or n equal payment A .

Present Worth/Net Present Worth (NPW) or Net Present Value (NPV): -

It is the Present Worth of a given series of cash flow is the equivalent value/worth at the end of year zero or beginning of year 1.

Annual Worth (AW): -

Annual worth of a project is a uniform series of amount for a stated study period.

Future Worth (FW): -

Future worth of a project is the equivalent worth of all cash flows at the end of study period.

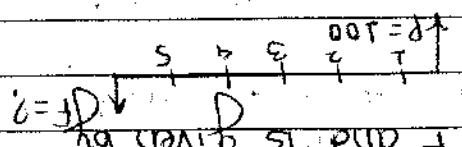
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A.

Development of formulas for Equivalence Calculation:-
Single cash flow:-
It involves the equivalence of a single present amount and its future worth. So, it deals with only a single present amount P and its future worth F and is given by

$$F = P(1+i)^N$$

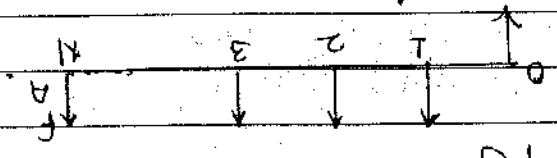


B.

Equal payment series:-

This describes the cash flow of the common installment loan contract, which arranges repayment of the loan in equal periodic installments.

$$i.e. F = A \left[\frac{(1+i)^N - 1}{i} \right]$$



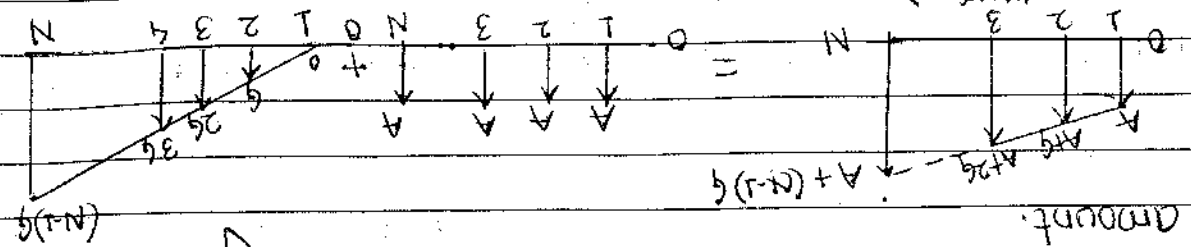
where A = Annual payment/income
fig: even series

C.

Linear gradient series:-

The amounts are not always uniform, when many transactions involve series of cash flows but they vary in some regular differences.

One common pattern of variation occurs when each other cash flow in a series increases or decreases by a fixed amount.



$$F = \left[\frac{A}{i} \left((1+i)^N - 1 \right) + \frac{g}{i} \left((1+i)^N - N - 1 \right) \right]$$

$$P = \left[\frac{A}{i} \left((1+i)^N - 1 \right) + \frac{g}{i} \left((1+i)^N - N - 1 \right) \right] \cdot \frac{1}{(1+i)^N}$$

For withdrawal: If i = interest rate per year and withdrawals monthly then $i_{month} = (1 + i_{year})^{1/12} - 1$

$$i_{year} = (1 + i_{month})^{12} - 1$$

$$eg: i_{semi} = (1 + i_{quarter})^2 - 1$$

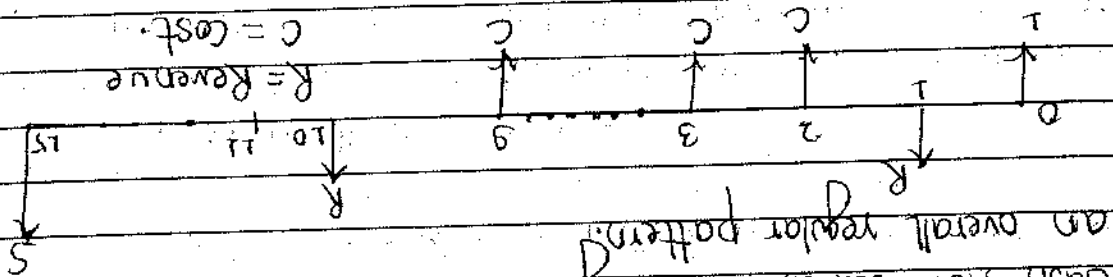
Rule 2:- From lower compounding period to higher compounding period.

$$eg: i_{quarter} = i_{year} \div 4$$

Rule 1:- From higher compounding period to lower compounding period.

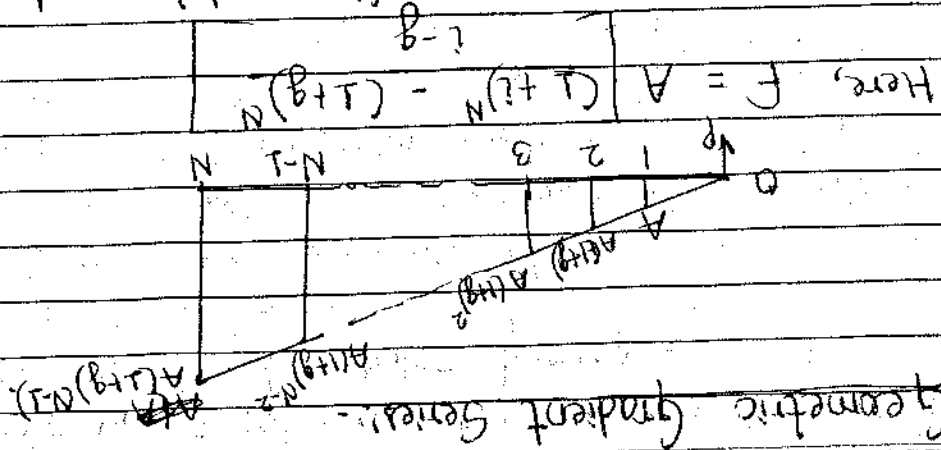
For Compounding:-

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F. Unequal payment series/irregular series:- A series of cash flow which are irregular and doesn't exhibit an overall regular pattern.

When the series in cash flow is determined not by a fixed amount like above but by some fixed rate, then the series is called geometric gradient series.



a)

Continuous compounding and discrete cash flow:-

Here, cash flows occurs at discrete interval (eg. once per year) but it is compounded continuously throughout interval. Let the nominal rate of interest per year be r , and the use of continuous compounding,

We have, $F = P(1+i)^n$

$F = P e^{rn}$

$F = P \{F/P, r\%, N\}$

Interest Factors and Symbols:-

To find	Given	Factor	Factor Name	Functional Symbol
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Single Cash Flow:-

F	P	e^{rn}	Continuous compounding	compound factor
P	F	e^{-rn}	Continuous compounding	present equivalent

Uniform Series:-

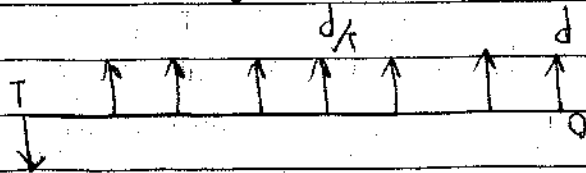
F	A	$e^{rn} - 1$	Continuous compounding	compound amount
P	A	$e^{rn} - 1$	Continuous compounding	present equivalent
A	F	$e^{rn} - 1$	Continuous compounding	sinking fund
A	P	$e^{rn} - 1$	Continuous compounding	capital recovery

b) Continuous compounding and continuous cash flow:-

It means a series of cash flows occurring infinitesimally short interval.

Let r = Nominal interest per year,

If there are P numbers of payment per year which amount to a total of one unit per year, then,



$$P = A \left\{ \frac{(e^{rN} - 1)}{e^{rN}} \right\} / \frac{1}{e^{rN}}$$

$$F = A \left\{ \frac{e^{rN} - 1}{r} \right\}$$

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Summary:-
Single Cash Flow:-

Finding F when given P $F = P(1+i)^N$

Finding P when given F $P = F(1+i)^{-N}$

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Uniform Series:-

Finding F when given A $F = A \left[\frac{(1+i)^N - 1}{i} \right]$

Finding P when given A $P = A \left[\frac{i}{(1+i)^N - 1} \right]$

Finding A when given F (Sinking Fund) $A = F \left[\frac{i}{(1+i)^N - 1} \right]$

Finding A when given P $A = P \left[\frac{i}{(1+i)^N - 1} \right]$

Q. Explain the importance of Time Value of Money.

Ans: Time Value of Money is an important concept to investors because a rupee on hand today is worth more than a rupee promised in the future. The rupee on hand today can be used to invest and earn interest or capital gain. A rupee promised in the future is actually worth less than a rupee today because of inflation. The time value of money can be broken up into two areas:

$$\text{Present value} = \frac{\text{Future value}}{(1 + \text{rate of return})^{\text{no. of periods}}}$$

$$\text{Future value} = \text{Present value} \times (1 + \text{rate of return})^{\text{no. of periods}}$$

Basic Methods of Engineering Economic Studies

Minimum Attractive Rate of Return - MARR

The minimum attractive rate of return (MARR) is that interest rate at which an investor can accept to earn or borrow money easily. It is regarded as a minimum rate of return that required to invest the money. In engineering economic analysis, ~~MARR~~ MARR is an interest rate and an indispensable factor to calculate time value of money. Generally, MARR is determined by top management from time to time and firm to firm i.e. MARR is determined from the opportunity cost viewpoint.

- Factors responsible to the determination of MARR:
1. The amount of fund (money) available for investment, and the source and cost of these funds.
 2. The number of good projects available for investment.
 3. The perceived risk that is associated with investment opportunities.
 4. The purpose of investment alternatives and their financial attractiveness.
 5. The type of organization involved (i.e. government, public, private).

MARR: $\text{government} < \text{public} < \text{private}$.
In practice, MARR is a decision criterion that is provided to the analyst for purpose of evaluating the economic merits of alternative in the course of action.

4.2

Payback Period Method - Simple and Discounted.
Payback Period is the length of time needed before an investment makes enough to recover the initial investments. The payback method calculates the number of years required for cash inflows to just equal cash outflows. A low-valued payback period is considered for the better project when choosing among mutually exclusive projects.

On the basis of way to compute, payback period method can be classified into two groups:-

(a)

Simple Payback Period: - Simple payback indicates the required time period to break even on an investment without considering time value of money. It is computed as follows:

Simple payback period (SP) = $\frac{\text{Initial Investment}}{\text{Expected Savings or Net cash inflow per period}}$

Mathematically,

$$\text{Simple payback period (SP)} = \sum_{k=1}^{\theta} (R_k - E_k) - I \geq 0$$

Where, I = Capital investment.

R_k = Revenues at the end of year k.
 E_k = Expenses at the end of year k.

Advantages:-

- 1.
- 2.
- 3.

1. Easy to calculate.
2. It is interpreted in terms of time.
3. It does not require any assumption about the project in terms of timing like useful life, interest rate, etc.

Disadvantages:-

1. Fails to measure profitability i.e. no profit is made during the payback period.
2. It ignores time value of money.
3. It doesn't account for savings after the payback period.

(b) Discounted Payback Period:-
 This method includes time value of money for determining payback period. Hence, it is defined as the number of years required to recover the investment from discounting cash flow i.e. considering time value of money.

It is computed as follows.

$$\theta' = \sum_{k=1}^n (R_k - E_k) \cdot (P/F, i\%, k) - I = 0$$

Where, θ' = Discounted payback period,

R_k = Revenues at the end of year k
 E_k = Expenses at the end of year k
 I = Minimum Attractive Rate of Interest
 I = Capital investment.

Steps:-

1. Discount each of the future cash flows into present.
2. Calculate the required number of years to recapture initial investment.

It should be noted that, it can overcome one major shortcoming of simple payback period that is consideration of time value of money but all other shortcomings are still found in this method.

Advantages:-

- Consider the time value of money
- Consider the riskiness of the project's cash flow.

Disadvantages:-

- Ignore cash flows beyond the discounted payback period.
- Requires an estimate of the cost of capital in order to calculate payback.
- No concrete decision criteria that indicate whether to investment increases the firm's value.

4.3 Equivalent Worth Methods: Present Worth Method, Future Worth Method and Annual Worth Method:-

Equivalent worth method convert all cash inflows into equivalent worth at some point of time present or annual (or future) by using an interest rate equal to minimum attractive rate of return (MARR).

(a) Present Worth Method:-

Present worth method is based on the concept of equivalent worth of all cash flows to the base point or beginning point in time called the present. Present Worth (PW) or Net Present Worth (NPW) or Net Present Value (NPV) of a given series of cash flow is the equivalent value/birth at the end of year zero or beginning of year 1. Hence, according to this method, all cash inflows and outflows are discounted to the base point at an interest rate for the economic study period in the following manner.

$$PW(i\%) = F_0(1+i)^0 + F_1(1+i)^{-1} + F_2(1+i)^{-2} + \dots + F_N(1+i)^{-N}$$

$$\text{or, } PW(i\%) = \sum_{k=0}^N F_k(1+i)^{-k}$$

Where,

i = Effective interest rate or MARR per compounding period.
 k = Index for each compounding period
 F_k = Future cash flow at the end of period k .
 N = No. of compounding periods in economic study.

Decision Rule:

While using the PW method for economic evaluation of any project, we should consider the following decision rule.

- If $PW(i\%) > 0$, accept the project.
- If $PW(i\%) = 0$, remain indifferent.
- If $PW(i\%) < 0$, reject the project.

(b) Future Worth Method (FW):

Future worth of a project is the equivalent worth of all cash flows at the end of study period. Hence, according to this method all cash inflows and outflows are compounded forward to a reference point in time called future worth. It has given MARR to obtain the future worth. It has become more popular in recent years. It can be calculated as:-

$$FW(i\%) = \sum_{k=0}^N F_k(1+i)^{N-k}$$

Where

i = Effective interest rate

k = Index for each compounding period.

$FV =$ Future cash flows at the end of year k .
 $N =$ Number of compounding periods

Decision rule:-

If $FV(i\%) > 0$, accept the project
If $FV(i\%) = 0$, remain indifferent
If $FV(i\%) < 0$, reject the project

(c)

Annual Worth Method:-

Annual worth of a project is a uniform series of

amount for a stated study period. It provides

the basis for measuring investment worth into a

series of equal payments at the end of each period.

AW of a project is its annual equivalent receipts (R)

minus annual equivalent expenses (E) less its annual

equivalent capital recovery (CR) amount with given

MARR.

ie. ~~AW~~ $AW(i\%) = R - E - CR$

Decision Rule:-

If $AW(i\%) > 0$, accept the project
If $AW(i\%) = 0$, remain indifferent
If $AW(i\%) < 0$, reject the project

Application of AW:-

Asset replacement.

Break-even analysis

Make or buy decision.

Studies dealing with manufacturing costs.
Economic value added analysis.

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Capital Recovery (CR):-

The annual equivalent of the capital cost covering

(a) Internal Rate of Return (IRR): - It is that interest rate (return rate) which equates the equivalent worth of an alternative's cash inflows to the equivalent worth of cash outflows. In other words, IRR is the break-even interest rate at which equivalent worth of a project's cash flow is zero. At this particular rate of return, the equivalent worth of revenue generated by the project is enough to bear the equivalent worth. IRR is the term used for rate of interest return that stresses the interest earned by project that is internally invested.

4.4 Rate of Return Methods: Internal Rate of Return (IRR), Method and External/Modified Rate of Return (ERR/MIRR) Method: - Rate of return is defined as an annual rate of profit resulting from an investment. It is especially relevant to evaluate mutually exclusive alternatives.

where,
 I = initial investment for the period.
 S = Salvage value at the end of study period
 N = project study period

$CR(i) = I(A/P, i\%, N) - S(A/F, i\%, N)$	or,	$CR(i) = (I-S)(A/P, i\%, N) + S(i\%)$
$CR(i) = I(A/P, i\%, N) + I(i\%)$	or,	$CR(i) = (I-S)(A/F, i\%, N) + I(i\%)$

The loss in the value of the asset and interest on invested capital is known as capital recovery amount. It is calculated as,

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Based on PW formulation:-

$$PW(i\%) = 0$$

$$PW_{\text{inflow}} - PW_{\text{outflow}} = 0$$

$$PW(i\%) = \sum_{k=0}^N R_k (P/F, i\%, k) - \sum_{k=0}^N E_k (P/F, i\%, k) = 0$$

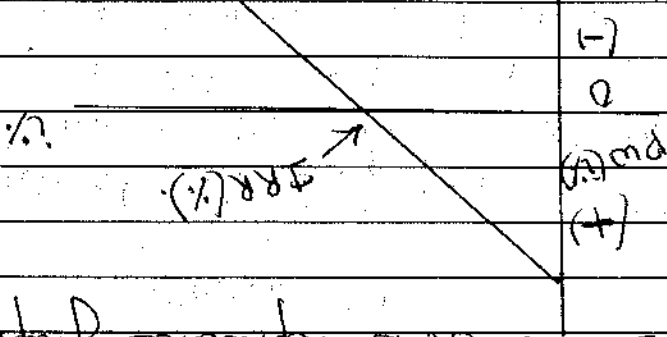
Where,

R_k = revenues for the k^{th} year.

E_k = expenditure for the k^{th} year.

N = Project life or study period.

If we plot a curve of PW for different values of $i\%$, the IRR can be computed by spotting where the PW is zero. It is represented graphically as:-



Although, IRR can be computed by equating any equivalent worth to zero, it is advisable to use the PW formulation as it is found extensively used.

Based on FW formulation:-

$$FW(i\%) = \sum_{k=0}^N (F/P, i\%, N-k) - \sum_{k=0}^N E_k (F/P, i\%, N-k) = 0$$

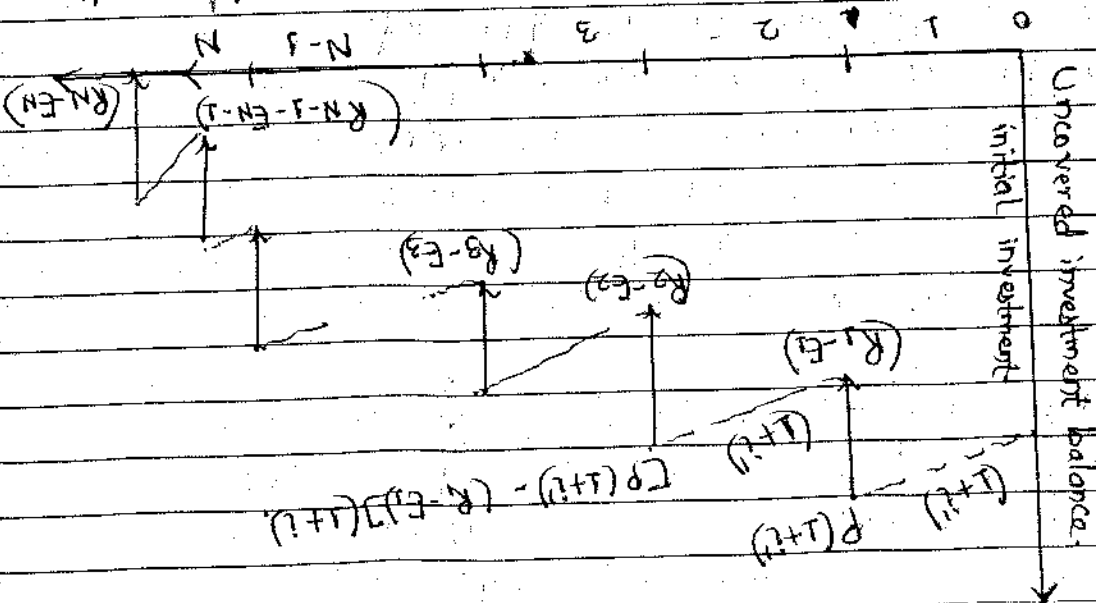
Based on AW formulation,

$$AW(i\%) = \sum_{k=0}^N R_k (F/P, i\%, N-k) (A/F, i\%, N) - \sum_{k=0}^N E_k (F/P, i\%, N-k) (A/F, i\%, N)$$

Decision Rule:-
 If $IRR > MARR$, accept the project.
 If $IRR = MARR$, remain indifferent.
 If $IRR < MARR$, reject the project.

1. Develop an equation for equivalent worth of any point of time indicating rate of interest by $i\%$ whose value is to be found out.
 2. Equate the developed equation as to zero.
 3. Solve it to get the value of $i\%$ which would be IRR .
- Method of calculating IRR:-**
 The following steps are followed to calculate the IRR of any project's cash flow:
 1. Develop an equation for equivalent worth of any point of time indicating rate of interest by $i\%$ whose value is to be found out.

Fig: An unrecovered investment balance diagram to illustrate IRR.



1. Drawbacks of IRR Method:-
The IRR method is based on the assumption that the recovered funds, if not consumed in each time period, are reinvested at i% rather than at MARR. This is not always practical.
When the algebraic sign of the cash flow changes more than one in the series, it is possible to obtain multiple rates of return.
IRR method does not consider the scale of investment.
Therefore, it may be misleading when choosing between mutually exclusive projects that have substantially different outlays.
IRR method involves linear interpolation for non-linear function and when solved manually by trial and error process, may not give accurate result and is more time consuming.
- (b) External Rate of Return (ERR):-
ERR method eliminates the drawback of reinvestment assumption to some extent. This method takes into account the external reinvestment rate (e) at which net cash flows generated by a project over its life can be reinvested outside the firm.
- Steps of ERR Calculation:-
1. All cash outflows are discounted to period zero (the present) at e% per compounding period.
$$\sum_{k=0}^N F_k (P/F, e\%, k)$$

2. All cash inflows are compounded to period N at e%
$$\sum_{k=0}^N R_k (F/P, e\%, N-k)$$

Benefit-cost ratio Method:-
Benefit-cost relation is a special tool of cost benefit analysis (CBA) and it is used to evaluate public sector project, although it can be extended to any private sector project. As the name implies B-C relation, it involves the calculation of a ratio of benefit to cost. Benefit-cost (B/C) ratio can be defined as the ratio of the equivalent worth of benefits to the equivalent worth of costs. Whether evaluating a project in the private sector or in a public sector, the time value of money must be considered to account for the timing of cash flows occurring after inception of

Advantages of ERR over IRR:-
1. It does not need trial and error process to solve for i%.
2. There is no possibility of multiple rates of return.

Decision rule:-
If $ERR > MARR$, accept the project.
If $ERR = MARR$, remain indifferent.
If $ERR < MARR$, reject the project.

Where, R_k = net revenue in period k
 E_k = net expenses in period k
 N = Project's life.
 $E\%$ = External reinvestment rate.

2. ERR is the interest rate that equivalence between the quantities.
i.e. $\sum_{k=0}^N E_k (P/F, E\%, k) (E/P, i\%, N) = \sum_{k=0}^N R_k (F/P, E\%, N-k)$

The project. Thus, B/C ratio is actually a ratio of discounted benefits to discounted cost. It is also known as the saving investment ratio. Two commonly used formulation of B/C ratio are as follows:-

A. Conventional B/C ratio:-

It is the ratio of gross benefits to costs and expressed as:-

(a) With PW formulation

$$\text{B/C ratio} = \frac{PW(B)}{PW(I) - PW(S) + PW(0 \neq M)}$$

(b) With FW formulation:-

$$\text{B/C ratio} = \frac{FW(B)}{FW(I) - FW(S) + FW(0 \neq M)}$$

(c) With AW formulation

$$\text{B/C ratio} = \frac{AW(B)}{AW(I) - AW(S) + AW(0 \neq M) + CR + AW(0 \neq M)}$$

B. Modified B/C ratio:-

It is the ratio of net benefits to costs and expressed as:-

(a) With PW formulation

$$\text{B/C ratio} = \frac{PW(B) - PW(0 \neq M)}{PW(I) - PW(S)}$$

(b) With FW formulation

$$\text{B/C ratio} = \frac{FW(B) - FW(0 \neq M)}{FW(I) - FW(S)}$$

(c) With AW formulation,

$$B/C \text{ ratio} = \frac{AW(B) - AW(0 \& M)}{AW(I) - AW(S)} = C.R.$$

where,
 B = Benefits of the proposed project.
 $0 \& M$ = Operation and maintenance cost
 S = Salvage value of the proposed project.
 I = Initial investment in "
 $C.R.$ = Capital recovery amount.

Decision Rule:-

If $B/C \text{ ratio} > 1$, accept the project
 If $B/C \text{ ratio} = 1$, remain indifferent.
 If $B/C \text{ ratio} < 1$, reject the project.

Project is feasible if $Benefit(B) > Cost(C)$, or $B/C > 1$.

CHAPTER - 5

Comparative Analysis of Alternatives

Some terms:-

1.

Independent projects:-

A project is said to be independent if its selection is free from the acceptance or rejection of any other projects in a group.

2.

Dependent projects:-

The projects related to one another in such a way that the acceptance or rejection of one project influences the acceptance of others are called dependent projects. The dependencies among these projects may be classified into following two types:-

a

Mutually exclusive projects:-

When the selection of one of the feasible design or alternatives excludes the choice of any of the others, the alternatives are called mutually exclusive. In other words mutually exclusive projects are those when one project is chosen all other are excluded in the group.

b

Contingent projects:-

Two or more projects are said to be contingent if the acceptance of one project requires the acceptance of another. For example, the decision to connect internet is dependent upon the purchase of a computer however, the computer can be purchased without internet connection.

b) Investment alternatives: - If the first alternative capital investment is base.

g) Cost alternative: - The first alternative (Least Establish a base alternative.

1. Arrange the feasible alternative, based on increasing capital investment.

Procedure for incremental analysis: - For this purpose, we adopt incremental analysis in the way as (PW, AW and FV) method.

The projects having highest value of IRR, ERR and BCR may not be the preferred alternatives they are just relative measure and cannot be method:-

Rate of Return method and Benefit cost ratio

alternative with highest equivalent worth is selected as best alternative. (PW, AW or FV) i.e. equivalent worth. And the alternatives are compared on the basis of In equivalent worth method, two or more

Annual worth method:- Equivalent one with least payback period is selected. or more alternative projects are compared and to recover the cost of an investment for two In payback period, the length of time required Payback period:-

Return Methods and Benefit Cost Ratio Method:- Method, Equivalent worth Method, Rate of having some useful life by Payback Period

5.1. ~~Cost~~ Comparing Mutually Exclusive Alternatives

life are changed into project having same useful
 two or more alternatives having different useful
 Repeatability Assumption: -

multiple of the useful lives
 period can be infinite in length or a common
 may be used in their comparison if the study
 alternatives are different, the repeatability assumption
 when the useful lives of mutually exclusive
 Co-terminated Assumption, Capitalized Worth Method:
 different useful lives by Repeatability Assumption,
 Comparing Mutually Exclusive Alternatives having

5.2

cash flow was acceptable.
 the last one for which the incremental
 b. Repeat, and select the preferred alternative

alternative as the current best.
 alternative. Otherwise, retain the last acceptable
 choose the next alternative as the current best
 the current selected alternative is acceptable.
 next (higher capital investment) alternative and
 2. If the incremental cash flow between the
 alternative until no more alternative exist.
 Use iteration to evaluate differences between

3.

selected.
 obtained, then do nothing alternative is
 acceptable alternative is obtained. If none is
 criteria (PW, etc) values. Continue until an
 capital investment and check the profitability
 choose the next alternative in order of increasing
 base. If the first alternative is not acceptable,
 is acceptable ($IRR > MARR$), select it as the

life by expanding their life upto atleast common year
 The economic consequences that are estimated
 to happen in all succeeding life spans replacement
 Actual situations in engineering practice seldom
 meet both conditions.

Co-terminated Assumptions: -
 A finite and identical study period is used for
 all alternatives.

This planning horizon, combined with appropriate
 adjustments to the estimated cash flows, puts the
 alternatives on a common and comparable basis.
 Used when repeatability assumption is not applicable.
 This is the approach most frequently used in
 engineering practice.

1) Useful life shorter than study period
 A) Cost alternative: Each cost alternative must provide
 same level of service as study period;
 Control for service or least equipment for remaining time.
 Repeat part of useful life of original alternative until
 study period ends.

B) Investment alternative: Assume all cash flows reinvested
 in other opportunities of MARR to end of study period.
 This is mostly applicable in our syllabus numerical
 problem

b) Study period shorter than useful life: -
 When the study period is shorter than the useful
 life of any project or an asset, we truncate the

alternative at the end of study period using an estimated market value. This assumes that disposable assets will be sold at the end of study period at that value.

Capitalized worth method:-

The process in which present worth of all revenues and for expenses over an infinite length of time is calculated is called capitalized worth (CW) method.

This is a convenient basis for comparing mutually exclusive alternatives when the period of needed service is indefinitely long and repeatability assumption is applicable.

The CW of a perpetual series of end of period uniform payments A , with interest at $i\%$ per period is $A(P/A, i\%, \infty)$.

$$\therefore CW(i\%) = \frac{A}{i}$$

Hence, the CW of a project with interest $i\%$ per year is the annual equivalent of the project over its useful life divided by i .

Chapter 5: Comparative Analysis of Alternatives

Meaning of Independent Project

Most firms have a number of unrelated investment opportunities available.

An **independent project** is a project that may be accepted or rejected without influencing the accept-reject decision of another independent project. For example, the purchase of a milling machine, office furniture, and a forklift truck constitutes three independent projects.

Only projects that are economically independent of one another can be evaluated separately.

In other words, the decision regarding any one project has no effect on the decision to accept or reject another project. Such projects are said to be **independent**.

For a single project, we have two investment alternatives: to accept or reject the project.

For two independent projects, we can have four investment alternatives: (1) to accept both projects, (2) to reject both projects, (3) to accept only the first project, and (4) to accept only the second project.

Meaning of Dependent Project

In many decision problems, several investment projects are related to one another such that the acceptance or rejection of one project influences the acceptance or rejection of others.

Two such types of dependencies are **mutually exclusive** projects and **contingent** projects.

a. Meaning of Contingent Project

We say that two or more projects are **contingent** if the acceptance of one requires the acceptance of the other.

For example, the purchase of a computer printer is dependent upon the purchase of a computer, but the computer may be purchased without purchasing the printer.

Suppose the acceptance of C is contingent on the acceptance of both A and B, and the acceptance of B is contingent on the acceptance of A. Then the number of decision alternatives can be formulated as follows:

Alternative	X_A	X_B	X_C
1	0	0	0
2	1	0	0
3	1	1	0
4	1	1	1

b. Meaning of Mutually Exclusive Project

Many engineering situations we face are with selecting the most economically attractive project from a number of alternative projects, all of which solve the same problem or meet the same need.

It is unnecessary to choose more than one project in this situation, and the acceptance of one automatically entails the rejection of all of the others. Such projects are said to be **mutually exclusive**. Several alternatives are **mutually exclusive** when any one of them will fulfill the same need and the selection of one of them implies that the others will be excluded.

Take, for example, buying versus leasing an automobile for business use; when one alternative is accepted, the other is excluded.

Comparing Mutually Exclusive Alternatives having same useful life: PW, AW, FW and B/C Ratio

Compute the $PW(i)$ for each alternative and select the one with the largest $PW(i)$.

When you compare mutually exclusive alternatives with the *same revenues*, they are compared on a *cost-only basis*. In this situation (because you are minimizing costs, rather than maximizing profits), you should accept the project that results in the *smallest, or least negative, NPW*.

Similarly, use same decision criteria for AW, FW, and B/C Ratio method while comparing and selecting between multiple alternatives.

Comparing Mutually Exclusive Alternatives having different useful lives by Repeatability Assumption

Two projects will have two different useful life, so comparing is difficult for them, Thus:

Assume that projects will be repeated for certain times so both projects will have same useful life

Times to be repeated for any project can be accessed by computing LCM of useful life of both projects

Their overall PW/FW will be accessed (including their repeated project cycle assuming they are repeated at the end of each cycle, till their useful life becomes same)

Compare PW/FW with each other and select best one.

Comparing Mutually Exclusive Alternatives having different useful lives by Co-terminated Assumption

Two projects will have two different useful life, so comparing is difficult for them, Thus:

Access shorter project's PW/FW (Project A)

For projects with longer useful life (Project B), assume this project will be terminated on shorter projects useful life, Thus:

Find CR of initial investment and salvage value of Project B

Find PW of CR (computed on above step) for remaining useful life of Project B beyond short project (Project A)

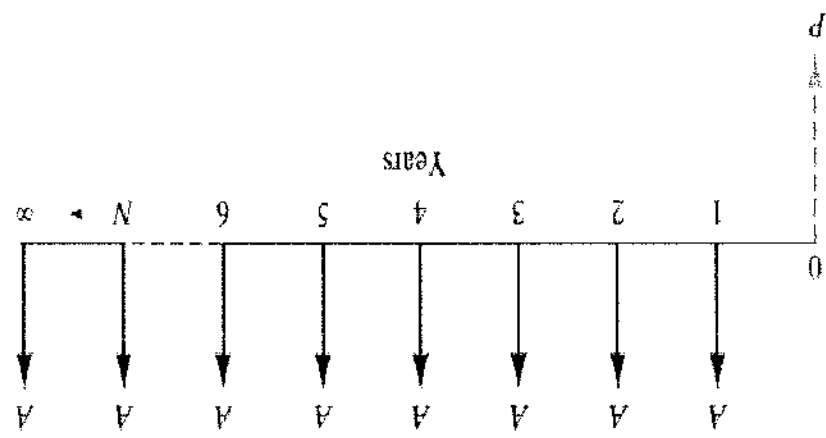
Find PW/FW of overall project including PW of CR (accessed in step 2)

Compare PW/FW with each other and select best one.

Comparing Mutually Exclusive Alternatives having perpetual lives by Capitalized Equivalent Method

Another special case of the PW criterion is useful when the life of a proposed project is **perpetual** or the planning horizon is extremely long (say, 40 years or more).
 Many public projects, such as bridges, waterway structures, irrigation systems, and hydroelectric dams, are expected to generate benefits over an extended period (or forever).

We will use the **capitalized equivalent** (CE(*i*)) method for evaluating such projects.

$$PW(i) = A(P/A, i, N \rightarrow \infty) = \frac{A}{i}$$


Origin/Sources of Project Risks

In previous chapters, cash flows from projects were assumed to be known with complete certainty; our analysis was concerned with measuring the economic worth of projects and selecting the best ones to invest in.

Although that type of analysis can provide a reasonable basis for decision making in many investment situations, we should certainly consider the more usual *uncertainty*.

In this type of situation, management rarely has precise expectations about the future cash flows to be derived from a particular project.

In fact, the best that a firm can reasonably expect to do is to estimate the range of possible future costs and benefits and the relative chances of achieving a reasonable return on the investment.

The decision to make a major capital investment such as introducing a new product requires information about cash flow over the life of a project. The profitability estimate of an investment depends on cash flow estimations, which are generally uncertain.

The factors to be estimated include the total market for the product; the market share that the firm can attain; the growth in the market; the cost of producing the product, including labor and materials; the selling price; the life of the product; the cost and life of the equipment needed; and the effective tax rates. Many of these factors are subject to substantial uncertainty.

We use the term **risk** to describe an investment project whose cash flow is not known in advance with absolute certainty, but for which an array of alternative outcomes and their probabilities (odds) are known.

We will also use the term **project risk** to refer to variability in a project's NPV.

A greater project risk usually means a greater variability in a project's NPV, or simply that the *risk is the potential for loss*. This chapter begins by exploring the origins of project risk.

In particular, managers have no way of determining either the probability that a project will lose money or the probability that it will generate large profits.

Because cash flows can be so difficult to estimate accurately, project managers frequently consider a range of possible values for cash flow elements.

If a range of values **Risk**: The chance that an investment's actual return will be different than expected.

Boston Metal Company (BMC), a small manufacturer of fabricated metal parts, must decide whether to enter the competition to become the supplier of transmission housings for Gulf Electric, a company that produces the housings in its own in-house manufacturing facility, but that has almost reached its maximum production capacity. Therefore, Gulf is looking for an outside supplier. To compete, BMC must design a new fixture for individual cash flows is possible, it follows that a range of values for the NPV of a given project is also possible. Clearly, the analyst will want to gauge the probability and reliability of individual cash flows and, consequently, the level of certainty about the overall project worth.

A common approach is to make single-number "best estimates" for each of the uncertain factors and then to calculate measures of profitability, such as the NPV or rate of return for the project.

This approach, however, has two drawbacks:

1. No guarantee can ever ensure that the "best estimates" will match actual values.
2. No provision is made to measure the risk associated with an investment, or the project risk.

Method of Describing Project Risks; Sensitivity Analysis, Breakeven Analysis, Scenario Analysis

We may begin analyzing project risk by first determining the uncertainty inherent in a project's cash flows. We can do this analysis in a number of ways, which range from making informal judgments to calculating complex economic and statistical quantities.

In this section, we will introduce three methods of describing project risk:

(1) sensitivity analysis

One way to glean a sense of the possible outcomes of an investment is to perform a sensitivity analysis. This kind of analysis determines the effect on the NPW of variations in the input variables (such as revenues, operating cost, and salvage value) used to estimate after-tax cash flows.

A **sensitivity analysis** reveals how much the NPW will change in response to a given change in an input variable.

In calculating cash flows, some items have a greater influence on the final result than others. In some problems, the most significant item may be easily identified.

For example, the estimate of sales volume is often a major factor in a problem in which the quantity sold varies with the alternatives. In other problems, we may want to locate the items that have an important influence on the final results so that they can be subjected to special scrutiny.

Sensitivity analysis is sometimes called "what-if" analysis, because it answers questions such as "What if incremental sales are only 1,000 units, rather than 2,000 units? Then what will the NPW be?"

Sensitivity analysis begins with a base-case situation, which is developed by using the most likely values for each input. We then change the specific variable of interest by several specified percentage points above and below the most likely value, while holding other variables constant.

Next, we calculate a new NPW for each of the values we obtained.

A convenient and useful way to present the results of a sensitivity analysis is to plot **sensitivity graphs**. The slopes of the lines show how sensitive the NPW is to changes in each of the inputs: The steeper the slope, the more sensitive the NPW is to a change in a particular variable.

Sensitivity graphs identify the crucial variables that affect the final outcome most.

Graphic displays provide a useful means to communicate the relative sensitivities of the different variables to the corresponding NPW value.

However, **sensitivity graphs** do not explain any interactions among the variables or the likelihood of realizing any specific deviation from the base case.

Certainly, it is conceivable that an answer might not be very sensitive to changes in either of two items, but very sensitive to combined changes in them.

(2) Break-even analysis

When we perform a sensitivity analysis of a project, we are asking how serious the effect of lower revenues or higher costs will be on the project's profitability.

Managers sometimes prefer to ask instead how much sales can decrease below forecasts before the project begins to lose money. This type of analysis is known as **break-even analysis**.

In other words, break-even analysis is a technique for studying the effect of variations in output on a firm's NPW (or other measures). We will present an approach to break-even analysis based on the project's cash flows.

To illustrate the procedure of break-even analysis based on NPW, we use the generalized cash flow approach.

We compute the PW of cash inflows as a function of an unknown variable (say, x), perhaps annual sales. For example,

$$\text{PW of cash inflows} = f(x)_1.$$

Next, we compute the PW of cash outflows as a function of x :

$$\text{PW of cash outflows} = f(x)_2$$

NPW is, of course, the difference between these two numbers. Accordingly, we look for the break-even value of x that makes Note that this break-even value is similar to that used to calculate the internal rate of return when we want to find the interest rate that makes

$$F(x)_1 = f(x)_2.$$

the NPW equal zero. The break-even value is also used to calculate many other similar "cutoff values" at which a choice changes.

(3) scenario analysis.

Although both sensitivity and break-even analyses are useful, they have limitations. Often, it is difficult to specify precisely the relationship between a particular variable and the NPW. The relationship is further complicated by interdependencies among the variables.

Holding operating costs constant while varying unit sales may ease the analysis, but in reality, operating costs do not behave in this manner. Yet, it may complicate the analysis too much to permit movement in more than one variable at a time.

Scenario analysis is a technique that considers the sensitivity of NPW both to changes in key variables and to the range of likely values of those variables. For example, the decision maker may examine two extreme cases: a "worst-case" scenario (low unit sales, low unit price, high variable cost per unit, high fixed cost, and so on) and a "best-case" scenario.

The NPWs under the worst and the best conditions are then calculated and compared with the expected, or base-case, NPW.

The need to estimate probabilities leads us directly to our next step: developing a probability distribution (or, put another way, the probability that the variable in question takes on a certain value).

If we can predict the effects on the NPW of variations in the parameters, why should we not assign a probability distribution to the possible outcomes of each parameter and combine these distributions in some way to produce a probability distribution for the possible outcomes of the NPW?

CHAPTER - 6 RISK ANALYSIS

The chance that an investment's actual return will be different than expected is called risk. In another words risk is a condition where there is a possibility of deviation between desired and expected outcome but for which an array of alternative outcomes and their probabilities are known.

6.1

Origin/Source of Project Risks:-

Risks are caused by the lack of precise knowledge regarding future business condition, technological development, market demand and so on. It is useful to consider some of the factors/sources that affect the risk and uncertainty involved in the analysis of the future consequences of an engineering project. There are four major sources of risk, and uncertainty that affect project's NPV i.e. engineering economy studies.

1.

Cash Flow estimate:-

The first source is the possible inaccuracy of the cash flow estimates used in the study. The accuracy of the cash-inflow estimate is difficult to determine. If they are based on past experience, a fair degree of reliance may be placed on estimates. Otherwise, the estimate would become more uncertain.

2.

Nature of business:-

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- Another source of risk and uncertainty is the type of business involved in the economy. All business are not same nature because some types of business operations are less stable than others. For example, mining enterprises/projects are more risky than farming projects with the expectation of stability and income generation.
3. Rate of interest / Rate of inflation:-
Current health of economy and future expectations of economic conditions are responsible for determining market rate of interest and its stability. Hence, whenever capital is to be invested in an engineering project, an investor should be considered in deciding what risk is present on rate of interest.
4. Study period:-
Another risk source is the length of study period used in the analysis. A long study period normally decreases the probability of all the factors of estimated i.e. long study period generally increases the uncertainty of a capital investment and economic return.
Other risks are:-
5. Unclear specification
6. Volatile and unpredictable future
7. Social risks
8. Variability in output

6.2 Methods of Describing Project Risks: Sensitivity Analysis, Break-even Analysis, Scenario Analysis

#

Sensitivity Analysis:-

Sensitivity analysis reveals how much the NPV of a project will change in response to a given change in one input variable/parameter. It is a general non probabilistic methodology readily available, to provide information about potential impact of uncertainty in selected factor estimates. Its routine use is fundamental to developing economic information useful in the decision process. In general, sensitivity means the relative magnitude of change in the measure of merit caused by one or more changes in estimated study factor values.

#

Steps for sensitivity graph analysis:-

1. Plot PV , FV , AW , ERR or BRR on y-axis.
2. % error in estimate of parameter value is plotted on x-axis.
3. The slope of the line shows level of sensitivity.
4. The steeper the slope, more sensitivity.

Sensitivity graphs identify the critical variables that affect the final economic outcome most.

#

Break-even Analysis:-

Break-even analysis is a most common methodology used for economic evaluation of new investment alternatives. It determines the value of a critical factor at which economic trade-off are balanced. When the equivalent worth of cash inflows and outflows are dependent upon a single factor, the value of this factor for which

equivalent worth of cash outflows equals the equivalent worth of cash inflows is called breakeven point and its analysis is called breakeven analysis.

Break-even Analysis for a single Project:-

The costs of producing a good can be split into two main parts:-

fixed costs (CF):- fixed costs are those that remain the same for the firm however many goods are produced. All start-up costs such as land, building, rent, insurance, etc. are considered as fixed cost.

Variable costs (CV):-

Variable costs vary with the number of items produced. It includes cost of wages, raw materials, electric, fuel, etc.

Total cost is the sum of fixed cost (CF) and variable cost (CV).

$$CT = CF + CV$$

The linear relationship assumed for variable costs, $CV = VC(X)$

$$\therefore CT = CF + VC(X)$$

where VC = variable cost per unit output.

X = quantity of production.

$$\text{Also, } ST = SP(X)$$

from break-point even point,

$$ST = CT$$

$$\text{or, } SP(X) = CF + VC(X)$$

$$\text{or, } CF = X(SP - VC)$$

$$\text{or, } X = \frac{CF}{SP - VC}$$

$$SP - VC$$

Obviously, to reach a break-even position, the

Break-even analysis for comparing two alternatives:-

$$EWA = f_1(y) \quad \text{and} \quad EWB = f_2(y)$$

alternative 8.

as to determine Break-even point of

Define the common variable and its dimensional unit

4. If the anticipated level is below than break-even

If the anticipated level is below than break-even value, select alternative with the higher variable cost.

Scenario Analysis:-

Sensitivity and break-even analysis are useful to show the effect of one common variable and NPW. But it is difficult to specify precisely the relationship between a particular variable and the NPW because in reality there may be inter-dependent between the variables. Hence, the relationship is further complicated by inter-dependencies among the variables at a time.

Scenario Analysis is a technique that considers that sensitivity of NPW due to change in key variables at a time and the range of likely values of those variables. For example:- the decision maker may examine two extreme case: a 'worth-case' scenario (low sales/price/high cost and so on) and a 'best-case' scenario (high sales/price/low cost and so on).

The NPWs under the worse and the best scenarios are then calculated and compared with the expected (or base case) NPW.

#

Certainty:-

It is defined as a state of knowledge in which the decision maker knows in advance the specific outcome to which each alternative will invariably lead. So, under certainty, the decision maker has perfect knowledge about future result of whatever decision he might make.

Risks:-

It is defined as a state of knowledge in which

#

Uncertainty:-

decision maker can predict different alternatives and each alternative leads to one set of outcome and probability of their occurrence is known. So risk refers to a situation where a project has a number of possible alternative outcomes, but the probability of each occurring is known.

It is a state in which there are so many alternatives whose probability can not be estimated i.e. it is decision situation where several states/alternatives are possible but sufficient information is not available to assign probability values to their occurrence.

Ecological Limits and Economic Development.

CHAPTER - 7

The chapter heading deals about the role/issue of ecological constraints for the economic development and resulting environmental degradations.

Economic Theory and Ecological Limit.

Economic theory is a way to explanation of economic activities and its impact on various grounds.

After 1930s, due to the event of great depression in the world, economic theory have shifted to deal economic behaviour by explaining in aggregate form which refers macro economic theory and it is also called Keynesian economic theory. The micro economic theory is specially focus to achieve help level of employment and income.

Similarly, after 1980s, economic thoughts have

growing concern for ecological constraints that means

limiting availability of natural resources for economic

development and resulting environmental degradation.

Hence, at present the theory of ecological economics

has become more important issue throughout the world.

Ecological economics point out that except

sunlight all other resources like air, water, food,

wood, fiber, minerals, and energy sources are the

foundation of the economy and come from the earth.

Our lives and economy are completely dependent upon

Ecological Limit:-
Ecological limit refers functioning ecological mechanism of the earth to keep economic development with sustainable manner that must be limiting availability of natural resources technically. Hence, Ecological limit is to keep intact ecological services which is need to keep the safe of economy within the capacity of global ecosystem to sustain it. The result of growing economic activity has been increasing costs in many areas of economy like atmosphere, environment, minerals, use of land, forest, water and so on. There are limits to the earth's capacity to act as a source of material inputs feeding for economic system and sink for wastage products.

and bounded by the ecosystem and materials of this planet (earth) that means in reality our economy is a subsystem of the earth. Modern economic theory has tended to over look about the importance of ecosystem, the use of land and environment for balancing ecological constraints that refers ecological economics. It is a new academic field that attempts to correct critical environment, social and economic problems over time. Similarly, it is more responsible for modern realities like ecological deterioration & record inequality.

4.2 Concept of sustainable development.

Since 1980s, the growing concern for environment has given birth to the concept of sustainable development. The term 'sustainable development' was brought into common use by the report "Our common future" in 1987 A.D.

This report defines, "sustainable development is meeting the needs of present generation without compromising the needs of future generations".

The concept of sustainable development focus the principle of resource management and carrying capacity of ecosystem. This principle is specially associated with the consumption pattern and recreation of renewable resource by human being according to natural life cycle. In another word, the resources should be utilized by keeping intact the natural life cycle so that the future should create natural resources continuously.

Hence, sustainable development is an art of resource management that fulfills the need of present generation and also guarantees the fulfillment needs of the future generation.

4.3 Ecological Footprint.

It refers the total direct and indirect requirement of land use for the given unit of crop production with global average i.e. primary productivity for resource generation and waste absorption for the agricultural and industrial production and infrastructure development of an economy. It also describes how the trade flows of

Open economic would become adjustment by the land requirement for imports and exports. Ecological footprint would be different for different regions of the world depending upon the dominance of carbon footprint i.e. require for recycling ~~car~~ carbon-dioxide by the forests, ocean and soil.

7.4 Overcoming Ecological Limits:
This chapter discusses how technology and human values can play a significant role in creating space for economic development by relaxing the ecological constraints. Hence, overcoming ecological limits point out the scope maintaining between economic growth and environment through the development of renewable energy sources, decarbonization, recycling of wastes, taking organic view of technology and resource development. Hence, each and every development institutions should follow the implementation policy of sustainable development by realizing the relationship between economic growth and natural environment.

Economic Theory and Ecological Limit:

Ecological economics point out that except sunlight all other resources like air, water, food, wood, fiber, minerals & energy sources are the foundation of the economy and come from the earth. our lives & economy are completely dependent upon & bounded by the ecosystem & materials of this planet (earth) that means in reality our economy is a subsystem of the modern economic has tended to overlook about the importance of ecosystem, the use of land and environment for balancing ecological constraints that refers ecological economics. It is a new academic field that attempts to correct critical environment, social & economic problems over time. Similarly, it is more responsible for modern realities like ecological deterioration & record inequality.

Ecological Limit:

The result of growing economic activity has been increasing costs in many areas of the economy like atmosphere, environment, minerals, use of land, forest, water & so on. There are limits to the earth's capacity to act as a source of material inputs feeding for the economic system & sink for the wastage products. Hence, functioning ecological mechanism of the earth to keep economic development with sustainable manner there must be limiting (minimum) availability of natural resources that technically refers ecological limit.

Hence, ecological limit is to keep intact ecological services which is need to keep the sale of economy within the capacity of global ecosystem to sustain it.

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Sustainable development:

Sustainable development is defined as a process of meeting human development goals while sustaining the ability of natural systems to continue to provide the natural resources and ecosystem services upon which the economy and society depends.

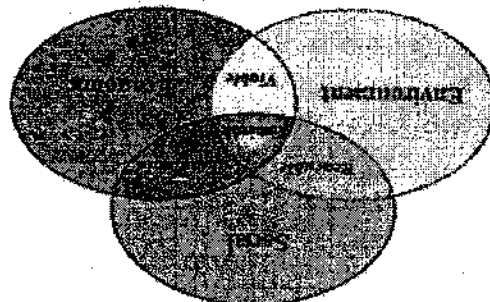


Fig: scheme of sustainable development

The concept of sustainable development has developed beyond the initial intergenerational framework to focus more on the goal of "socially inclusive and environmentally sustainable economic growth

Three aspects of sustainable development

Economic: An economically sustainable system must be able to produce goods and services on a continuing basis, to maintain manageable levels of government and external debt, and to avoid extreme sectorial imbalances which damage agricultural or industrial production

Environmental: An environmentally sustainable system must maintain a stable resource base, avoiding over-exploitation of renewable resource systems or environmental sink functions,

and capturing non-renewable resources only to the extent that investment is made in adequate substitutes. This includes maintenance of biodiversity, atmospheric stability, and other ecosystem functions not ordinarily classed as economic resources.

Social: A socially sustainable system must achieve distributional equity, adequate provision of social services including health and education, gender equity, and political accountability and participation.

Ecological Footprint:

The ecological footprint measures how much nature it takes to support people. It is an ecological accounting system. It contrasts how much biologically productive area people use for their consumption to how much biologically productive area is available. More specifically, the ecological footprint is the biologically productive area needed to provide for everything people use: fruits and vegetables, fish, wood, fibers, absorption of carbon dioxide from fossil fuel use, and space for buildings and roads. Biocapacity is the productive area that can regenerate what people demand from nature. Footprint and biocapacity can be compared at the individual, regional, national or global scale. Both footprint and biocapacity change every year with number of people, per person consumption, efficiency of production, and productivity of ecosystems.

The original ecological footprint is defined as the land area that would be needed to meet the consumption of a population and to absorb all their waste (Wackernagel and Rees 1995). Consumption is divided into 5 categories: food, housing, transportation, consumer goods, and services. Land is divided into 8 categories: energy land, degraded or built land, gardens, crop land, pastures and managed forests, and 'land of limited availability', considered to be untouched forests and 'non-productive areas', which the authors defined as deserts and ice-caps.

Overcoming Ecological Limits:

Another major justification for a sustainability criterion relates to ecological complexity and irreversibility. Current ecological systems have evolved over many centuries to achieve a balance involving interactions among thousands of species of plants and animals (the total number of species is unknown, but is in the millions), as well as delicately balanced physical and chemical relationships in the atmosphere, oceans, and in freshwater and terrestrial ecosystems. Extensive exploitation of natural resources permanently alters these ecological balances, with effects not fully predictable. In some cases, upsetting the ecological balance can lead to disaster—desertification, collapse of ocean food systems, destruction of the ozone layer, pollution of aquifers, outbreaks of super-pests resistant to insecticides, and the like. Species extinction is a clear example of irreversible damage, imposing unknown economic and ecological costs in the future.

Ecological economists therefore argue for a precautionary principle—we should strive for minimum interference with the operation of natural systems, especially where we cannot predict long-term effects. This principle obviously defies easy definition in economic calculations of resource value and use. Such calculations, therefore, are of value only if we can place them in the broader ecological context, whose priorities should sometimes override market equilibrium logic.

In addition, before addressing specific topic areas, we will examine some new analytical techniques developed by ecological economists.

One important issue is adding an environmental dimension to the measurement of economic output or GDP—or perhaps replacing GDP with a more inclusive measure of human well-being and ecosystem health. Another approach is to model the interrelationship between economic and environmental systems, focusing on flows of energy, resources, and wastes.

Nepal has a long history in the taxation. Land tax has been found to be practiced already in Lichhavi period and excise duty in Rana regime. Interestingly, these taxes are still in existence. Upto the Rana regime, tax system was not institutionalized and method of collection too was not systematic. For example, custom duties used to be collected through contractors, land revenues by Mukhiya, Zimwal or Zamindar in different parts of the country and they were authorized by the then power centres.

Modern taxation system was rationalized by introducing the Income Tax in Nepal by the first Parliamentary Government in 1959. Income Tax Act 1962 was enacted in 1962 replacing business, Profit and Remuneration Tax Act of 1959. The Income Tax Act, 1962 was replaced by Income Tax Act, 1974, which was amended for eight times and existed for a period of 28 years. The Income Tax Act, 1974 and all the income tax related provisions made under other special enactment have been repealed and the existing Income Tax Act, 2058 became effective since Chaitra 19, 2058 (01 April 2002). The Act governs all income tax matters and is applicable throughout the Kingdom of Nepal. It is also applicable to residents residing wherever outside Nepal. The Inland Revenue Department (IRD) is currently responsible for the enforcement of Tax Laws and administration of the following taxes: Income Tax, Value Added Tax (VAT), Excise Duty, Vehicle Tax and certain fee and duties like Entertainment fee (Film Development fee), Tourism Development Fee, Special fee, etc. Likewise, the Department is also responsible for monitoring the non-tax revenue such as dividends, royalties etc. of the Government.

Tax Structure in Nepal

The tax structure in Nepal is divided into two types

1, Direct Tax and

2. Indirect Tax.

1. Direct Tax

Direct tax is levied on individual person or corporations and tax burden ultimately lie on taxpayer. This tax has further been divided into:

- (i) Personal Tax, and
- (ii) Company or Corporate Tax.

Personal Tax

Profits from a business, salaries and benefits, income from house rent, interest and dividends are taxable income. Income tax in case of individuals is the most encountered tax system and has the following rates:

The taxable income of a resident individual for an income-year will be taxed at the following rates:

Up to Rs. 300,000 - Not taxable

From Rs. 300,000 - upto Rs. 500,000 - @15%

Above Rs. 500,000 - @ 25%

- The taxable income of a couple, if they chose to be treated as a couple will be taxed at the following rates; Up to Rs. 400,000 - Not taxable

From Rs. 400,000 to Rs.600,000 IQs. - @ 15%

Above Rs. 600,000 - @ 25%.

Company or Corporate Tax

This tax is exempted on dividends declared by the company, Companies listed in the stock exchange are subject to 33% corporate tax. Expenses wholly and exclusively incurred in the generation of income are deducted from taxable corporate and personal income, 5 % of total gross income of industrial units may be deducted for advertisement, entertainment and other contingencies. Depreciation on all plant and machinery equipment ranges between 15 %/0-25%/0 per annum and industrial and commercial buildings between 5%7% per annum. Transport vehicles may be depreciated between 15%-25% per annum and furniture between 100/0-15% per annum.

2. Indirect tax

Indirect tax is imposed on goods manufactured in or imported in to Nepal. The rate of tax varies depending on the nature of goods manufactured or imported. There are several types of indirect tax:

1. Customs Duty:

A customs duty is imposed on any goods imported. The duty from 0% to 110%. Live animals, fish and most primary products are exempt from import duties. Machinery or goods related to basic needs are charged 5% duty. For products hazardous to health such as cigarettes and liquor, duty is 110%. For luxury goods such as cosmetics, air conditioners etc. the duty is 40%. Any goods imported from The South Asia Association for Regional Co-operation(SAARC) which includes Nepal, India, Pakistan, Bangladesh, Bhutan, Sri Lanka and the Maldives will be charged 10% and 5% less on total customs duty respectively. In an effort to help combat air pollution in the Kathmandu Valley, the Government recently approved temporary reduction of duty to 10% and 5% for vehicles and parts respectively.

1. Sale Tax:

There are two levels for sales tax: 10% and 20%. The tax is 10% on basic needs products like tea, cooking, oil, etc, And 20% on luxury products like cigarettes, TV, VCR, watches, vehicle, liquor, etc.

iii. Excise Duty or Countervailing Tax:

Goods imported are charged a countervailing tax whereas goods manufactured in Nepal are charged an excise duty only. The rate of excise duty or the countervailing tax varies depending on the nature of the goods in question. For example, cigarettes are taxed on the number of sticks and the duty on television is charged as a total percentage of the price.

iv. Value Added Tax (VAT):

VAT is a new tax system for Nepal. VAT was justified in the light of government fiscal imbalances and need for extra revenue mobilization through an efficient tax system. Although the VAT Act was passed in 1995 and the VAT regulation was approved in 1996, the VAT could only be introduced from November 16, 1997. The VAT replaced the then Sales tax, Hotel tax, Contract tax and Entertainment tax. There is persistent increase in the number of VAT registrants. As the taxpayers are increasing, the amount of revenue collection and the level of tax compliance are improving too. 13% rate of VAT is nowadays adopted in Nepal

CHAPTER - 8 Depreciation and its causes, Asset Depreciation and Accounting Depreciation:-

The reduction in value of fixed capital assets like plant and machinery, buildings, equipments, vehicles, etc. is technically known as depreciation. It is loss in value of capital goods due to wear and tear, ageing, or obsolescence. The value of capital goods is assumed to decrease each year for accounting purpose.

According to R.N. Carter, "Depreciation is the gradual and permanent decrease in the value of an capital asset from any cause."

According to Institute of Cost and Management accountants, England, "Depreciation may be defined as the permanent decrease in the value of assets due to use and or the lapse of the time."

- Depreciation reveals the following facts:-
1. It is a gradual decrease in the value of capital asset. It is treated as a non-cash expense or loss.
 2. The gradual diminution in the utility value of a capital asset is due to wear and tear or from any other reasons.
 3. It aims at distributing the net cost of tangible asset over the estimated useful life of the asset.
 4. It is a process of allocation, not of valuation.

Causes of Depreciation:-

1. **Wear and tear:-** The continuous use of an asset makes it old and decrease in working capacity and hence the value of asset depreciates. It is also called physical depreciation.

2. **Effusion of time:-** With the passage of time, the value of some asset diminishes, even if they are not used in the business.

3. **Obsolescence:-** An existing asset may lose its usefulness due to an improvement in technology, new invention, change in style, etc. It is also called functional or technological depreciation.

4. **Exhaustion or depletion:-** An asset may get exhausted through working. It applies to mines and quarries.

5. **Accident:-** Due to accident, it also may cause depreciation in assets. Such accidental causes may be natural disasters or man-made disasters.

6. **Fall in market value:-** The value of an asset may decrease due to fall in market price of the assets.

Advantages / Importance of Depreciation:-

1. **Ascertainment of cost of production.**
2. **Correct ascertainment of net profit or loss.**
3. **Replacement of asset.**
4. **To keep capital safe.**
5. **For correct ascertainment of income tax.**
6. **Fair financial position.**

Factors affecting the amount of Depreciation:-

1. **Total cost of asset.**

$$\begin{aligned} \text{Economic depreciation} &= \text{Purchase price} - \text{current market value} \\ &= \text{Purchase price} - \text{Salvage value} \\ &= \text{Initial cost} - \text{Salvage value} \end{aligned}$$

It can be mathematically vehicle is worth less with each passing year. and efficiency of our car decline with age i.e. the describe the decreasing value of our car. The appearance familiar with the term economic depreciation to for eg: If we buy a car, we must probably and computer, desktop and a laptop. quantity and quality demand of market eg: Typewriter need for an asset due to inability to meet increased change in technology. It decrease or eliminate the functional depreciation:- It occurs as a result of

a) Physical depreciation:- It can occur in any fixed asset due to wear and tear and deterioration due to interaction with the new environment. It results decline in performance and require high maintenance costs.

b) Economic depreciation:- It is of like equipment, real estate, vehicles with passing the time and wear and tear. So, it is also called in the economic value/utility of fixed assets. Asset depreciation refers the gradual decrease

- # Asset, Depreciation (Economic depreciation)
2. Estimated useful service life.
3. Estimated salvage value.
4. Estimated expenditure for repair.
5. Legal provision.

The selection of method for providing depreciation depends upon types of asset, nature of asset use, management policy, legal provision, etc. Some important

There are several methods of depreciation.
 Recovery System (MACRS): -
 Production Method, Modified Accelerated Cost
 Method, Sum of the Year Digit Method, Unit of
 Method, Declining Balance Method, Sinking Fund
 Method, Straight Line

8.2

= B.T.C.F - Income tax amount for particular year.

4.

Find A.T.C.F (After tax cash flow)
 = B.T.C.F - Depreciation amount for particular year.

3.

Find taxable income

2.

Find book value of asset = Initial cost - Accumulated depreciation

1.

Steps to calculate Accounting Depreciation:-

Calculation of depreciation of each year.

flow of any project.

Basically, it is used to estimate after tax cash flow of any project.

project undertaken.

determining the income taxes associated with any project undertaken.

analysis because it provides a basis for determining the income taxes associated with any project undertaken.

It is mostly used in engineering economic analysis because it provides a basis for determining the income taxes associated with any project undertaken.

proportion of its total depreciable life.

of asset's value (initial cost) into book value in proportion of its total depreciable life.

Accounting depreciation is the systematic allocation of asset's value (initial cost) into book value in proportion of its total depreciable life.

#

Methods of depreciation are as follows:-

1. Straight line method.
2. Declining balance Method.
3. Sinking fund Method.
4. Sum of the year digit method.
5. Service output method (Unit of Production Method)
6. Modified accelerated cost recovery system (MACRS)
7. Double rate declining balance method
8. Straight line plus average interest Method.

Straight Line Method:-

It is the simplest method for charging depreciation. It is also known as fixed installments method. Under this method, the same or equal or fixed amount of depreciation is charged every year throughout the life of the asset so that the book value of asset may be reduced to zero. It is calculated by considering original cost, useful life, salvage value of the asset. This method is suitable for those asset whose working life can be easily estimated.

$$\text{Annual depreciation} = \frac{\text{Original cost of asset} - \text{Estimated salvage value}}{\text{Estimated life of the asset}}$$

$$= \frac{\text{Total depreciation value}}{\text{Life of asset}}$$

If Rate is given,

$$\text{Annual depreciation} = \frac{\text{Total depreciable value} \times \text{Rate (\%)}}{100}$$

$$\text{Rate of depreciation} = \frac{\text{Amount of annual depreciation} \times 100}{\text{Total depreciable value}}$$

$$\text{Total depreciable value} = \text{Initial cost} + \text{All other expenses - Salvage value}$$

Advantage:-

1. It is simple to calculate and easy to understand.
2. It can write-off the total cost of asset completely at the end of life.
3. It makes easy for valuation of cost.

Disadvantages:-

1. It is not proper to provide equal depreciation when asset is getting old.
2. It ignores interest on capital invested in fixed asset.
3. This method is not recognized by income tax authorities.

2. Declining Balance Method / Diminishing / Matheson / Constant Percentage Method / Uniform / Average:-

This method is assumed that the annual cost of depreciation is a fixed or constant percentage of the book value at the beginning of the year. It is based on the assumption that large amount of depreciation should be provided in the earlier year of use than later years because the maximum loss is occurred in the early years of use. We cannot reduce the book value of asset to zero under this method.

(Book value = Original cost - Accumulated depreciation)

We can use following formula to calculate depreciation

$$d_k = B(1-R)^{k-1}(R)$$

$$B_k = B[1 - (1-R)^k]$$

$$B_{V_k} = B(1-R)^k$$

where, $R = 1 - \sqrt[N]{\frac{S}{B}}$

R = rate of depreciation
 N = Estimated life of asset
 S = Salvage value
 B = Original cost

DATE: _____
 PAGE NO.: _____

- Advantages:-**
1. High amount of depreciation for earlier years of assets and as assets getting old, depreciation goes on diminishing is more scientific.
 2. It is suitable for assets having long life such as building, plant and machinery.
 3. It is recognized by income tax authorities.

- Disadvantages:-**
1. It is difficult to decide the rate of depreciation if salvage value is not given.
 2. The book value of asset cannot be reduced to zero.
 3. It is applicable only for long life asset.
 4. It ignores interest on capital invested in fixed assets.

3. Sinking Fund Method:-

Sinking Fund Method of depreciation ensures that the full capital invested in a project is recovered at the end of project's life. This method provides fixed periodic charges (constant amount) at the end of each period that earns compound interest over the life of the asset and would be equal to the original cost of asset. According to this, the book value of asset decreases off increasing rate with respect to the life of the asset.

Let $P = \text{Initial/Original cost}$
 $S = \text{Salvage value}$
 $N = \text{Life of the asset}$
 $i = \text{Compound interest rate}$

5. Unit of Production Method (Service-Output Method)

All the depreciation methods, discussed above are

Where I = Initial Investment

S = Salvage Value

SOYD for total useful life

SOYD for particular year = Remaining useful life at the beginning of particular year $\times (I - S)$

the beginning of particular year

from the ratio of the sum of the years digit for the total useful life and remaining useful life at the beginning of particular year

to this method, per year depreciation charge is calculated from the ratio of the sum of the years digit for the total useful life and remaining useful life at the beginning of particular year

asset nears the end of its estimated life. According to this method, per year depreciation charge is calculated from the ratio of the sum of the years digit for the total useful life and remaining useful life at the beginning of particular year

depreciation charges as asset getting old i.e. the the beginning years of asset and smaller

method results larger depreciation charges during the beginning years of asset and smaller

number of estimated life years of an asset. This calculated on the basis of proportion of total

Under this method, depreciation per annum is

Sum of the Year Digit (SOYD) Method:-

4.

$$Bvk = I - (I - S) \times (A/F, \frac{1}{N}) \times (F/F, \frac{1}{N})$$

To find book value at the end of year/period k .

$$dk = (I - S) \times (A/F, \frac{1}{N}) \times (F/F, \frac{1}{N})$$

To find net depreciation charges in the year k .

$$A = (I - S) \times (A/F, \frac{1}{N})$$

To find fixed annual equivalent amount.

Now, dk = Depreciation charges in the year k .

Bvk = Book value at the end of year k .

A = Annual equivalent amount

6. Modified Accelerated Cost Recovery System (MACRS)

It is the principal method for computing depreciation in engineering projects. Historically, for tax purpose and accounting, an asset's depreciable life was determined by its estimated useful. But the MACRS scheme totally abandon this practice and simpler guidelines were set which created several classes of assets each with more or less arbitrary life called recovery period. The salvage value of property is always zero. MACRS scheme includes 8 categories of assets: 3 years, 5 years, 7 years, 10 years, 15 years, 20 year, 27.5 year and 39 year. Half year convention:- It is assumed that all assets are placed at mid-year and they have zero salvage value. Half year depreciation is allowed for the first year and full year's depreciation is allowed in each of the remaining years of the assets recovery.

based on elapsed time (year). But if the decrease in value of any asset is based on total working hour or production unit, then it is called service output method or unit of production method. This method gives equal depreciation charge for each unit of output or working hour regardless of the lapse of time involved.

Depreciation rate per unit of output/working hours

$$R = \frac{I - S}{I}$$

Total working hour or Production Unit.

period and finally remaining half year depreciation in the end of recovery period. The MACRS asset is depreciated initially by the declining balance and then to straight line method.

Introduction to Corporate Income Tax, Taxation Law, Depreciation Rates, Personal Tax, Corporate Tax, VAT.

Individuals as well as corporations are also subjected to income tax. Income tax having on corporation is known as corporate income tax or corporate tax. It was first introduced in America in 1909 AD. Corporation is a word used to cover a variety of business enterprises having a legal personality and distinct from their owners like public corporations and joint stock companies.

In the view of corporate income tax, it is a way of including all corporate source of income in the personal income tax base. Nepal govt. has followed book profit/net profit as the tax base for the purpose of corporate income tax. According to Income Tax Act 2058 BS, there are different tax rates from 20% to 30% for different types of corporation.

Taxation Law: - Tax: - It is a compulsory contribution/payment to government by the people and institution without any expectation of direct benefit.

In Nepal, modern taxation system was introduced only after the introduction of multiparty

8.3

Introduction to Corporate Income Tax, Taxation Law, Depreciation Rates, Personal Tax, Corporate Tax, VAT.

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According to Income Tax Act 2058 (2002), depreciation allowance rates have been granted on various assets like buildings, furniture, vehicles, etc. which are as follows:-

#

Depreciation Rates:-

VAT Act - 2052.

Excise Tax Act - 2058

Custom Tax Act - 2064

Income Tax Act - 2058

enacting in Nepal. At present, following tax acts are

the geography of Nepal. income tax matters and it is applicable throughout the geography of Nepal. Income Tax Act, 2002. This act governs all enactment have replaced with the existence of tax related provisions made under other special The Income Tax Act, 1974 and all the income eight times and existed for a period of 28 years. 'Income Tax Act 1974', which was amended for Nepal Income Tax Act 1962 was replaced by business profit and remuneration tax Act of 1959. Income Tax Act' was introduced by replacing excise duty and registration fee. In 1962, 'Nepal were existence of custom duty, entertainment tax, Parliamentary government in 1959. Before that, there profit and income was first introduced by the first by introducing Excise Act 1958. Tax on corporate system in 1951. The tax system was rationalized

If the tax is levied on various sources of individual income, then it is known as personal income tax. It refers to the economic gain by the person during the particular accounting period. Its basic objectives are revenue collection and redistribution of social income.

Personal Tax:
Personal tax is a direct tax and it is generally imposed by the central authority and collected through the local authorities. Personal tax is mostly levied on income and property in Nepal.
Income Tax:-

Block	Assets Included	Rate Applied
A.	Building, structures and similar assets of permanent nature.	5% per year on base amount
B.	Office equipment, fixture, furniture, computer and data processing equipments.	25% per year on base amount.
C.	Automobiles, buses, minibuses and all other transport assets.	20% per year on base amount
D.	Construction and earth moving equipments and any other depreciable assets that are not included in another block.	15% per year on base amount.
E.	All intangible assets including software.	Cost price - Salvage value Useful life in years

(1) House and Land Tax: -
In case of VDCs, Rs. 5 to Rs. 100 per year on the basis of number, structure area and location of house.

tax: In Nepal, there are two types of property

provide services of public goods. The revenue generated from it is normally used to social services by the rural and urban authorities. System is to raise revenue for the provision of the main objective of any property taxing authorities.

Property tax: -
Property taxes are compulsory levied on the ownership of land, building, vehicles, weapons and so on. They are mostly levied on capital value or annual rental value and are collected by local

If income is more than 25 lakhs, total tax will be $25\% + 0.4 \times 25\% = 35\%$.

Income limit per year	Tax rate/year
upto 2,50,000	1%
2,50,000 - 3,50,000	15%
3,50,000 - 25,00,000	25%
above 25,00,000	35%
upto 3,00,000	1%
3,00,000 - 4,00,000	15%
4,00,000 - 25,00,000	25%
above 25,00,000	35%

According to existing Income Tax Act (2002), the taxable income of a resident will be taxed at the following rates: -

In case of municipalities:-	
Value of house and Land	Rate of Tax/Year
upto Rs. 10 lakhs	NIL
Net upto Rs. 10 lakhs	Rs. 300
upto 30 lakhs	0.05%
upto Rs. 50 lakhs	0.25%
upto Rs. 1 crore	0.50%
Above Rs. 1 crore	1.5%
(ii) Unified Property Tax:-	
Value of Property	Rate of Tax/Year
upto Rs. 10 lakhs	Rs. 50 to Rs. 200
Rs. 10 lakh - Rs. 20 lakh	Rs. 200 to Rs. 400
Rs. 20 lakh - Rs. 30 lakh	Rs. 400 to Rs. 750
Rs. 30 lakh - Rs. 50 lakh	Rs. 750 to Rs. 1000
More than Rs. 50 lakhs	Rs. 1000 to Rs. 1500
# Corporate Tax:- (Discussed earlier)	
According to Income Tax Act (2002), there are different flat rates for different types of companies.	
Corporation/Entity Tax Rate	
1. The income from export and special industry.	20%
2. The income from Petroleum industry, banks and financial institutes.	30%
3. Income from industrial enterprises or related with infrastructure projects like Road, Bridge, Tunnel, Electric Power generation, etc.	20%
4. Other general corporate bodies.	25%

#

VAT:

VAT is a complete new system of taxation for Nepal. VAT is activated along with the replacement of sales tax, hotel tax, contract tax and entertainment tax in Nepal.

VAT is a broad based indirect tax system, so that it is levied on sales of all goods and services excluding those that have been exempted by the law. Since VAT is levied only on value addition made at each stage in the process of production (import) to distribution, this tax system is more neutral, efficient, elastic and fair and it is also said to be an improved version of the sales tax.

Objective to implement VAT in Nepal.

- To increase government revenue
- To make flexible tax system.
- To make transparent tax system.
- To avoid tax evasion.
- To make efficient tax system.
- To promote volume of export trade, etc.

Main Features of VAT:-

1. Indirect tax levied on goods and services.
2. Self-policing in nature i.e. catch-up effect.
3. Tax credit and tax refund system.
4. Export promotional (due to zero VAT on export)
5. Based on accounting system and transparent.
6. Less chances of cascading effect.
7. Wider scope/have.
8. More popular throughout the world.

8.4 After Tax Cash flow Estimate, General Procedure

for Making After Tax Economic Analysis

After Tax Cash flow Estimate:-

It is measure of a company's ability to generate positive cash flow after deducting taxes. The general formula for After tax cash flow is,

$$ATCF = \text{Net Income} + \text{Depreciation} + \text{Amortization}$$

Sometimes, analysis also add back other non-cash

items and proceeds from debt or equity issuance.

The amount remaining after expenses, not

mortgage payments and income taxes have been

deducted from the gross income of an investment

property is the after tax cash flow:-

General Procedure for Making After Tax Economic

Analysis:-

After tax economic analysis refers to the profitability measurement of any project including all income taxes. In another words, it is the analysis of after tax cash flows (ATCFs) estimate.

Net, R_k = Revenue from the project during period k .

E_k = Expenses from the project during period k .

d_k = Depreciation amount of project during period k .

t = Income Tax rate.

T_k = Income tax amount during period k .

∴ Taxable income = $R_k - E_k - d_k$ ∴ where $R_k > E_k + d_k$ → no tax liability
Tax amount (T_k) = $t(R_k - E_k - d_k)$ | $R_k < E_k + d_k$ → no tax liability

In the economic analysis of engineering

projects, ATCFs in year k (ATCFs) can be

computed in terms of BTCFs in year k (BTCFs)

$$BTCF_k = R_k - E_k$$

Year	BTCF	Depreciation	Taxable Income	Cash flow for Income tax purposes	ATCF
K	$R_k - E_k$	d_k	$R_k - E_k - d_k$	$(R_k - E_k - d_k)(1 - t)$	$(R_k - E_k) + t d_k$

Tabular form to compute ATCFs may be given as:-

$$\begin{aligned}
 ATCF_k &= BTCF_k - T_k \\
 &= R_k - E_k - t(R_k - E_k - d_k) \\
 &= (1 - t)(R_k - E_k) + t d_k
 \end{aligned}$$

~~Enterprise * Financing * and * Capital * Investment.~~

Method of financing: Equity financing, Debt financing and Capital structure

When project is selected by making investment decision, the next step is to decide/choose method of financing i.e. source of financing. The methods of financing in any project are :-

Equity financing:

It refers financing by share increment. It has two forms.

The use of retained earnings into company development.

If a company does not have sufficient cash on hand to make investment and not wish to

bottom (loan). The company can arrange money by selling common stock to raise the required fund. Once the company has decided to issue common

stock it must estimate flotation costs (cost incurred in share issue like banker's fee, accountant charges, etc.) if the company is already publicly owned, the offering price commonly will be based on the existing market price of the stock.

A company usually establishes a target capital structure or debt ratio in the process of financing decision. This target may change over time as business condition vary, if debt ratio is below target level, company increases capital by issuing debt and vice-versa.

It is the percentage of total capital provided by debt or borrowed funds. A debt ratio of 0.4 indicates that 40% of the capital is borrowed and 60% funds are provided by share. This type of financing is called mixed financing.

$$\text{Capital structure} = \frac{\text{Total debt}}{\text{Total capital}} = \text{Debt ratio}$$

Capital structure:-
A method of financing, the ratio of total debt to total capital represents capital structure or debt ratio.

Debt financing is money borrowed from investors for a fixed period of time at fixed interest. One of the primary methods of debt financing is issuing bonds, that needs interest paid for each period yearly or semi-annually regular basis but the principal is paid in a lump sum when the bond matures. Bond financing also needs flotation costs.

Debt financing:-
It is the second major type of financing for any company which includes both short term borrowing from financial institutions and sale of long-term bonds.

9.2 Cost of capital: Cost of Equity, Cost of debt and

Calculating cost of capital.

Most of the business firms arrange their capital

budget with stock (share) and long-term debt (bond)

as a source of capital.

A firm's cost of capital reflects the average

cost of various sources of funds that the firm

uses. Hence, the ways in which the cost of each

type of financing are as follows:

A. Cost of equity (share)

B. Cost of debt (bond)

C. Calculating cost of capital.

A. Cost of equity:

In principle, cost of equity capital involves an

opportunity cost because the firm's after-tax

cash flow (profit) belongs to the stockholders.

Management may either pay out these earnings in

the form of dividends or reinvest in the

business i.e. retain earnings.

If management decides to retain earnings i.e.

reinvest, an opportunity cost is involved because

shareholders could have received the earnings as

dividends and get chance to invest other financial

assets or projects.

Therefore, the firm should earn on its retained earnings

at least as much as the shareholders could earn by

other alternative/project.

When investors are considering to buy a firm's

stock, they have two things in mind: first one is

cash dividends and second one is gains of the time

By using these both components, an investor determines rate of return or present market value of stock (share) by discounting expected future dividends.

Cost of retained earnings: $\frac{P_0}{P_0 + P_1} = \frac{P_0}{P_0 + P_1}$

$P_0 = \text{Present cost of stock}$

Assuming new common stock. $k_e = D_1 + g$.

$\text{D}^* = \text{paspikip} \text{ londa} \text{ pasy}$

Marginal average cost of equity

Cr = amount of equity financed from retained earnings

CP = " " " " " "
preferred stock

Cost of Equity, Calculation, Based on Financial Market:

Cost of equity is the risk free cost of debt i.e.

return on treasury bills plus a premium for taking a risk as to whether a return will be received. The following formula quantifies the cost of equity (i.e.,

$$i_e = r_f + \beta(r_m - r_f)$$

where, r_f = risk free interest rate

r_m = market rate of return

β = stock price volatility that quantifies risk

= measures one firm's stock price relative

to the market stock price as a whole.

If $\beta > 1$, stock is more volatile than the market

If $\beta < 1$, stock is less volatile than the market.

B.

Cost of debt:-

Calculation of cost of debt as in the form of

weighted average cost of capital is relatively

forward and simple. There are two types of debt

financing are loans and bonds. Because the

interest payments on both are tax deductible,

the effective cost of debt will be reduced.

To determine the after tax cost of debt

(i) use formula,

$$i_d = \left(\frac{C_d}{C_s} \right) K_s (1 - t_m) + \left(\frac{C_d}{C_b} \right) K_b (1 - t_m)$$

where, i_d = Cost of debt

C_s = amount of short term loan

C_b = amount of bond financing.

$C_d = C_s + C_b$ = total amount of debt

K_s = before tax interest rate on short term loan

K_b = before tax interest on bond

t_m = firm's marginal tax rate.

Marginal cost of capital:-
It is defined as the cost of obtaining another dollar of new capital and the marginal cost rises as more and more capital is raised during a given period.

Thus, 14.70% would be the marginal cost of capital that a company with the given financial structure would expect to pay to raise Rs 10 million.

$$K = 0.0685 \times 4 + 0.1993 \times 6 = 14.70\%$$

For eg: $C_d = Rs 4 \text{ million}$ $C_e = Rs 6 \text{ million}$
 $V = Rs 10 \text{ million}$ $i_d = 6.85\%$ $i_e = 19.93\%$

Where, $V = C_d + C_e$
 $C_d = \text{Total debt capital } (C_s + C_b)$
 $C_e = \text{Total equity capital } (C_{cr} + C_{cp})$
 $i_e = \text{average equity interest rate per period.}$
 $i_d = \text{after tax average borrowing interest rate per period.}$

∴ Cost of capital, $K = i_d C_d + i_e C_e$
 ∴ Cost of capital, $K = i_d C_d + i_e C_e$
 We can determine or calculate tax-adjusted weighted average cost of capital (simply the cost of capital) by reflecting the cost of raising funds from different sources.
 A firm raises capital on the basis of the target capital structure and the target structure remains unchanged in the future.
 We can determine or calculate tax-adjusted weighted average cost of capital (simply the cost of capital) by reflecting the cost of raising funds from different sources.

Private investment is another project funding mechanism which includes domestic and foreign people or organization.

Through PPP, government can reduce cost and risk and private sector can generate business opportunities and general people can receive better and more accessible services.

1. Improve service delivery
2. Cost - effectiveness
3. Increase investment in public infrastructure
4. Reduce public sector risk.
5. Better use of resource.
6. Project acceleration.

Benefits from PPP:-

- In PPP, government remains actively throughout the life of project and private sector is responsible for the more commercial functions like project design, construction, finance and operation.
- The main goal of PPP is to combine the best capabilities of the public and private sectors for mutual benefit.
- PPP is widely accepted project funding mechanism throughout the world specially in the area of hydropower, water and sanitation, refuse disposal, hospitals, stadiums, railway, roads and so on.
- Nepal has also taken it as a means of development since 1990s. Some
- Nepal has separately introduced regulatory framework for managing PPP projects at local government level such as DDC, VDCs and municipalities.

I_0 = Investment expenditure for the entire project life

$$\sum_{n=0}^m \frac{I_n}{(1+r)^n} = \sum_{n=0}^m \frac{B_n}{(1+r)^n}$$

It is an indicator to measure the financial return on investment of an income generation project and used to make the investment decision. FIRR is obtained by equating the present value of investment costs (cash outflows) and present value of net incomes (cash inflows).

#

Financial Internal Rate of Return (FIRR): -

9.4 FIRR, EIRR and Return on Equity:-

Having financial and technical capabilities. All the private companies (Pvt. Ltd) and business projects are established and run by the private investment. Generally, private sector choose low risk with high return, low investment and short term life of the projects. Market oriented development strategies encourage private sector involvement in development process. At present, private sectors has been the engine of growth in most of the countries. Private investment refers domestic and foreign investment by individuals or institutions in any economic areas. At present, the government has emphasized globalization and liberalization approach by formulating proper policy and plans that has provided potential room for the development of private sector investment.

period from year 0 to m.
 B_n = Annual net incomes for the entire project life
 period from year 0 to m.
 By solving the above equality, we can obtain the value of r which is called FIRR.
 There may be various types of FIRR like:
 → FIRR of project entity (for total investment)
 → FIRR of shareholders (for share capital)
 → FIRR of lenders/bank (for debt capital)
 In practice, it is not necessary to obtain all FIRRs. Rather, it would be important to identify what kinds of investment are involved and calculate FIRR accordingly for the decision making purpose.
 Generally, FIRR of the project entity namely considers FIRR on total investment which is satisfactory level for the investment decision.
 # Economic Internal Rate of Return (EIRR):
 Profits are essential signaling mechanism for investment decision. However financial profitability (private profitability) are important in investment decisions by the firms, they are not good signaling for the whole i.e. national economy. So that, there is essence of economic analysis of project by using FIRR.
 Following 4 points departure economic analysis from financial analysis:
 1. Project boundary is financial but economic profitability will be differ.

2. Market price may underestimate economic benefit

i.e. opportunity cost may be higher than real cost.

Project may create externalities.

Distortion in environment (policy distortion).

If there is:-

a) No project boundary and externalities

b) Price and policy distortion is zero,

ETRR can be used as proxy for EIRR.

Formula to calculate EIRR:-

$$EIRR = FIRR$$

Profitability associated with consumers due to market price distortion

Profitability associated with externalities expressed in market price

Profitability associated with environment distortion.

NOTE:- According to ADB report,

The rate of return that would be achieved on all

project resource where all benefits and costs are

measured in economic prices. The EIRR is calculated

as the rate of discount for which the present value

of the net benefit stream becomes zero, or at which

the present value of the benefit stream is equal to the

present value of the cost stream. For a project to be

acceptable, the EIRR should be greater than the

economic opportunity cost of capital.

#

Return * of Equity (ROE) :-

ROE is also measurement of profitability that calculates how many rupees of profit a company generates with each rupee of shareholders equity. So, it is the amount of net income returned as a % of shareholders equity.

ROE is computed for common shareholders so that preferred dividends are not included in the calculation of ROE. Hence, preferred dividends are taken out from the net income for ROE calculation.

$$ROE = \frac{\text{Net income} - \text{preferred dividends}}{\text{shareholders equity or common equity}} \times 100$$

$$= \frac{\text{Net income}}{\text{shareholders equity}} \times 100$$

Analysis:-

ROE measures how efficiently a firm can use the money from shareholders to generate profit and grow the company. Hence, ROE is a profitability ratio from the investors point of view not the company.

Investors (shareholders) always want to see high return on equity ratio because it indicates that the company is using its investor's fund effectively.

Many investors choose to calculate the ROE at the beginning and end of period to see the change in return ratio. It shows the track a company's progress and ability to maintain a positive earnings trend.

CHAPTER - 10 Basic Accounting Procedure

10.1 Accounting Terminologies: Asset and liabilities: Fundamental equation of accounting:-

Accounting:-

It is a system of collecting, summarizing, analyzing, classifying and interpreting business transactions in financial or monetary terms. It is a science of recording transactions of economic nature in a systematic manner and also art of analyzing and interpreting them.

#

Assets:-

The resources or properties owned by the business organizations to generate economic benefit is known as assets. Assets provide current and future economic benefits like cash, land, plant and machinery, building etc.

Assets can be tangible or intangible which solely or combine contribute in earning capacity of the firm. Such assets are acquired by the firm in measurable monetary units.

Assets can be classified into four groups:-

1.

Fixed assets:-

fixed assets are also said to be permanent resource or properties because such assets acquired by a business for long last. They are not purchased for resale but used to increase earning capacity of business eg: land & building, plant & machinery, vehicle, furniture, etc.

2. Current assets :-
They are also known as floating assets or liquid assets because these are converted into cash in normal course of business. Generally, current assets are not kept by the business firm for more than one year.
Examples :-
Cash in hand and cash at bank
Bill receivable :- part of debtor converted into it with exchange of bills
Stock-in-trade :- closing stock or the value of stock not sold during the year
Sundry debtors :- Amount owed to business by its customers
Short-term investment :- via marketable securities which are expected to realise into cash in their value.
Accrued income :- incomes already earned but not received.

3. Fictitious assets :-

They do not have any real value so that they are assets for namesake. They are taken as assets on legal and technical ground.
for example :- Preliminary expenses, advertisement expenses, discount on shares and debentures, etc.

4. Intangible asset :-

These are assets which cannot be touched and do not have any physical form. for example :-
Patent :- right brought from parent company to produce some goods
Goodwill :- reputation of the firm helps in doing business
Copyright :- right to publish books in different countries
Trademark :- term, design, and symbol & their combination

- Capital:-**
- The amount invested by the trader in his trade.
 - Capital means owner's equity which is changing due to every transactions in business.
 - Capital denotes the seed invested in the beginning and surplus earned afterward.
 - Capital or owner's equity = Assets - Liabilities.
- Liability**
- Liabilities are borrowed sums from the third parties to the business.
 - Liabilities are debts payable in the future by the firm to its creditors. It represents economic obligation to pay cash or to provide goods or service in future.
 - Generally, liabilities are created by borrowing money or purchasing goods and services on credit.
 - Liabilities can be of two types:-
- A. Current Liabilities:-**
- They are repayable within one year i.e. an accounting period. Current assets are converted into cash to pay current liabilities.
 - Examples:-
- Sundry Creditors:-** i.e. trade liabilities arises due to credit purchase of goods.
 - Bill payable:-** Part of sundry creditors evidenced by a written acknowledgement of debt.
 - Bank overdraft:-** Amount drawn from bank more than the amount deposited.
 - Outstanding expenses:-** Expenses incurred already

Assets, liabilities and capitals are the three basic elements of every business transaction. Here, assets denotes resource and capital & liabilities are source of finance. The value of resource and sources must be equal. The relationship between these three elements remains unaltered that can be expressed by using a mathematical form which is called accounting equation. The change in one element results in corresponding change in the same item

#

Fundamental Equation of Accounting:-

Mortgage loan: - Loan taken against the security of property, generally long-term loan.

d.

Band: - Similar to debenture, issued by government or corporations acknowledging that money has been lent to it and will be paid back with interest.

c.

Bank loan: - Loan taken other than short term.
Debenture: - Written document of loan issued by company.

a.

b.

Long term liabilities are also known as permanent liabilities which are raised for more than one year i.e. they are the obligations or debts repayable for more than one accounting period.

B.

Long term liabilities:-

Provisions: - An estimated amount set aside to meet a loss which is likely to occur but is uncertain in amount.

g.

Short-term loan: - Loan taken other than long term

f.

Advance income: - Income already received but not yet earned.
but not yet paid.

e.

Or in other element. It can be expressed as,

$$\text{Assets} = \text{Capital} + \text{Liabilities}$$

$$\text{or, Capital} = \text{Assets} - \text{Liabilities}$$

$$\text{or, Liabilities} = \text{Assets} - \text{Capital}$$

Hence, fundamental accounting equation explains double entry system of book keeping which emphasises on the total assets being always equal to the total equities (capital + liabilities). An accounting equation is also a foundation to double entry system of book-keeping.

The concept of double entry system refers, for every debit, there is credit, which has resulted in accounting equation.

Accounting equation is also known as balance sheet equation.

Procedure for developing an accounting equation: The steps required for developing an accounting equation are as under:

1. To find out variables: - Assets, liabilities and capital in accounting equation are called variables. The first step is to find out the variables affected by a transaction.

2. To find out effects on variables: After finding the variables affected by a transaction, the second step for developing an equation is to find out effects on variable. The effects may be increased or decreased. To show the effect on equation: -

After ascertaining the variables and effect on variable of a transaction, it is necessary to show the effect on appropriate side of equation.

50.2 Financial statements: The Balance sheet, Income Statement, and Cashflow statements:-

The summary information about financial performance and position of a concern is known as financial statement.

The purpose of preparing financial statement is for periodical review of the activities of organization. Financial statements are prepared at the end of accounting period.

Features of financial statement:-

1. It is concerned with past records maintained by the firm so it is historical.
2. It is financial nature and expressed in terms of money.
3. It presents the summary of revenue and expenses and net income or loss of a firm.
4. It shows the financial position through the help of balance sheet.

Objectives of financial statement:-

1. To know about business activities whether running at a profit or loss.
2. To ascertain financial position of the business.
3. To provide meaningful information about the financial activities of a business to different persons or parties.
4. To provide information about the capacity of the business of paying loan and interest there on.
5. To provide information about economic resources and obligations and their changing pattern.

Income Statement:-
 Income statement shows the net result of the business operation during an accounting period which includes trading account and P/L account. It presents the summary of revenues.

Liabilities and capital	Assets	
Rs.	Rs.	

Balance Sheet:-
 Balance sheet contains Assets and liabilities. Statement so it is called balance sheet. account which are not included in income statement and balance of ledger. It is the preparation of trading & P/L account. end of accounting period after completing position of a firm which is prepared at the. It is a statement summarizing the financial given date. of assets and liabilities of a business at a. It is not an account but is a statement.

- The Balance Sheet:-**
1. The management can get lot of information that helps decision making.
 2. Gives information related to profitability and operation costs.
 3. Shows the financial position of a business.
 4. Management can take decision regarding sanctioning of loans and future planning.

expenses and net income or loss of a firm, so it serves as a measure of the firm's profitability. The amount received by ordinary course of business is known as revenue income like:

- Earning from selling products.
- Earning from discount, commission, interest, transfer fees, etc.

Expenditure includes operating and maintenance cost that yields benefit in an accounting period use:

- Expenditure to acquire raw materials or stock of finished goods
- Expenditure whose benefit will expire within the year
- Expenditure to maintain assets within a year, etc.

#

Cash Flow Statement: -

The statement which shows the source and uses of cash is known as cash flow statement.

Cash flow is the statement of source and uses of cash funds that shows where from cash

came and where cash was used during the given period

A statement which explains the changes in cash

position from one balance sheet date to next balance

sheet date is known as cash flow statement.

Cash flow = Cash + flow

= Cash or cash equivalent + changes or movement.

= highly liquid instrument + inflow or outflow

* Cash inflow: - issue of share, debenture, sale of fixed

asset, sale of investment, receiving loan, etc.

* Cash outflow: - redemption of share, debenture, repayment

loan, purchase of fixed assets, etc.

* Highly liquid instrument - money market funds, commercial papers, treasury bills, etc.

The difference between inflow and outflow of cash and cash equivalent is termed as net increase or decrease in cash and cash equivalent. The statement which reports cash flow during the period classified by operating, investing and financial activities is known as cash flow statement.

Nepal accounting standard has used the following terms on cash flow statement.

1. Cash: - Cash comprises cash on hand & demand deposit. Cash equivalent: These are highly liquid investments that are readily convertible and low risk. Change in value.

3. Cash flows: - Inflows and outflows of cash and cash equivalents.

4. Operating activities: - Principal revenue producing activities except investing and financing activities.

5. Investing activities: - Acquisition of long term assets. Financing activities: - The activities relating to changed in the size & composition of the equity capital and borrowings of the enterprises.

Importance of cash flow statement:

1. Cash position of a firm can be easily evaluated.
2. Helpful for planning and co-ordinating financial operation properly.
3. Assist for the effective steps to strengthen the internal financial position.

1. To provide information about cash receipt and cash payment of a company during an accounting period.
2. To provide information about a company's operating, investing and financing activities during accounting period.
3. To provide information about the changes in cash position of the company.
4. To evaluate the financial policies of the company.
5. To help in understanding liquidity position of the firm.
6. To understand the reasons for variation in cash position.
7. To assist the short-term cash planning of firm.

Objectives of Cash Flow Statement:-

4. Helpful in planning the repayment of loan, replacement of fixed assets and other long-term cash planning.
5. Useful for both internal and external users.
External users:- bank & other financial institution.

10.3	Using Ratios to make Decision: Debt Ratio, Current Ratio, Quick Ratio - Acid Test Ratio, Inventory Turnover Ratio, Total Asset Turnover, Profit Margin on sales, Return on Total Assets, Price Earnings Ratio and Book Value per Share:-
	In accounting, ratio is a technique of analysis and interpretation of financial statement through mathematical expression. Hence, it is defined as the mathematical expression of the relationship between two accounting figures.
	Ratio analysis can evaluate the performance of any business organization, so that ratio analysis is an analysis of financial statement by the help of ratio between two accounting figures.
	Importance of Ratio analysis:-
1.	Helpful in assessing operating efficiency of the business.
2.	Helpful in measuring financial solvency.
3.	Helpful in decision making.
4.	Helpful in future forecasting.
5.	Helpful in corrective action.
6.	Helpful in comparing inter-firm performance.
7.	Helpful in cost control.
#	Debt Ratio: (Solvency/ Leverage/ capital structure Ratio):- Debt consists short term and long term debt of organization.

- The present financial condition of the firm for paying long term loan is tested by debt ratio. It measures the ability of firm for paying the interest regularly and to repay the principal on maturity i.e. agreed time.
- It helps to know about the ability of firm for paying the interest on loan and utilization of firm's fixed asset as well as resource on productive sector.

Following ratios are commonly used for this:-

1. Debt to equity ratio (DER)
It is the relationship between long-term debt and owner's equity which is a popular measure of the long term solvency of a firm.

$$\text{Debt-equity ratio} = \frac{\text{long term debt}}{\text{Shareholder's equity}} = \frac{\text{Total debt}}{\text{Shareholder's equity}}$$

- Interpretation of DER:-
Higher DER indicates larger amount of debt in comparison equity.
- Creditors prefer low DER which secures their investment.
- Higher DER is most risky due to more liabilities/debt.
- Lower DER indicates more capital invested by shareholders or owner.

$$2. \text{Debt to total capacity ratio (DTCR):}$$

$$\text{DTCR} = \frac{\text{Long term debt}}{\text{Total debt}} = \frac{\text{Capital employed}}{\text{Capital employed}}$$

Capital employed = share holder equity + long term debt
 = Net fixed assets + working capital.
 $\therefore \text{Total debt} = \text{long term debt} + \text{current liabilities.}$

Interpretation of DTOR:-
 It shows the ratio
 Lower ratio indicates the more security to the creditors.
 Higher ratio represents more risk to the creditors and even to the owners.

Current Ratio:-
 It shows the ratio of current assets to current liabilities which is calculated by dividing current assets by current liabilities.
 The objective of this ratio is to measure the ability of the firm to meet its short-term obligation.
 $\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current liabilities}}$

Interpretation:-
 Higher the current ratio, the greater ability of the firm's to pay its short-term debt and vice-versa.
 Generally, current ratio of 2:1 is regarded as an acceptable standard.
 If the current ratio of a firm is less than 2:1, that indicates the firm has not paid its urgent debt on time, which shows the bad

business performance. If it is more than 2:1, the firm has more investment in current assets, that do not produce satisfied return.

Quick Ratio or Acid-test Ratio or Liquid Ratio:-

A relation between quick (liquid) assets and current liability is termed as quick ratio.

Quick assets includes all the current assets other than stock and prepaid expenses.

It measures the ability of firm to pay current liability immediately.

$$\text{Quick Ratio} = \frac{\text{Quick assets}}{\text{current liabilities}}$$

$$[\text{Quick assets} = \text{Current assets} - \text{stock} - \text{prepaid expenses}]$$

Interpretation:-

Higher the quick ratio, better the liquidity position. An ideal quick ratio is said to be 1:1.

Inventory Turnover Ratio (Stock Turnover ratio)

It is the relationship between cost of goods sold and average inventory. It is calculated by dividing the cost of goods sold by average inventory.

It indicates as to how fast the goods are sold and also shows the speed with which stock is rotated into sale.

It is also called stock turnover ratio.

$$\text{Inventory turnover ratio} = \frac{\text{Cost of goods sold}}{\text{Average inventory}}$$

$$\text{Cost of goods sold} = \text{Opening stock} + \text{Purchase} + \text{Direct expenses} - \text{Closing inventory} = \text{Sales} - \text{Gross profit}$$

$$\text{Average inventory} = \frac{\text{Opening stock} + \text{closing stock}}{2}$$

Direct expenses = Wages + Carriage Inwards + Freight Inwards

Generally the cost of goods sold may not be available in the published financial statements.

$$\text{Inventory turnover ratio} = \frac{\text{Sales}}{\text{Average inventory}}$$

Closing inventory = stock or inventory of balance sheet

Interpretation:-

- High inventory turnover indicates efficient inventory management.
- Low inventory turnover indicates stock are kept for long term in store.
- With this, management can assess whether stock has been more efficiently used or not.

Total Asset Turnover (Investment turnover ratio)

It is also known as investment turnover ratio that shows the relationship between sales and total assets.

It measures the efficiency of a firm in managing and utilizing its assets.

$$\text{Total Asset Turnover Ratio (TART)} = \frac{\text{Net sales}}{\text{Total Assets}}$$

$$[\text{Total assets} = \text{fixed assets} + \text{investment} +$$

current assets, loans & advance - frictions assets).
 [Net sales = Total sales - sales Return]

Interpretation:-

Higher the ratio implies better utilization of total assets for making the sales and vice-versa.

Profit margin on sales (Ratio of net profit to net sales quantity):-

- Maximization of profit is the main objective of each business concern. Keynes said that "Profit is the engine that drives the business organization".
- A firm should be able to earn profit sufficiently from each unit sold.
- The following ratio/margin can be ascertained from sales basis:-

A. Gross profit margin (ratio) on sales:-
 It is the direct expression of gross profit in relation of sales.

$$\text{Gross profit margin} = \frac{\text{Gross profit Amount}}{\text{Sales quantity}}$$

$$\text{Gross profit} = \text{Sales amount} - \text{cost of goods sold}$$

Interpretation:-

Higher ratio is a sign of efficient management that reflects lower cost of goods sold and maximum profit.

B. Net profit margin on sales:-
 It is the ratio between sales quantities and net profit amount.

Net profit margin = $\frac{\text{Net profit amount}}{\text{Sales quantity}}$
 [Net profit = Gross profit - selling expenses - interest - taxes]

Interpretation:-
 Higher ratio shows efficient use of resources and making profit.

Return on Assets:-
 It is the relationship between net profit and total assets.
 It is also called profit to assets ratio.

$$\text{Return on assets} = \frac{\text{Net profit}}{\text{Total assets}}$$

Interpretation:-
 Higher ratio implies available resources are employed efficiently.

Price Earning Ratio:-

It is the ratio between market price per share and earning per share.
 It shows that how much investors are ready to pay per rupee of reported profit.
 It is the best indicator of ongoing performance of a company as well as future prospects.

$$\text{Price Earning ratio} = \frac{\text{Initial market price per share}}{\text{Current earning per share}}$$

#

Book value per share:-

It is also called net worth per share.

It is the ratio between common shareholder's equity and number of common shares outstanding.

It refers to the value of a share which is calculated on the basis of data recorded on the balance sheet.

Book value per share = Common shareholder's equity

No. of common shares outstanding.

Note:-

Ratio Analysis:-

Ratios are used for efficiency and profitability

measurement of any organization by the relationship between figures and evaluating physical performance.

They can be based on balance sheet and income

statement. The present popularly used ratios are:-

1)

Liquidity Ratios:-

They measure ability to pay back short term debts. Current ratio is a good indicator of liquidity.

2)

Debt Ratios:-

They measure ability to meet long term financial obligations. They are also known as solvency ratios.

3) Operating Ratios:-

They measure the efficiency of operations. Inventory turnover ratio is an example of it.

RAKESH SAH

Tutorial

TIME VALUE OF MONEY:-

Simple interest
If Rs. 100 is invested for five years at a simple interest rate of 15%, what will be the interest and maturity value?

Solution:-

Given that,

Invested value (P) = Rs. 100

Simple interest rate (r) = 15% = 0.15

Period (N) = 5 years.

Interest (I) = ?

Maturity value (F) = ?

We know that,

$$I = P \times N \times r$$

$$= Rs. 100 \times 0.15 \times 5$$

$$= Rs. 75$$

And,

$$F = P + I$$

$$= Rs. 100 + Rs. 75$$

$$= Rs. 175$$

Hence, the interest will be Rs. 75 and maturity value will be Rs. 175.

If Rs. 1,000 is loaned for three years at a simple interest rate of 10% per year, what will be the interest and total amount at the end of three years?

Solution:-

Given that,
 Loaned amount (P) = Rs. 1,000
 Period (N) = 3 years.
 Simple interest rate (r) = 10%
 Interest (I) = ?
 Total amount (F) = ?
 We know that,

$$I = P \times N \times r$$

$$= \text{Rs. } 1000 \times 3 \times 0.10$$

$$= \text{Rs. } 300$$

Also,

$$F = P + I$$

$$= \text{Rs. } 1000 + \text{Rs. } 300$$

$$= \text{Rs. } 1300$$

Hence, the interest will be Rs. 300 and total amount at the end of three years will be Rs. 1300.

Compound interest

3. If Rs. 1000 is loaned for three years at interest rate of 10% compounded each year, what will be the interest and total amount at the end of three years?

→ Solution:-

Given that,

$$\text{Loaned amount (P)} = \text{Rs. } 1,000$$

$$\text{Period (N)} = 3 \text{ years.}$$

$$\text{Interest rate (r)} = 10\%$$

$$\text{Compound interest (C.I.)} = ?$$

$$\text{Total amount (F)} = ?$$

We know that,

Deposit Rs. 5000 today in an account, paying 12% interest for six years, how much is simple interest? How much is compound interest?

Hence, the total amount due after three years will be Rs. 1157.63

$$= \text{Rs. } 1157.63$$

$$= \text{Rs. } 1000 (1 + 0.05)^3$$

$$F = P(1+i)^N$$

We know that,

Total amount (F) = ?

Interest rate (i) = 5% = 0.05

Period (N) = 3 years.

Borrowed amount (P) = Rs. 1000

Given that,

Solution:-

If an engineer borrows Rs. 1,000 from the company credit union at 5% per year compound interest, compute the total amount due after three years.

Hence, The compound interest will be Rs. 331 and the total amount at the end of three years will be Rs. 1331.

$$= \text{Rs. } 1331 - \text{Rs. } 1000$$

$$C.I = F - P$$

Also,

$$= \text{Rs. } 1331.$$

$$= \text{Rs. } 1000 (1 + 0.05)^3$$

$$F = P(1+i)^N$$

→ Solution:-
Given that,

Today's amount (P) = Rs. 5000

Interest (I) = 12% = 0.12

Period (N) = 6 years.

Simple interest (I) = ?

Compound interest (C.I) = ?

We know that,

$$I = P \times I \times N$$

$$= Rs. 5000 \times 0.12 \times 6$$

$$= Rs. 3600$$

Also,

$$\text{Compound interest (C.I)} = P [(1+I)^N - 1]$$

$$= Rs. 5000 [(1+0.12)^6 - 1]$$

$$= Rs. 4869.11$$

Hence, simple interest is Rs. 3600 and compound interest is Rs. 4869.11

6. Devarmo: Problem 3: Compare the interest earned by Rs. 9,000 for five years at 8% simple interest per year with the interest earned by the same amount for five years at 8% compounded annually. Explain why a difference occurs? → Solution:-

Given that:-

Amount (P) = Rs. 9000

Period (N) = 5 years.

Interest rate (I) = 8% = 0.08

Simple interest (I) = ?

compound interest (C.I) = ?

We know that,

$$I = P \times N \times r$$

$$= \text{Rs. } 9000 \times 0.08 \times 5$$

$$= \text{Rs. } 3600$$

$$\text{Also, } F = P + I$$

$$= \text{Rs. } 9000 + \text{Rs. } 3600$$

$$= \text{Rs. } 12600$$

Similarly,

$$C.I = P [(1+r)^N - 1]$$

$$= \text{Rs. } 9000 [(1+0.08)^5 - 1]$$

$$= \text{Rs. } 4223.7$$

$$\text{And, } F = P(1+r)^N$$

$$= \text{Rs. } 9000 (1+0.08)^5$$

$$= \text{Rs. } 13,223.70$$

Now,

The compounded interest is greater than the simple interest for the same period of time. This difference occurs because compound interest is calculated on the basis of the total amount at the end of the previous period and this total amount includes the original principal plus the accumulated interest that has been left in the account.

Nominal interest rate

For an interest rate of 0.50% per month, determine the nominal interest rate per (a) quarter (b) year and (c) two year.

Solution:-

Given that,

Interest rate (r) = 0.50%

Nominal rate per quarter (quarterly) = ?

Nominal rate per year (yearly) = ?

Nominal rate per two year (two year) = ?

We know that,

$$i_{\text{eff}} = (1 + i_m)^{12} - 1$$

$$= (1 + 0.005)^{12} - 1$$

$$= 0.0616$$

Now,

We know,

$$i_{\text{eff}} = (1 + i_m)^m - 1$$

$$\text{or, } 0.0616 = (1 + i_m)^4 - 1$$

$$\text{or, } 1 + i_m = (1.0616)^{1/4}$$

$$\text{or, } i_m = 0.015$$

Quarterly = ?

$$= (1 + 0.015)^4 - 1$$

$$= 0.0616$$

Quarterly = 1.5%

$$i_{\text{eff}} = (1 + i_m)^m - 1$$

$$\text{or, } 0.0616 = (1 + i_m)^{12} - 1$$

or, $r = 6\%$ per year.

For 1 two year,

$$1 - 1 = \left(1 + \frac{r_{\text{two year}}}{m}\right)^m - 1$$

or, $0.0616 = \left(1 + \frac{r_{\text{two year}}}{24}\right)^{24} - 1$

Also, or, 1 two year = 12% per two years.

$$1 \text{ two year} = 1 \text{ year} \times 2 = 6\% \times 2 = 12\%$$

[Nominal interest rate per year (r) = monthly interest $\times$ month in year]

For an interest rate of 18% per year compounded monthly, determine the nominal interest rate per

a) two months
b) six months and
c) two years.

Solution:-

Given that,

$$\text{Interest rate (1 year)} = 18\%$$

Nominal interest rate per two months,

$$r = \left(1 + 0.18\right)^{\frac{1}{12}} - 1 \times 2$$

$$1 - 1 = \left(1 + 0.18\right)^{\frac{1}{12}} - 1$$

$$= 6.2875$$

Now, $1 - 1 = \left(1 + \frac{r}{m}\right)^m - 1$

or, $6.2875 = \left(1 + \frac{r}{2}\right)^2 - 1$

$$\text{Monthly interest (monthly)} = \frac{\text{Interest rate}}{\text{month in year}} = \frac{1.8}{12} = 1.5\%$$

Now,

$$\text{Nominal interest rate per two month} = 1.5 \times 2 = 3\%$$

$$\begin{aligned} \text{b) Nominal interest rate for six month} &= \text{Monthly interest} \times \text{six month} \\ &= 1.5\% \times 6 = 9\% \end{aligned}$$

$$\begin{aligned} \text{g) Nominal interest rate for 2 years,} &= \text{Monthly interest} \times \text{month in two year} \\ &= 1.5\% \times 24 = 36\% \text{ per two year.} \end{aligned}$$

Effective * interest rate:-

9. A bank offers 8% nominal rate of interest with quarterly compounding. What is the effective rate of interest?

Given that:-

$$\text{Nominal rate of interest (r)} = 8\% = 0.08$$

Effective rate of interest (i_{eff}) = ?

We know that,

$$i_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$= \left(1 + \frac{0.08}{4}\right)^4 - 1$$

$$= 0.0824$$

$$= 8.24\%$$

Hence, the effective rate of interest is 8.24%

Hari obtained a new credit card from a national bank, with a stated rate of 18% per year compounded monthly. For a Rs. 1000 balance at the beginning of the year, find the effective annual rate and the total amount owed to bank after one year provided no payments are made during the year.

Solution:-

Given that,

Stated rate per compounded monthly (r) = 18% per year = 0.18

Beginning balance (P) = Rs. 1000

Effective annual rate (?) = ?

Total amount after 1 year (F) = ?

We know that,

$$F = P(1 + r)^n$$

$$= \left(1 + \frac{0.18}{12}\right)^{12} - 1$$

$$= 19.56\%$$

Also,

$$F = P(1 + r)^n$$

$$= Rs. 1000 [1 + 0.1956]^1$$

$$= Rs. 1195.62$$

Hence, the effective annual rate is 19.56% and the total

amount owed to bank after one year is Rs. 1195.62.

11. R. Panneerselvam: Example 3.9: A person invests a sum of Rs. 5,000 in a bank at a nominal interest rate of 12% for ten years. The compounding is quarterly. Find the maturity amount of the deposit after ten years.

Ans: Solution:-

Given that,
Invested sum (P) = Rs. 5000

Nominal interest rate (r) = 12%.

Period (N) = 10 years.

Maturity amount (F) = ?

We have,

Effective rate of interest,
 $r_e = (1 + \frac{r}{m})^m - 1$

$$= (1 + \frac{0.12}{4})^4 - 1$$

Now, = 12.55% compounded annually.

$$F = P(1+i)^N$$

$$= 5000(1 + 0.1255)^{10}$$

$$= \text{Rs. } 16310.19$$

Hence, The maturity amount of the deposit after ten years will be Rs. 16310.19.

12. An engineer deposits Rs. 30,000 into an account which pays interest at 8% per year, compounded quarterly. What will be the total amount at the end of two years?

Solution:-

Given that:-

Deposited amount (P) = Rs. 30,000

Interest rate (r) = 8% compounded quarterly.
Total amount after two year (F) = ?
We know that,

$$\text{Left} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$= \left(1 + \frac{0.08}{4}\right)^4 - 1$$

$$= 0.0824 = 8.24\%$$

Now,

$$F = P(1 + i_{\text{eff}})^N$$

$$= \text{Rs. } 30,000 (1 + 0.0824)^2$$

$$= \text{Rs. } 35147.69$$

Hence, The total amount at the end of two years will be Rs. 35147.69.

De Garmo: Example 3-20: A credit card company charges an interest rate of 1.375% per month on the unpaid balance of all accounts. The annual interest rate, they claim is $12 \times (1.375\%) = 16.5\%$. What is the effective rate of interest per year being charged by the company?

Given that,

$$\text{Interest rate (r)} = 1.375\% \text{ per month}$$

$$= 16.5\% \text{ per year.}$$

Effective rate of interest (i) = ?

We know that,

$$1 + i_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$= \left(1 + 0.165\right)^{12} - 1$$

$$= 0.1780$$

$$= 17.8\% \text{ per year.}$$

Hence, the effective rate of interest per year being charged by the company will be 17.8%.

14. Example 3-21, DeGarmo: Suppose that a Rs. 100 lump-

sum amount is invested for 10 years at a nominal interest rate of 6% compounded quarterly. How much is it worth at the end of the tenth year?

Solution:-

Given that,

Lump sum amount (P) = Rs. 100

Period (N) = 10 years.

Nominal interest rate (r) = 6% compounded quarterly

Total amount after 10 years (F) = ?

We have,

$$1 + i_{\text{eff}} = \left(1 + \frac{r}{m}\right)^m - 1$$

$$= \left(1 + 0.06\right)^4 - 1$$

$$= 0.0613 = 6.13\%$$

Now,

$$F = P(1 + i_{\text{eff}})^N$$

$$= \text{Rs. } 100 (1 + 0.0613)^{10}$$

$$= \text{Rs. } 181.29$$

Hence, it worth Rs. 181.29 at the end of the tenth year.

Present worth

If Rs. 500 is received for five years, what will be its present value at a 20% discount rate?

Solution:-

Given that,

$$A = 500$$

$$P = 500$$

$$i = 20\%$$

$$P = A(P/A, 20\%, 5)$$

$$P = 500 \times \frac{1 - (1 + i)^{-N}}{i}$$

$$= 500 \times \frac{(1 - 20\%)^5 - 1}{20\%}$$

$$= \text{Rs. } 1495.30$$

Hence, its present value will be Rs. 1495.30.

17. R. Panneerselvam: Example 3.2:- A person wishes to have a future sum of Rs. 1,00,000 for his son's education after 10 years from now. What is the single payment that he should deposit now so that he gets the desired amount after 10 years? The bank gives 15% interest rate compounded annually.

→ Solution:-

Given that,

Future sum (F) = Rs. 1,00,000

Period (N) = 10 years.

Interest rate (i) = 15%, compounded annually.

We know that,

$$P = F = F(P/F, i, n).$$

$$(1+i)^n$$

$$= 1,00,000 (P/F, 15\%, 10)$$

$$= 1,00,000 (1+i)^{-n}$$

$$= 1,00,000 (1+0.15)^{-10}$$

$$= 1,00,000 (1+0.15)^{-10}$$

$$= 1,00,000 \times 0.2472$$

$$= \text{Rs. } 24,720.$$

Hence, the person has to invest Rs. 24,720 now so that he will get a sum of Rs. 1,00,000 after 10 years at 15% interest rate compounded annually.

18. R. Panneerselvam: Example 3.5: A company wants to set up a reserve which will help the company to have an annual equivalent amount of Rs. 1,00,000

for the next 20 years towards its employees welfare measures. The reserve is assumed to grow at the rate of 15% annually. find the single payment that must be made now as the reserve amount.

> Solution:-

given that:

Annual equivalent amount (P) = Rs. 1,00,000.

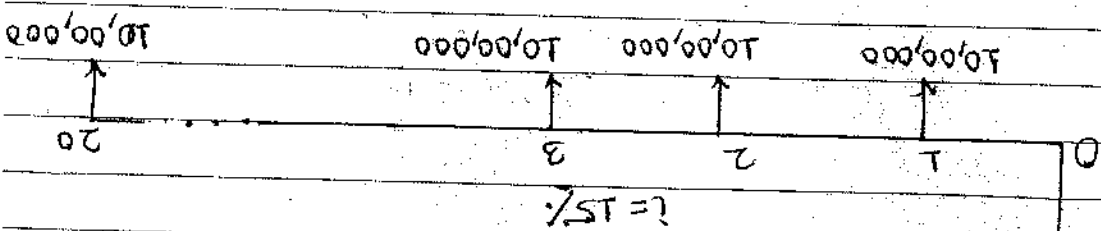
Period (N) = 2 years.

Interest rate (i) = 15%.

Single Payment (P) = ?

We know that,

The corresponding cash flow diagram is given below.



(Now,

$$P = A \left[\frac{P/A, i, n}{(1+i)^n - 1} \right]$$

$$= A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$= 10,00,000 \left[\frac{(1+0.15)^{20} - 1}{0.15(1+0.15)^{20}} \right]$$

$$= \text{Rs. } 10,00,000 \times 6.2593$$

$$= \text{Rs. } 62,59,300$$

Hence, the single payment that must be made now as the reserve amount is Rs. 62,59,300.

Future Worth

o. R. Panneerselvam: Example 3.1. A person deposits a sum of Rs. 20,000 at the rate of 18% compounded annually for 10 years. Find the maturity value after 10 years.

Solution:-

Given that,
Deposited sum (P) = Rs. 20,000

Hence, the investor should be willing to pay Rs. 6301.69 now for his property.

$$= \text{Rs. } 6301.69$$

$$(1+i)^n$$

$$= F$$

$$P = F(P/F, i, n)$$

We know that,

Present worth (P) = ?

Increasing rate (i) = 8%

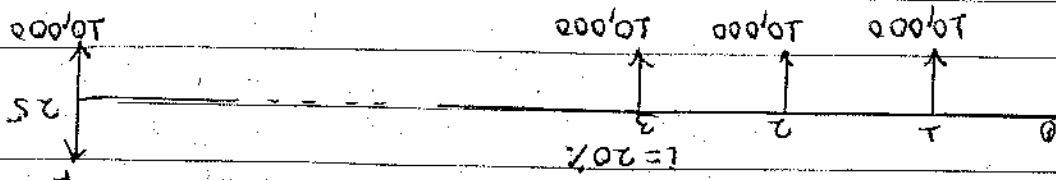
Period (N) = 6 years.

Future Worth (F) = Rs. 10,000

Given that:-

Solution:-

9. Example 3.4, DeGarmo: An investor has an option to purchase a tract of land that will be worth Rs. 10,000 in six years. If the value of the land increases at 8% each year, how much should the investor be willing to pay now for this property?



We have, the corresponding cash flow diagram is

Maturity value after 25 years (F) = ?

Interest rate (i) = 20%, compounded annually.

Period (N) = 25 years.

Present investment (P) = Rs. 10,000.

Given that,

Solution:-

he is 60 years old.

annually. Find the maturity value of his account when

next year. The bank gives 20% interest rate, compounded

year for the next 25 years starting from the end of the

invest an equal sum of Rs. 10,000 at the end of every

35 years old is planning for his retired life. He plans to

1. R. Panneerselvam: Example 3.3. A person who is now

Hence, The maturity value after 10 years will be Rs. 1,04,680.

$= \text{Rs. } 1,04,680.$

$= \text{Rs. } 20,000 \times 5.234$

$= \text{Rs. } 20,000 (1 + 0.18)^{10}$

$= P(1+i)^N$

$F = P(F/P, i, n)$

We know that,

Maturity value after 10 years (F) = ?

Period (N) = 10 years

Rate of interest (i) = 18%, compounded annually.

Now,

$$F = P(F/P, i, n)$$

$$= P \frac{(1+i)^n - 1}{i}$$

$$= Rs. 10,000 \frac{(1+0.2)^{25} - 1}{0.2}$$

$$= Rs. 10,000 \times 471.981$$

$$= Rs. 47,19,810.$$

Hence, the future sum of the annual equal payments after 25 years is equal to Rs. 47,19,810.

Annual worth:-

22. R. Panneerselvam: Example 3.4: A company has to replace a present faulty after 15 years at an outlay of Rs. 5,00,000. It plans to deposit an equal amount at the end of every year for the next 15 years at interest rate of 18% compounded annually. Find the equivalent amount that must be deposited at the end of every year for the next 15 years.

→

Solution:-

Given that,

$$\text{Future price (F)} = Rs. 5,00,000.$$

$$\text{Period (N)} = 15 \text{ years.}$$

$$\text{Interest rate (i)} = 18\% \text{ compounded annually.}$$

$$\text{Equivalent amount at present (P)} = ?$$

We know that,

$$P = F(P/F, i, N)$$

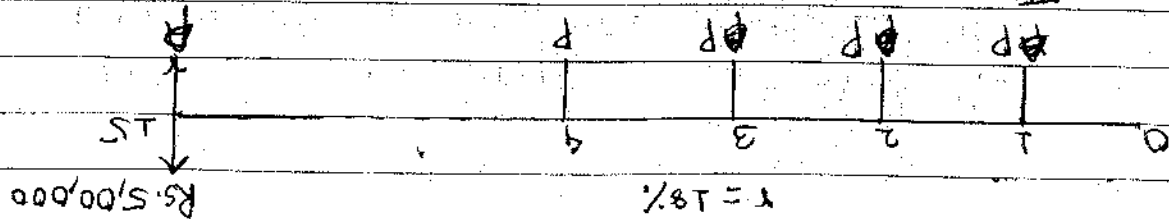
$$= F \frac{(1+i)^N - 1}{i}$$

$$= \text{Rs. } 5,00,000 \left\{ \frac{0.18}{(1+0.18)^{15} - 1} \right\}$$

$$= \text{Rs. } 5,00,000 \times 0.0164$$

$$= \text{Rs. } 8200.$$

The corresponding cash flow diagram is shown below.



Hence, The equivalent amount that must be deposited at the end of every year for the next 15 years is Rs. 8200.

R. Panneerselvam: Example 3.6. A bank gives a loan to a company to purchase an equivalent equipment worth Rs. 10,00,000 at an interest rate of 18%. compounded annually. This amount should be repaid in 15 yearly equal installments. Find the installment amount that the company has to pay to the bank.

Solution:-

Given that,
 Worth of equipment (F) = Rs. 10,00,000
 Interest rate (?) = 18%, compounded annually.
 Period (n) = 15 years.
 Installment amount (P) = ?

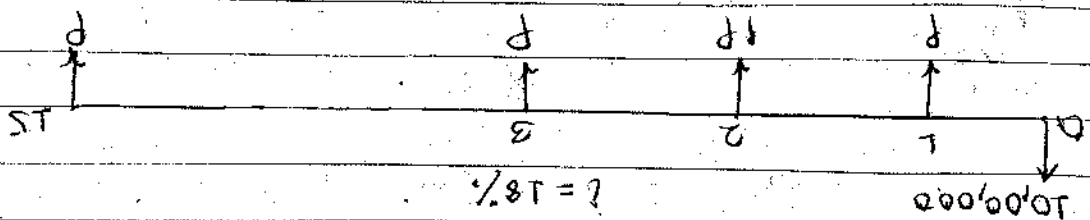
$$P = F \frac{i(1+i)^n}{(1+i)^n - 1}$$

$$= 10,00,000 \times \frac{0.18(1+0.18)^{15}}{(1+0.18)^{15} - 1}$$

$$= \text{Rs. } 10,00,000 \times 0.1964$$

$$= \text{Rs. } 196,400$$

The corresponding cash flow diagram is shown below.



Hence, the installment amount that the company has to pay to the bank is Rs. 196,400.

_____ x _____ x _____

Tutorial

CHAPTER - 4

Basic Methods of Engineering Economic Studies

Payback * Period * Method

1. Example: P. 287: Chain S. Park.

Compute discounted payback period of a project considering 15% rate of return from the following data:

Year	Cash flow (\$)
0	-85,000
1	30,000
2	30,000
3	30,000
4	1,000
5	1,000
6	1,000

→ Solution:-

Given that,

Rate of return (r) = 20% = 0.20, 15% = 0.15

$$\text{Discounted cash flow into present} = A \left[\frac{1}{(1+r)^N} \right] = \frac{A}{(1+r)^N}$$

Discounted * Payback Period.

Year	Cash flow (\$)	Discounted cash flow present	Cumulative cash f
0	-85,000	-85,000	-85,000
1	30,000	26,086.95	-58,913.05
2	30,000	22,771.26	-36,141.79
3	30,000	19,796.75	-16,345.04
4	1,000	684.97	-15,660.07
5	1,000	595.21	-15,064.86

Given that,

Solution:

Year	Net cash flow Rs.
0	-25,000
1	+8,000
2	+8,000
3	+8,000
4	+8,000
5	+8,000
6	+18,000

2. Example: P.137: Santosh K. Shrestha.
Compute discounted payback period at $i = 20\%$ from the following data.

From above diagram or table, the cumulative cash flow does not turn positive at any period. Hence, we can conclude that the payback period of the given project is more than 6 years.

Year	Cash flow \$	PM at 15%	Cumulative PM
0	-85,000	-85,000	-85,000
1	30,000	$30,000 / (1+0.15)^1 = 26,086.95$	-58,913.05
2	30,000	$30,000 / (1+0.15)^2 = 22,684.31$	-36,228.74
3	30,000	$30,000 / (1+0.15)^3 = 19,725.48$	-16,503.26
4	1,000	$1,000 / (1+0.15)^4 = 571.75$	-15,931.51
5	1,000	497.17	-15,434.34
6	1,000	432.32	-15,002.02

Rate of Return (i) = 20%

Now, Discounted Payback Period.

Year	Net cash flow (Rs)	Discounted cash flow into present.	Cumm. cash flow
0	-25,000	-25,000	-25,000
1	+8,000	$8000/(1+0.2)^1 = 6,666.67$	-18,333
2	+8,000	$8000/(1+0.2)^2 = 5,556$	-12,777
3	+8,000	$8000/(1+0.2)^3 = 4,630$	-8,147
4	+8,000	$8000/(1+0.2)^4 = 3,858$	-4,289
5	+13,000	$13,000/(1+0.2)^5 = 5,224$	+935

Here, the cumulative cash flow turns into positive in period 5. Therefore, payback period lies between period 4 and 5.

By interpolating, we get the required payback period,

$$T = 4 + \frac{4289}{5224} = 4 + 0.821$$

$$\therefore T = 4.82 \text{ years.}$$

Hence, discounted payback period at $i = 20\%$ is 4.82 years.

Example 5.13: P.210: William G. Sullivan:-

A new equipment has been proposed by engineer to increase productivity. Investment cost is \$25,000 and the equipment will have a market (salvage) value at the end of its expected life of five years. Increased productivity attributable to the equipment will amount to \$8,600 per year after subtracting extra operating cost from the value of the additional production. Compute single payback period and discounted payback period at $MARR = 20\%$

→ Solution:

Given that,

$$MARR = 20\%$$

$$\text{Capital investment cost} = \$25,000$$

$$\text{Market value} = \$8,000$$

$$\text{Useful life (n)} = 5 \text{ years.}$$

Now,

Simple Payback Period.

Year	Net cash flow (\$)	Cumulative cash flow (\$)
0	-\$25,000	-\$25,000
1	\$8,000	-17,000
2	\$8,000	-9,000
3	\$8,000	-1,000
4	\$8,000	+7,000
5	\$8,000	+15,000

Here, cumulative cash flow turns to positive in year 4. Therefore, payback period lies between 3 and 4. By interpolating we get payback period.

$$= 3 + \frac{1000}{8000} = 3.125 \text{ years.}$$

$$\therefore T = 3.125 \text{ year.}$$

Discounted Payback Period.

Year	Net cash flow (\$)	Discounted cash flow (PW) \$	Cumulative (\$)
0	-\$25,000	-\$25,000	-\$25,000
1	+8,000	+6,667	-18,333
2	+8,000	+5,556	-12,777
3	+8,000	+4,630	-8,147
4	+8,000	+3,858	-4,289
5	+8,000	+3,224	+935

Here, the cumulative cash flow turns to positive in period 5. Therefore, payback period lies between period 4 and 5.

By interpolating, we get the required payback period.

$$4 + \frac{4289}{5224} = 4.82 \text{ periods}$$

$$\therefore T = 4.82 \text{ period}$$

Equivalent Worth Methods:-

Present Worth:-

Example 4.3, Panneerselvam [PW]

Investment proposals A and B have net cash flows as follows:-

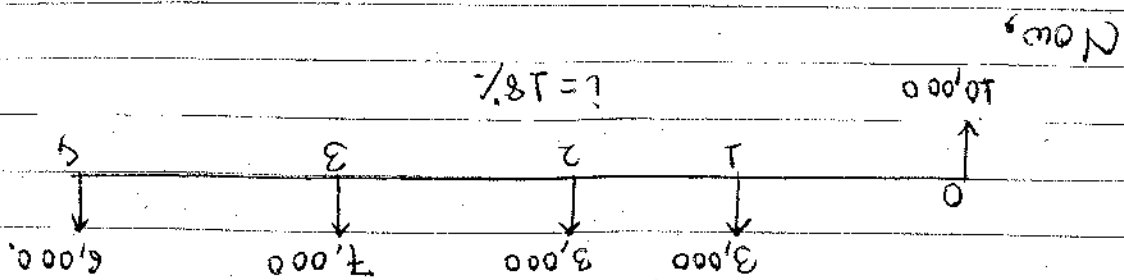
Proposals	0	1	2	3	4
A (Rs.)	-10,000	3,000	3,000	7,000	6,000
B (Rs.)	-10,000	6,000	6,000	3,000	3,000

Compare the present worth of A with that of B at $i = 18\%$. Which proposal should be selected?

> Solution:-

For A:-

Present worth of A at $i = 18\%$. The cash flow diagram of proposal A is given below.



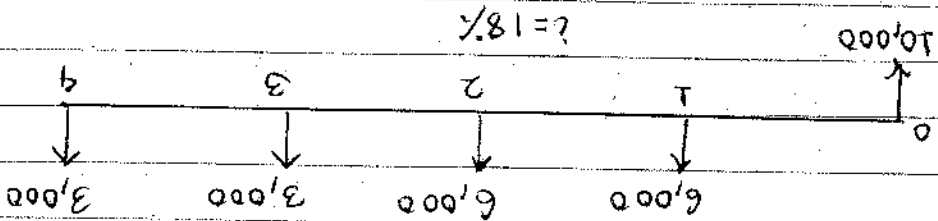
The present worth of the above cash flow diagram is computed as,

$$PW_A = -10,000 + \frac{3,000}{(1+0.18)^1} + \frac{3,000}{(1+0.18)^2} + \frac{3,000}{(1+0.18)^3} + \frac{7,000}{(1+0.18)^4}$$

$$= -10,000 + 3,000 \times (0.8475) + 3,000 \times (0.7182) + 7,000 \times 0.6086 + 6,000 \times 0.5158$$

$$= Rs. 2052.10.$$

For Proposal B,
Present worth of B at $i = 18\%$. The cash flow diagram of the proposal B is:-



The present worth of above cash flow diagram is,

$$PW_B = -10,000 + \frac{6,000}{(1+0.18)^1} + \frac{6,000}{(1+0.18)^2} + \frac{3,000}{(1+0.18)^3} + \frac{3,000}{(1+0.18)^4}$$

$$= -10,000 + 6,000 \times 0.8475 + 6,000 \times 0.7182 + 3,000 \times 0.6086 + 3,000 \times 0.5158$$

$$= Rs. 2767.40.$$

Since, the present worth of proposal B is higher than that of proposal A. Therefore, proposal B should be selected.

Example 4.4. Pannerebelvam [PW]

A granite company is planning to buy a fully automated granite cutting machine. If it is purchased under down payment, the cost of machine is Rs. 16,00,000. If it is purchased under installment basis, the company has to pay 25% of the cost at the time of purchase and remaining amount in 10 annual installment of Rs. 2,00,000 each. Suggest the best alternative for the company using the present worth basis at $i=18\%$ compounded annually.

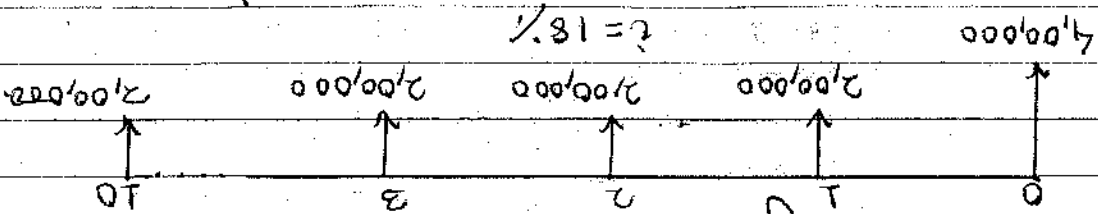
> Solution:-

Since there are two alternatives available for the company i.e.

- Alternative 1:- Down payment of Rs. 16,00,000
Alternative 2:- Down payment of Rs. 4,00,000 (25% of Rs. 16,00,000) and 10 annual equal installments of Rs. 2,00,000 each.

For Present worth of second alternative:-

The cash flow diagram of the second alternative is



$$\therefore PW_2 = 4,00,000 + 2,00,000 (P/A, 18\%, 10)$$

$$= 4,00,000 + 2,00,000 \times 4.4941$$

$$= \text{Rs. } 12,98,820.$$

Since, the cost of PW of alternative 2 is less than that of cost of PW of alternative 1, hence the company should select second alternative to buy the fully automated granite cutting machine.

3. Example 4.5 Pannerselvom (PW).

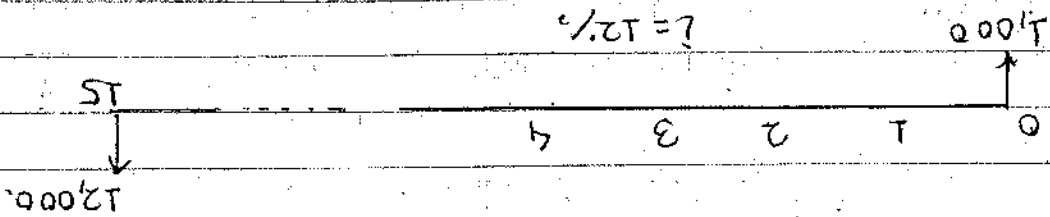
A finance company advertises two investment plans. In plan 1, the company pays Rs. 12,000 after 15 years for every Rs. 1000 invested now. In plan 2, for every Rs. 1000 invested, the company pays Rs. 4,000 at the end of 10th year and Rs. 2,000 at the end of 15th year. Select the best investment plan from the investors point of view at $i = 18\%$ compounded annually.

> Solution:-

Given that,

For plan 1,

The cash flow diagram for plan 1 is,



$$\therefore PW = -1,000 + 12,000 (P/F, 12\%, 15)$$

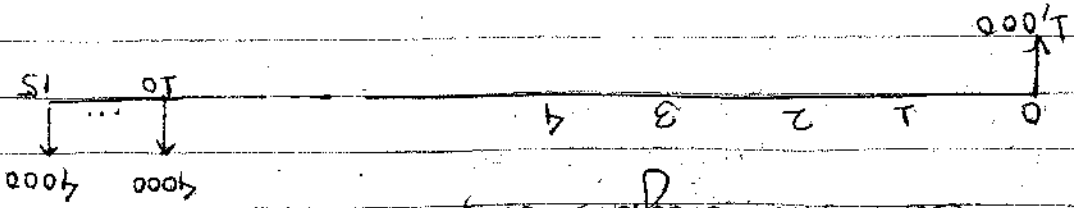
$$= -1,000 + \frac{12,000}{(1+0.12)^{15}}$$

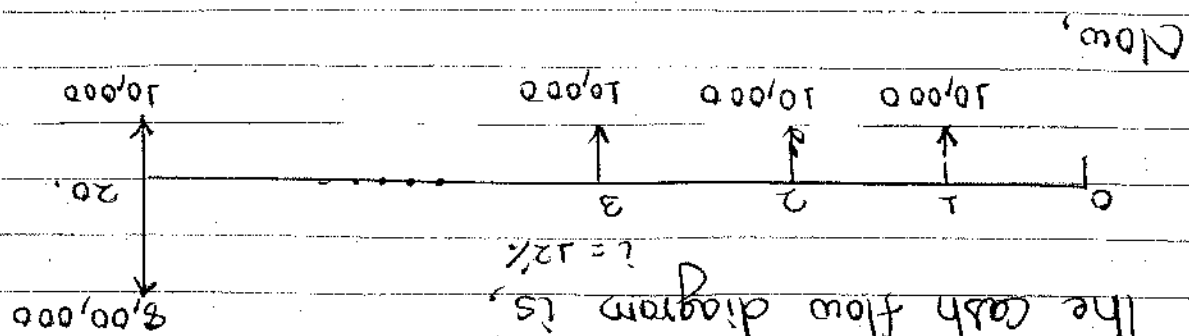
$$= -1,000 + 12,000 \times (0.1827)$$

$$= \text{Rs. } 1,192.40.$$

Similarly,
For plan 2,

The cash flow diagram is,





For Novel Investment Ltd,

Solution: -

basis with at $i = 12\%$

Novel Investment Ltd accepts Rs. 10,000 at the end of every year for 20 years and pays the investor Rs. 800,000 at the end of 20 years. Innovative Investment Ltd. accepts Rs. 10,000 at the end of every year for 20 years and pays the investors Rs. 15,00,000 at the end of 25th year. Which is the best investment alternative from the investor's point of view. Use present worth basis with at $i = 12\%$

1. Example 4.6 Panneerselvam [PM]

Since, the present worth of plan 1 is more than the present worth of plan 2, so the best investment plan from the investors point of view is plan 1.

$$= \text{Rs. } 1,018.80$$

$$= -1,000 + 4000 \times (0.3220) + 4000 \times 0.1827$$

$$= -1,000 + \frac{4000}{(1+0.12)^{10}} + \frac{4000}{(1+0.12)^{15}}$$

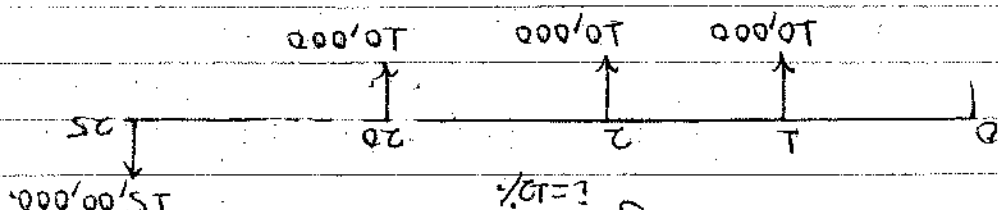
$$\therefore (PW)_2 = -1,000 + 4,000 (P/F, 12\%, 10) + 4,000 (P/F, 12\%, 15)$$

$$PW = -10,000 (P/A, 12\%, 20) + 8,00,000 (P/F, 12\%, 20)$$

$$= -10,000 \times 7.4694 + 8,00,000 \times 0.1037$$

$$= \text{Rs. } 8266.$$

Similarly,
for Innovative Investment Ltd,
The cash flow diagram is,



$$PW = -10,000 (P/A, 12\%, 20) + 15,00,000 (P/F, 12\%, 25)$$

$$= -10,000 \left(\frac{1 - (1 + i)^{-N}}{i} \right) + 15,00,000 \left(\frac{1}{(1 + 0.12)^{25}} \right)$$

$$= -10,000 \times 7.4694 + 15,00,000 \times 0.0588$$

$$= \text{Rs. } 13,506.$$

Since, the present worth of Innovative Investment Ltd's plan is more than that of Novel Investment Ltd's plan, therefore, Innovative Investment Ltd's plan is the best from investor's point of view.

Future Worth

Example 4.4: Santosh K. Shrestha [F.W.]

A company intends to mechanize its existing process by installing a machine. Machine is estimated to

cost of Rs. 1,00,000 and expected to save Rs. 15,000 per year for a period of 12 years. Is it worthwhile to install machine if salvage value is Rs. 25,000 at the end of 12 years. $MARR = 12\%$.

Solution:-

Given that:-

Cost of machine = Rs. 1,00,000.

Period (N) = 12 years

$MARR (r) = 12\%$

Salvage value = Rs. 25,000.

Future worth (FW) = ?

We know that,

using FW formulation,

$$FW = -1,00,000 (F/P, 12\%, 12) + 15,000 (F/A, 12\%, 12) + Rs. 25,000$$

$$= -1,00,000 \left[\frac{(1+12\%)^{12} - 1}{12\%} \right] + 15,000 \left[\frac{(1+12\%)^{12} - 1}{12\%} \right] + Rs. 25,000$$

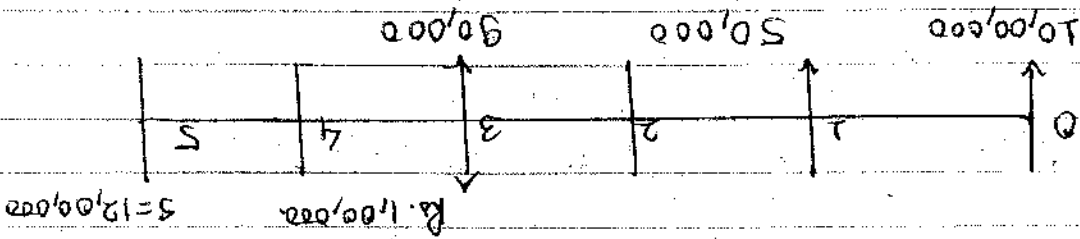
$$= -1,00,000 [1 + 0.12]^{12} + 15,000$$

1. Annual Worth: Example: 4.7 Santosh K. Shrestha [AW].

A company purchased a building five years ago for Rs. 10,00,000. Annual maintenance cost is Rs. 50,000 per year. At the end of 3 years, Rs. 90,000 was spent on roof repairs. At the end of 5 years, company sold building for Rs. 12,00,000. During the period of ownership, company rented the building for Rs. 1,00,000 per year paid off the beginning of each year. Use AW method to evaluate the investment if MARR = 12%.

→ Solution:-

Given that, The cash flow diagram is



$$AW = -10,00,000 (A/P, 12\%, 5) - 50,000 (F/P, 12\%, 2) + (A/F, 12\%, 5) + 12,00,000 (A/F, 12\%, 5) + 1,00,000$$

Benefit $\times$ Cost $\times$ Ratio $\times$ Method.

Example 11-2 Degarmo (B/C ratio)

Nepal government is considering extending the runways of Tribhuvan International Airport. The projected cost for Airport runway extension are of:-

Land can be purchased for Rs. 3,50,000

Construction cost for runway extension = Rs. 6,00,000

Additional annual maintenance cost for the extension = Rs. 22,500

Construction cost for the terminal building = Rs. 2,50,000

Annual operating and maintenance cost for extension = Rs. 75,000

Projected increase of two traffic controllers will require an annual cost = Rs. 1,00,000.

Annual benefits of runway extension have been estimated as follows:-

Rs. 3,25,000 - rental receipts from airlines leasing space at facility

Rs. 65,000 - airport tax charged to passengers.

Rs. 50,000 - convenience benefits for residents of Kathmandu.

Rs. 50,000 - Additional tourism benefits for residents of Kathmandu.

Apply the B/C ratio method with the study period of 20 years and a nominal interest rate of 10% per year to determine whether the Tribhuvan International Airport runway should be extended.

Solution:-

Given that,

Study period (N) = 20 years.

Nominal interest rate (r) = 10%.

Now,

Tutorial

CHAPTER - 5

Comparative Analysis of Alternatives

Payback Period Method:-

Example: P. 236. Radheyshyam Pradhan [Payback Period]
 Compute payback period of project X and Project Y from following data:

Particulars	Project (X) in millions	Project Y (Rs. in millions)
Original investment	- 12,000	- 12,000
Cash flow: Year 1	3,000	3,000
Year 2	3,000	3,500
Year 3	3,000	4,000
Year 4	3,000	4,500
Year 5	3,000	5,000

→ Solution:-

Considering $MARR = 15\%$
 for project X,

Discounted payback period.

Year	Cash flow (in millions)	PV	Cummu.
0	- 12,000	- 12,000	- 12,000
1	3,000	2,608.6	- 9,392
2	3,000	2,268	- 7,124
3	3,000	1,973	- 5,151
4	3,000	1,715	- 3,436
5	3,000	1,492	- 1,944

(P.T.O.)

for Project-X.

Simple Payback Period

Year	Cashflow (in Million)	Cumm.
0	-12,000	-12,000
1	3,000	-9,000
2	3,000	-6,000
3	3,000	-3,000
4	3,000	0
5	3,000	+3,000

Here, the cumulative cash flow turns to positive in year 4 i.e. it becomes zero. So, the payback period is 4 years.

for Project Y,

Simple Payback Period

Year	Cashflow (in million)	Cumm.
0	-12000	-12000
1	3000	-9000
2	3500	-5500
3	4000	-1500
4	4500	+3000
5	5000	+8000

Here, the cumulative cash flow turns to

positive in year 4. Therefore, payback period lies between year 3 and 4.

By interpolating, we get required payback period.

$$= 3 + \frac{1500}{5000}$$

$$= 3.33 \text{ year.}$$

Since, Project X has more payback period than the project Y, so we should select project X.

Example: P. 286: Chain S. Park [Payback Period]
 Consider original investment in year 0, and expected cash flows in other years at $MARR = 15\%$, compute payback period of Project A and Project B from the following data.

Year	Project A (\$)	Project B (\$)
0	-90,000	-90,000
1	30,000	25,000
2	30,000	25,000
3	30,000	25,000
4	1,000	25,000
5	1,000	25,000
6	1,000	25,000

→ Solution:

For Project A,
 Discounted Payback Period,

Year	Cashflow (\$)	Discount PW	Cumm. PW
0	-90,000	-90,000	-90,000
1	30,000	26086	
2	30,000		
3	30,000		
4	1,000		
5	1,000		
6	1,000		

Equivalent* Worth* Method:-
 1. Example 5.1 De Garmo (PW, FW and AW)
 Three mutually exclusive investment alternatives for implementing an office automation plan in an engineering design firm are being considered. Each alternative meets the same service (support) requirements, but differs in capital investment

Since, the cumulative cash flow turns to positive in year 6. Therefore, payback period lies between period 5 and 6.
 By interpolating, we get the required payback period.

$$= 5 + \frac{6198}{10808} = 5.57 \text{ years}$$

for Project B.

Year	Cash flow (\$)	Disc PW	Cum. PW
0	-90,000	-90,000	-90,000
1	25,000	21739	-68261
2	25,000	18904	-49357
3	25,000	16437	-32820
4	25,000	14293	-18627
5	25,000	12429	-6198
6	25,000	10808	+4610

amounts and benefits (cost savings) exist among them. The study period is 10 years, and the useful lives of all three alternatives are also 10 years. Market values of all alternatives are assumed to be zero at the end of their useful lives. If the firm's MARR is 10% per year, which alternative should be selected in view of the following estimates?

Alternative	Capital investment	Annual cost savings
A	390,000	69,000
B	920,000	167,000
C	660,000	133,500

↑ 0
↑ 1
↑ 2
↑ 9
↑ 10 years
r = 10% per year
Rs. 390,000 cash flow diagram of alt. A

Solution:
By PW method,

$$(PW)_A = -Rs. 390,000 + Rs. 69,000 (P/A, 10\%, 10)$$

$$= -390,000 + 69,000 \left[\frac{(1+0.1)^{10} - 1}{0.1} \right] \times \left[\frac{1}{(1.1)^{10}} \right]$$

$$\therefore F = A \left[\frac{(1+i)^n - 1}{i} \right] + P = F \left[\frac{1}{(1+i)^n} \right]$$

$$= -390,000 + 423975.13$$

$$= Rs. 33975.13$$

Similarly,

$$(PW)_B = -920,000 + Rs. 167,000 (P/A, 10\%, 10)$$

$$= -920,000 + 167,000 \left[\frac{(1+0.1)^{10} - 1}{0.1} \right] \times \left[\frac{1}{(1.1)^{10}} \right]$$

$$= Rs. 106,148$$

And,

$$(PW)_C = -Rs. 660,000 + Rs. 133,500 (P/A, 10\%, 10)$$

$$= Rs. 160,304$$

Since, alternative C has the largest PW value, so alternative C would be selected.

† By AW method:-

$$(AW)_A = -Rs. 3,90,000 (A/P, 10\%, 10) + Rs. 69,000$$

$$= -Rs. 3,90,000 \left[\frac{(1+0.1)^{10} \times 0.1}{(1+0.1)^{10} - 1} \right] + Rs. 69,000$$

$$\therefore F = P(1+i)^n \neq A = \left[\frac{F \times i}{(1+i)^n - 1} \right]$$

$$= -68470.7 + 69,000$$

$$= Rs. 5529.30$$

Similarly,

$$(AW)_B = -Rs. 920,000 (A/P, 10\%, 10) + Rs. 167,000$$

$$= Rs. 173161$$

$$(AW)_C = -Rs. 660,000 (A/P, 10\%, 10) + Rs. 1,33,500$$

$$= Rs. 26,118$$

Since alternative C has largest AW value, so alternative C should be selected.

By FW method:-

$$(FW)_A = -Rs. 3,90,000 (F/P, 10\%, 10) + 69,000 (F/A, 10\%, 10)$$

$$= -Rs. 3,90,000 \times (1.1)^{-10} + Rs. 69,000 \left[\frac{(1+0.1)^{-10} - 1}{0.1} \right]$$

$$= Rs. 88,138$$

> Solution:-

Which option would be selected based on equivalent worth method at $i = 20\%$.

	Option 1	Option 2	Option 3
Investment cost	2,69,000	3,19,000	3,39,000
Annual net savings	81,500	88,500	98,300
Useful lives	5	5	5
Salvage value	1,00,000	1,20,000	1,20,000

Consider the mutually exclusive alternatives:

Example 6.1, Santosh K. Shrestha (PW, FW & AW)

Since alternative C has largest FW value, so we should select alternative C as best option.

$$\begin{aligned}
 (FW)_B &= -Rs. 9,20,000 (F/P, 10\%, 10) + Rs. 16,70,000 (F/A, 10\%, 10) \\
 &= Rs. 2,75,342 \\
 (FW)_C &= -Rs. 6,60,000 (F/P, 10\%, 10) + Rs. 13,35,000 (F/A, 10\%, 10) \\
 &= Rs. 4,15,801
 \end{aligned}$$

Similarly,

Present Worth Method:

Example 4.3, Rameeselvam [PW]

Investment proposals of mutually exclusive alternatives A and B have net cash flows as follows:-

Proposal	End of years	A (Rs.)	B (Rs.)
0		-10,000	-10,000
1		3,000	6,000
2		3,000	6,000
3		7,000	3,000
4		6,000	3,000

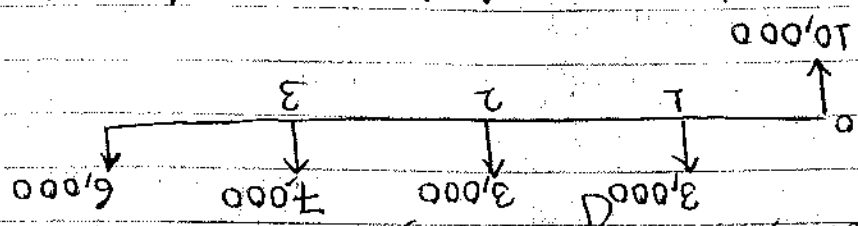
Compare present worth of A with that of B at $i = 18\%$. Which proposal should be selected?

→ Solution:-

Given that,

For alternative A:-

The cash flow diagram is,



∴ The present worth of above cash flow diagram is,
 $(PW)_A = -10,000 + 3,000 (P/F, 18\%, 1) + 3,000 (P/F, 18\%, 2) + 7,000 (P/F, 18\%, 3) + 6,000 (P/F, 18\%, 4)$

$$= -10,000 + 3,000 \times 0.8475 + 3,000 \times 0.7182 + 7,000 \times 0.6086 + 6,000 \times 0.5158$$

$$= \text{RS. } 2052.10.$$

Similarly,
 For alternative B,

→ Solution:-

	Electric-powered	Gas-powered	Solar-powered
First cost, \$	-2,500	-3,500	-6,000
Annual operating cost (\$)	-900	-700	-50
Salvage value \$ (\$)	200	350	100
Life, years	5	5	5

2. Example 5.1 Layland Blank [PW]
Perform a present worth analysis of mutually exclusive alternatives of equal service machines with the cost shown below, if MARR is 10% per year. Revenues of all the alternatives are expected to be the same.

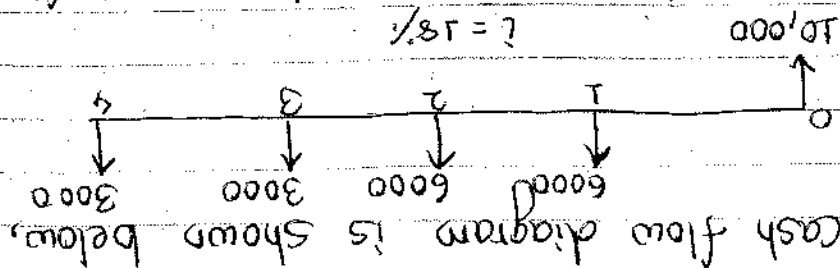
Since, the present worth of proposal B is higher than that of proposal A, therefore proposal B should be selected.

$$= \text{RS. } 2767.40$$

$$= -10,000 + 6,000 (0.8475) + 6,000 \times 0.7182 + 3,000 \times 0.6086 + 3,000 \times 0.5158$$

$$(PW)_B = -10,000 + 6,000 (P/F, 18\%, 1) + 6,000 (P/F, 18\%, 2) + 3,000 (P/F, 18\%, 3) + 3,000 (P/F, 18\%, 4)$$

The present worth of the above cash flow diagram is calculated as,



Given that,
For electric-powered,

RAKESH SHAH

Future Worth Method

1. Example 5.1, Panneerselvam [FW]

Consider the following two mutually exclusive alternatives-

Alternative	End of year	0	1	2	3	4
A(Rs.)		-50,00,000	20,00,000	20,00,000	20,00,000	20,00,000
B(Rs.)		-45,00,000	18,00,000	18,00,000	18,00,000	18,00,000

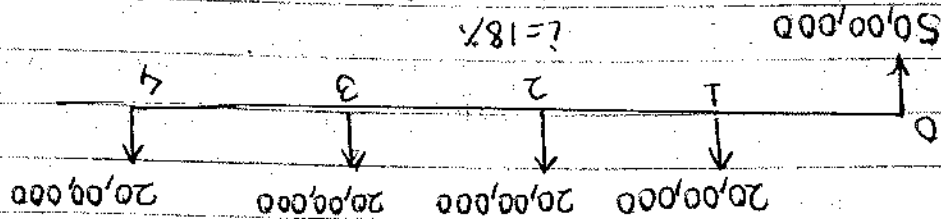
at $i = 18\%$, select the best alternative based on future worth method of comparison.

→ Solution:

Given that,

For alternative A,

The cash flow diagram is given below,



The future worth amount of above diagram can be computed as,

$$(FW)_A = -Rs. 50,00,000 (F/P, 18\%, 4) + 20,00,000 (F/A, 18\%, 4)$$

$$= -Rs. 50,00,000 \times 1.939 + Rs. 20,00,000 \times 5.215$$

$$= Rs. 7,35,000.$$

Similarly

for alternative B,
The cash flow diagram is given below,

Since, the future worth of alternative A is greater than that of alternative B, so the best alternative is ~~A~~ alternative A.

2. Example 5.2, Rameshwar [FW]

A man owns a corner plot. He must decide which of the several alternatives to select in trying to obtain a desirable return on his investment. After much study and calculation, he decides that the two mutually exclusive best alternatives are given in the following table.

Build gas station	Build soft icecream stand	
First cost (Rs.)	20,00,000	36,00,000
Annual property tax, Rs.	80,000	1,50,000
Annual income (Rs.)	8,00,000	9,80,000
Life of building (Year)	20	20
Salvage value, Rs.	0	0

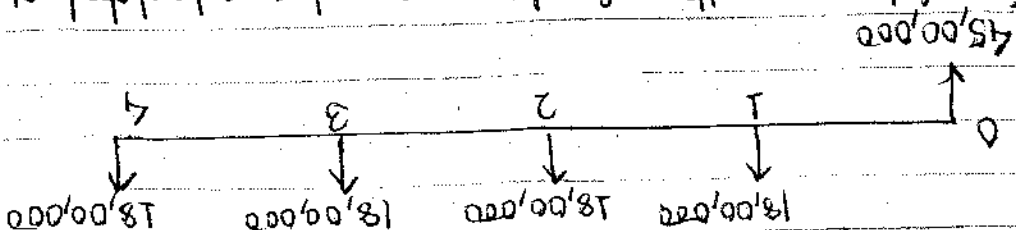
Evaluate the alternatives based on future worth

The future worth of above can be calculated as,

$$(FW)_B = -Rs. 45,00,000 (F/P, 18\%, 4) + Rs. 18,00,000 (F/A, 18\%, 4)$$

$$= -Rs. 45,00,000 \times (1.939) + Rs. 18,00,000 \times 5.215$$

$$= Rs. 6,64,500$$



Method at $i = 20\%$.

→ Solution:-

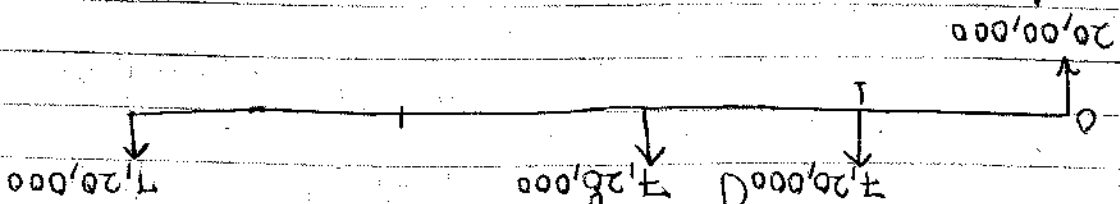
For alternative Build gas station,

$$\text{Net annual income} = \text{Annual income} - \text{Annual property tax}$$

$$= \text{Rs. } 80,000 - \text{Rs. } 80,000$$

$$= \text{Rs. } 7,20,000$$

The cash flow diagram for this is,



The future worth of alternative 1 is calculated as,

$$(FW)_1 = -\text{Rs. } 20,00,000 (F/P, 12\%, 20) + \text{Rs. } 7,20,000 (F/A, 12\%, 20)$$

$$= -\text{Rs. } 20,00,000 * 9.646 + \text{Rs. } 7,20,000 * 72.052$$

$$= \text{Rs. } 3,25,85,440$$

Similarly,

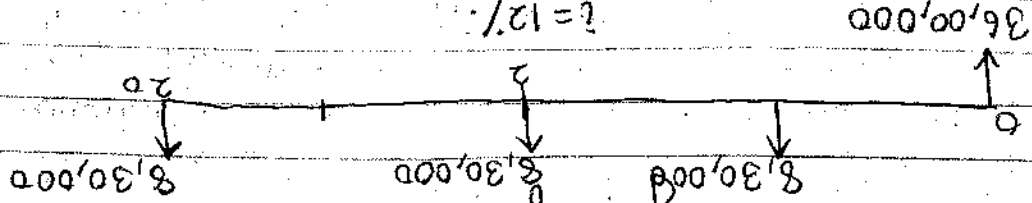
For alternative 2, build soft ice-cream stand,

$$\text{Net annual income} = \text{Annual income} - \text{Annual property tax}$$

$$= \text{Rs. } 9,80,000 - \text{Rs. } 1,50,000$$

$$= \text{Rs. } 8,30,000$$

The cash flow diagram for this is,



Now,

The future worth of this alternative can be calculated as,

$$(FW)_2 = -Rs. 36,00,000 (F/P, 12\%, 20) + Rs. 8,30,000 (F/A, 12\%, 20)$$

$$= -Rs. 36,00,000 \times 9.646 + Rs. 8,30,000 \times 72.052$$

$$= Rs. 2,50,77,560.$$

Since, the future worth of alternative 1 is greater than that of alternative 2, so building the gas station is the best alternative.

Annual Worth Method:-

Example 6.2, Pameer Selvam [AW]

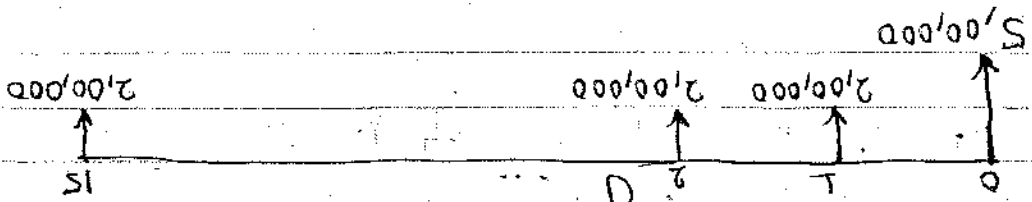
A company is planning to purchase an advanced machine center. Three original manufacturers have responded to its tender whose particulars are tabulated as follows:-

Manufacturer	Down payment (Rs)	Yearly equal installment (Rs)	No. of installment
1	5,00,000	2,00,000	15
2	4,00,000	3,00,000	15
3	6,00,000	1,50,000	15

Determine the best alternative based on the annual equivalent method by assuming $i = 20\%$, compounded annually.

Solution:-

For alternative 1,
The cash flow diagram is,



The annual equivalent cost expression of above cash flow diagram is,

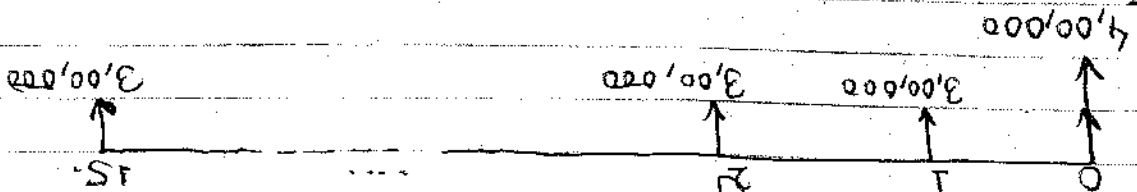
$$AE_1 = RS. 500,000 (A/P, 20\%, 15) + RS. 2,00,000$$

$$= RS. 5,00,000 \times 0.2139 + RS. 200,000$$

$$= RS. 3,06,950.$$

For alternative 2,

The cash flow diagram is,



The annual equivalent cost for above is,

$$AE_2 = RS. 4,00,000 (A/P, 20\%, 15) + RS. 3,00,000$$

$$= RS.$$

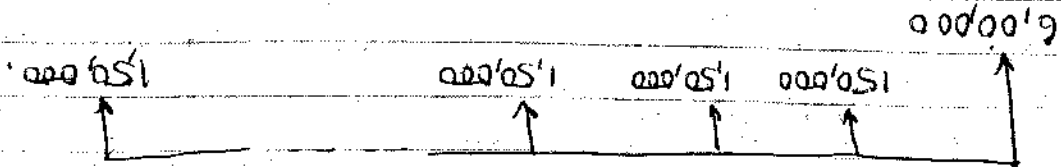
$$= RS. 4,00,000 \times 0.2139 + RS. 3,00,000$$

$$= RS. 3,85,560.$$

And,

For alternative 3,

The cash flow diagram is,



The annual equivalent cost expression is,

$$AE_3 = RS. 6,00,000 (A/P, 20\%, 15) + RS. 1,50,000$$

$$= RS. 6,00,000 \times 0.2139 + RS. 1,50,000$$

$$= \text{RS. } 2,78,340.$$

Since, the annual equivalent cost of manufacturer 3 is less than that of manufacturer 1 and manufacturer 2, so the company should buy the advanced machine from manufacturer 3.

2. Example 6.3. Panneseelvaru [AM]

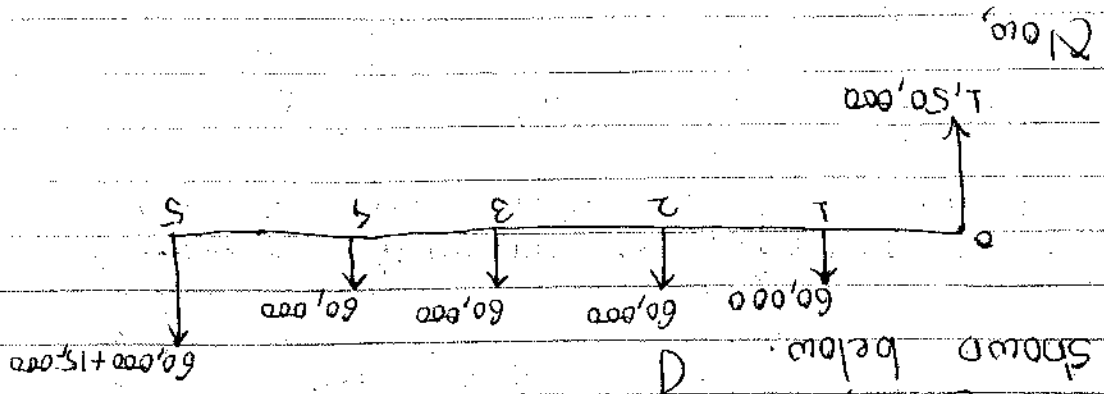
A company invests in one of the two mutually exclusive alternatives. The life of both alternatives is estimated to be 5 years with the following investment, annual returns, and salvage values.

Alternatives	Investment (Rs.)		
	A	B	
	-1,50,000	-1,75,000	
Annual equal return (Rs.)	+60,000	+75,000	
Salvage value (Rs.)	+15,000	+35,000	

Determine the best alternative based on annual equivalent worth method by assuming at $i = 25\%$

→ Solution:-
Given that,

The cash flow diagram of alternative A is shown below.



The annual equivalent revenue expression of above cash flow diagram is as follows:-

$$AC_A(25\%) = 60,000 + 15,000(A/P, 25\%, 5) - 1,50,000(A/P, 25\%, 5)$$

$$= 60,000 + 15,000(0.1218) - 1,50,000(0.3718)$$

$$= \text{Rs. } 6057.$$

Similarly,
For alternative B,

Initial investment, $P = \text{Rs. } 1,75,000$

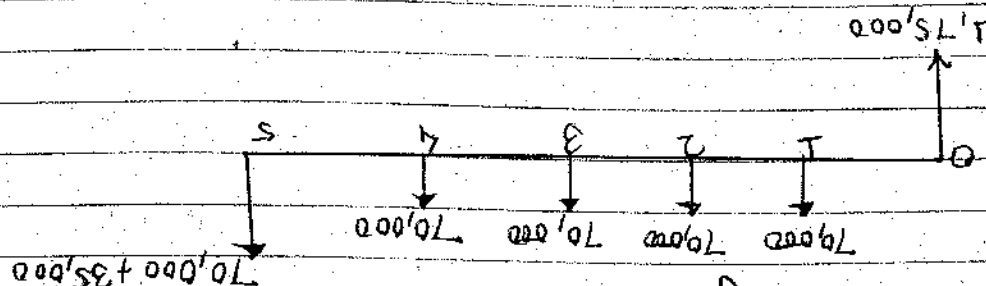
Annual equal return, $A = \text{Rs. } 70,000$

Salvage value at the end of machine life, $S = \text{Rs. } 35,000$.

Life $L = 5$ years.

Interest rate, $i = 25\%$, compounded annually.

The cash flow diagram for alternative B is shown below



The annual equivalent revenue expression of the above cash flow diagram is.

$$AC_B(25\%) = -1,75,000(A/P, 25\%, 5) + 70,000 + 35,000(A/P, 25\%, 5)$$

$$= -1,75,000(0.3718) + 70,000 + 35,000(0.1218)$$

$$= \text{Rs. } 9,198.$$

Since the annual equivalent net return of alternative B is more than that of alternative A. Thus, the company should select alternative B.

Rate-of-Return Method

1. Question 7.3, Pannierselvar (JRR)

A firm has identified three mutually exclusive investment proposals whose details are given below. The life of all the three alternatives is estimated to be five years with negligible salvage value. The minimum attractive rate of return for the firm is 12%.

Alternative	Initial investment, Rs.	Annual Revenue, Rs.	Life (years)
A1	5,00,000	1,00,000	10
A2	8,00,000	1,40,000	10
A3	3,00,000	70,000	10

Find the best alternative based on the rate of return method of comparison.

> Solution:-

Question 7.5, Panneerselvam [TRR]

P. R. S. SHAH

A company is planning to expand its business activity. It has two alternatives for the expansion program and the corresponding cash flow are tabulated below. Each alternative has a life of five years and a negligible salvage value. The minimum attractive rate of return for the company is 12%. Suggest the best alternative to the company.

Alternative	Initial investment, Rs.	Yearly revenue, Rs.
1	5,00,000	1,70,000
2	8,00,000	2,70,000

Find the best alternative based on the rate of return method of comparison.

>

Solution:

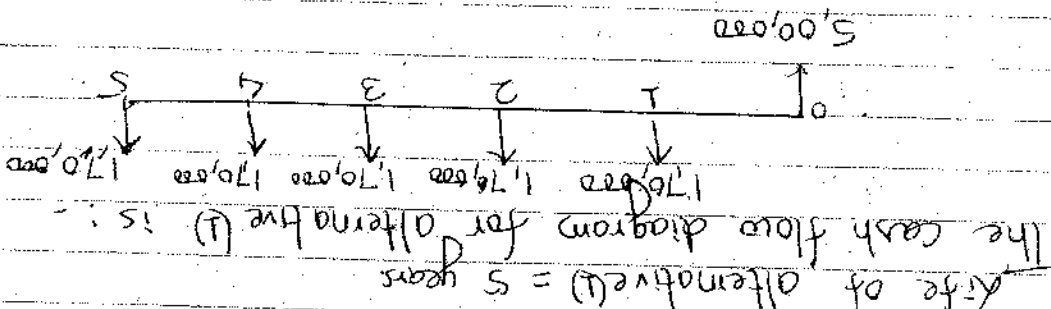
Given that:-

For alternative 1,

Initial outlay = Rs. 5,00,000.

Annual revenue = Rs. 1,70,000.

Life of alternative (1) = 5 years



The formulae for the net present worth of alternative (1) are:-

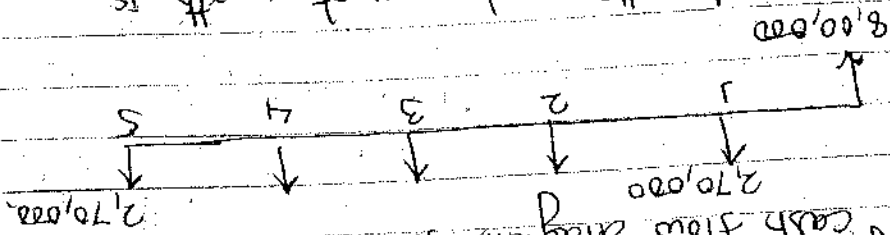
$$\begin{aligned}
 PW_1(i) &= -5,00,000 + 1,70,000(P/A, i, 5) \\
 &= -5,00,000 + 1,70,000(P/A, 15\%, 5) \\
 &= -5,00,000 + 1,70,000(3.3522) \\
 &= \text{Rs. } 69,874
 \end{aligned}$$

Also,

$$\begin{aligned}
 PW_2(22\%) &= -8,00,000 + 2,70,000(P/A, 22\%, 5) \\
 &= -8,00,000 + 2,70,000(2.8636) \\
 &= Rs. -26,828.
 \end{aligned}$$

$$\begin{aligned}
 \therefore PW_2(20\%) &= -8,00,000 + 2,70,000(P/A, 20\%, 5) \\
 &= -8,00,000 + 2,70,000(2.9906) \\
 &= Rs. 7,462.
 \end{aligned}$$

The formula for the net present worth is



Similarly, for alternative 2, the cash flow diagram for alternative 2 is

$$\begin{aligned}
 &= 20.78\% \\
 i &= 20\% + \frac{8,402 - 0}{8,402 - (-13,188)} \times 2\%
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{The rate of return of alternative (A) is} &= Rs. 13,188 \\
 &= -5,00,000 + 1,70,000(2.9636) \\
 &= Rs. 8402.
 \end{aligned}$$

$$\begin{aligned}
 PW_1(22\%) &= -5,00,000 + 1,70,000(P/A, 22\%, 5) \\
 &= -5,00,000 + 1,70,000(2.9906) \\
 &= Rs. 43,883.
 \end{aligned}$$

$$\begin{aligned}
 PW_1(17\%) &= -5,00,000 + 1,70,000(P/A, 17\%, 5) \\
 &= -5,00,000 + 1,70,000(3.1993) \\
 &= Rs. 43,883.
 \end{aligned}$$

$$\begin{aligned}
 \text{Annual equivalent benefit} &= \text{Benefit} - \text{cost ratio} = \text{Rs. } 9,00,000 = 1.137 \\
 &= \text{Rs. } 7,91,400 \\
 \text{Annual equivalent of initial cost} &= P(A/P, 10\%, 5) \\
 &= 30,00,000 \times 0.2638 \\
 &= \text{Rs. } 7,91,400
 \end{aligned}$$

Given that, for alternative (1)

Solution:

Two mutually exclusive projects are being considered for investment. Project A1 requires an initial outlay of Rs. 30,00,000 with net receipts estimated at Rs. 9,00,000 per year for the next five years. The initial outlay for the project A2 is Rs. 60,00,000 and net receipts have been estimated as Rs. 15,00,000 per year for the next seven years. There is no salvage value associated with the projects. Using benefit cost ratio, which project would you select? Assume an interest rate of 10%.

Benefit Cost Ratio Method.

Example 10.3

Since, the rate of return of alternative (1) is greater than that of the alternative 2, so alternative (1) is best option.

$$\begin{aligned}
 &= 20.435\% \\
 &= 20\% + 7,462 - 0 \times 2\% \\
 &= 7,462 - (-26,828)
 \end{aligned}$$

The rate of return of alternative (2) is

Similarly, for alternative (2),
Annual equivalent of initial cost

$$= P \cdot (A/P, 10\%, 7)$$

$$= ₹ 60,00,000 \times 0.2054$$

$$= ₹ 12,32,400$$

$$\therefore B.C. \text{ ratio} = \frac{₹ 15,00,000}{₹ 12,32,400} = 1.217$$

Since the benefit-cost ratio of alternative A2 is more than that of alternative A1, hence alternative A2 is to be selected.

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2. (b.) A now aged 40 years now had borrowed Rs. 50000 from a bank for the future step studies at the age of 20 yrs. Interest was charged at 11% per year compounded quarterly. He wished to pay back to bank annual equal installments with the first installment beginning 5 yrs after receiving loan. He just cleared that the loan now. what amount did he pay to end of following 50 yrs.

Given, $P = \text{Rs. } 50000$

$I = 11\%$ per year And Compounding quarterly.

$N = 20 \text{ yrs.}$

$A = ?$

Quarterly interest rate,

$$q = \frac{I}{4} \text{ per year}$$

$$= \frac{11\%}{4}$$

$$= 2.75\%$$

Self annual interest rate

$$! \text{ self} = (1 + q)^2 - 1$$

$$= (1 + 0.0275)^2 - 1$$

$$= 5.57\%$$

Using, single payment compounded amount factor

$$F = P (1 + i)^N$$

$$= 50000 (1 + 0.057)^{20}$$

Using, uniform periodic payments factor

$$F = A \frac{(1 + i)^N - 1}{i}$$

$\therefore A = \$61,217.32$ semi-annual payment

$$500000 * (1.05)^{40} = A \left[\frac{(1.05)^{40} - 1}{0.05} \right]$$

from eqn - (8) we get

$$\Rightarrow F = A \left[\frac{(1 + 0.05)^{30} - 1}{0.05} \right]$$

2015 Fall

1. (b) Following are the data for the production of a top radiator nacquet:-

Labour rate: Rs. 40/hr.

Leather: 50m at Rs. 250/metre.

Gut: 300m at Rs. 50/metre.

Graphe: 100kg at Rs. 200/kg.

Total Annual factory overhead: Rs. 500,000

Total Annual direct labor hours:- 2,50,000 hrs.

Labour hour needed: 250 hrs.

Break down the cost into components of prime cost and overhead cost and find out the manufacturing cost of each nacquet.

soln:-

Prime cost:-

Direct material cost:-

Leather = $50 \times 250 = \text{Rs. } 12,500$

Gut = $300 \times 50 = \text{Rs. } 15,000$

Graphe = $100 \times 200 = \text{Rs. } 20,000$

Direct labor cost = $250 \times 40 = \text{Rs. } 10,000$

Sub-total = Rs. 57,500

Overhead cost:-

Total Annual factory O/H = Rs. 50,00,000

Total Annual labor hrs = Rs. 2,50,000 hrs.

$\therefore$ factory overhead = $250 \times \frac{50,00,000}{2,50,000}$

= Rs. 4,000

Manufacturing cost of each package

$$= \$8,57,000$$

$$= \$8,53,000 + \$4,000$$

$$\text{Total cost} = \text{Prime cost} + \text{OH cost}$$

2015 Spring

2. (a) Mr. Sharma is planning for his retired life and has 10 more years of service. He would like to deposit 20% of his salary. Salary is Rs 10,000 at the end of the first year and thereafter be left to deposit the same amount (Rs 10,000) with an annual increase of Rs 2000 for the next 9 yrs with an interest rate of 15%. Find the total amount at the end of the 10th year of the above series.

Soln:-

From the question, it is clear, composite graded series of increasing type. In this series,

$$A_1 = \text{Rs. } 10,000$$

$$N = 10 \text{ yrs.}$$

$$r = 15\% \text{ per year}$$

$$P = \text{Rs. } 2000$$

$$F = ?$$

converting graded series into uniform series we get

$$A_2 = 2000 (A/P, 15\%, 10)$$

$$= \text{Rs. } 6766$$

So, the given graded series can be replaced by a uniform series having annual cash flow of A_2 , ~~calculated as~~

$$A = A_1 + A_2$$

$$= \text{Rs. } (10,000 + 6766)$$

$$= \text{Rs. } 16766$$

Now,

Future Amount, P is calculated as follows:-

$$P = 16766 (P/A, 15\%, 10)$$

$$= Rs. 390420$$

//

2013 Fall
2016 Fall
2019 Fall

2. (a) Santa wants to deposit \$10,000 in every year into a bank at an interest rate of 10% per year compounded annually & monthly.

What will be the maturity amount after 5 years?

50101

if interest is compounded semi-annually then $n=2$ and the effective interest rate is

$$I_{eff} = \left(1 + \frac{r}{n}\right)^n - 1$$

$$= \left(1 + \frac{0.10}{2}\right)^2 - 1$$

$$= 10.25\%$$

$$F = P \left(F/P, 10.25\%, 5 \right)$$

$$= 10000 (1.025)^2$$

$$= \$10,525.16$$

if compounded monthly

$$I_{eff} = \left(1 + \frac{0.10}{12}\right)^{12} - 1 = 10.47\%$$

50101

value of n

Annually = 1
Semi-Annually = 2
Monthly = 12
Quarterly = 4
Weekly = 52
Hourly = 365 * 24
Daily = 365
1 = 1

$$F = P \left(F/P, 10.47\%, 5 \right)$$

$$= 10000 * (1.1047)^5$$

$$= \$12,645.21$$

10

EST

6.11

$$19.721\% =$$

$$1 - 0.18 =$$

$$1 - 0.18 = R_{\text{continuously}}$$

$$19.721\% = 1 - \left(\frac{1 + 0.18}{365 \times 24} \right)^{365 \times 24} = R_{\text{Hourly}}$$

$$19.716\% = 1 - \left(\frac{1 + 0.18}{365} \right)^{365} = R_{\text{Daily}}$$

$$19.56\% = 1 - \left(\frac{1 + 0.18}{12} \right)^{12} = R_{\text{Monthly}}$$

2015

2. (a) Find effective interest rate when nominal rate of interest is 18% per year and compounded 8

of monthly R_{Monthly} R_{Daily} R_{Hourly} $R_{\text{continuously}}$

2015 Fall

Simple payback period

* Calculate the simple payback period for the given cash flow of the project:-

End of year (EoY)	Net cash flow (₹)
0	-25,000
1	8,000
2	8,000
3	8,000
4	18,000
5	13,000

Soln:-

End of year	Net cash flow (₹)	Cumulative cash flow (₹)
0	-25,000	-25,000
1	8,000	-17,000
2	8,000	-9,000
3	8,000	-1,000
4	18,000	+7,000
5	13,000	+20,000

Here

The cumulative cash flow turns to positive by 4 years. Therefore payback period lies betw

3 & 4.

By inter polation,

$$\frac{3}{-1,000} = \frac{4}{+7,000}$$

$$\therefore \text{payback period} = 3.125 \text{ years}$$

$$\text{Mode} \rightarrow 3(\text{stat}) \rightarrow 2(A+Bx) \rightarrow AC \rightarrow 55\%$$

$$\rightarrow 1(\text{stat}) \rightarrow 5(\text{reg}) \rightarrow 5(g) \rightarrow 3 \rightarrow 0.2125$$

discounted payback period.

* Evaluate the acceptability of the following proposal:
 If max allowable discounted payback is 6 yrs.

Assume MARR = 15%

Eoy. Amount (\$ thousands)

0	-15.00
1	200
2	400
3	450
4	450
5	600
6	900
7	1100

5010:-

$$P/(1+i)^n$$

End of year	Net cash flow	Present 15%	Cumulative P.V.
0	-1500	-1500	-1500
1	200	$200/(1+0.15)^1 = 173.91$	-1326.09
2	400	$400/(1+0.15)^2 = 302.46$	-1023.63
3	450	295.88	-727.74
4	450	257.29	-470.45
5	600	298.81	-172.15
6	900	389.09	216.94
7	1100	413.53	630.47

from above table,

Cumulative P.V. at 15% has -ve sign at the end of 5 years and +ve at the end of 6 years. Therefore, the discounted payback period lies between 5 & 6 years.

By interpolation,

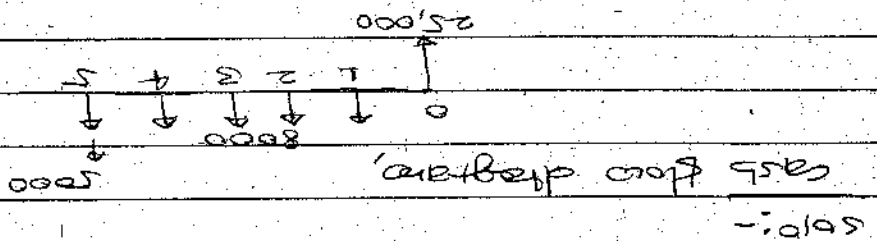
$$\frac{-172.15}{216.94} = \frac{5 - x}{6}$$

∴ discounted payback period = 5.44 < 6 years.

present worth method.

*) A piece of new equipment has been proposed by engineers to increase the productivity of certain manual welding operations. The forecasted cash flow is Rs. 25,000 and the equipment will have a market value of 5,000 at the end of a study period of 5 years. Increased productivity attributable to the equipment will amount to Rs. 8,000 per year after extra welding operations have been substituted from the revenue generated by the additional production. The proposal is found satisfactory.

use P.W. method.



$$\text{Net P.W.} = \text{P.W. of cash flow} - \text{P.W. of cash flow}$$

$$= 8000 \left(\frac{P/A, 20\%, 5}{P/A, 20\%, 5} \right) + 5000 \left(\frac{P/A, 20\%, 5}{P/A, 20\%, 5} \right) - 25000$$

$$= 8000 \left(\frac{(1+i)^n - 1}{i(1+i)^n} \right) + 5000 \left(\frac{1}{(1+i)^n} \right) - 25000$$

$$= 8000 \left(\frac{(1+0.2)^5 - 1}{0.2(1+0.2)^5} \right) + 5000 \left(\frac{1}{(1+0.2)^5} \right) - 25000$$

$$= 23924.89 + 2009.38 - 25000$$

$$= \text{Rs. } 934.27$$

$\therefore \text{P.W.} (20\%) > 0$, this equipment is economical. //

Net PW (20%) > 0, economically justified.

$$= \text{Rs. } 312.4$$

$$= 8000 - 25000 * 0.2 * (1 + 0.2)^5 + \frac{[T + 0.2]^5 - 1}{0.2} * 1500$$

$$\text{Net PW (20\%)} = 8000 - [25,000 (A/P, 20\%, 5) + 1500 (A/P, 20\%, 5)]$$

Annual worth method:-

Same result

$$= \text{Rs. } 934.29$$

$$= \frac{2324.8 * T}{(1 + i)^5}$$

$$\text{PW (20\%)} = 2324.8 (P/F, 20\%, 5)$$

economically justified.

As, Net Annual Worth (20%) > 0, the equipment is

$$= \text{Rs. } 2324.8$$

$$= -25000 * (1 + 0.2)^5 + 8000 * (1 + 0.2)^5 - 1 + 1500$$

$$= -25000 * (1 + i)^5 + 8000 * [(1 + i)^5 - 1] + 1500$$

$$\text{Net PW (20\%)} = -25000 (F/P, 20\%, 5) + 8000 (F/P, 20\%, 5) + 1500$$

Future worth method:-

IRR (Direct solo method)

Period	Project 1	Project 2
0	1500	-2000
1	0	1300
2	0	1500
3	0	-
4	1500	-

Solo:-

Project 1:-

$$1500 = 1500(P/F, i\%, 0)$$

$$1500 = 1500 * (P/F, i\%, 4)$$

$$1500 = 1500 * (1 + i)^{-4}$$

$$10(1 + i) = 1500 \left(\frac{1500}{1500} \right)$$

$$1 + i = 0.1013$$

$$\therefore i = 10.67\%$$

Project 2:-

$$PM(i) = -2000 + \frac{1300}{1 + i} + \frac{1500}{(1 + i)^2} = 0$$

$$\text{Let } (1 + i) = x \text{ then}$$

$$-2000 + 1300x + 1500x^2 = 0$$

Solving,

$$x = 0.8 \text{ or } -1.667$$

$$\therefore i = 25\% \text{ for } x = 0.8$$

$$\& i = -160\% \text{ for } x = -1.667 \text{ (Not possible)}$$

$$\therefore \text{The project, } i = 25\%$$

//

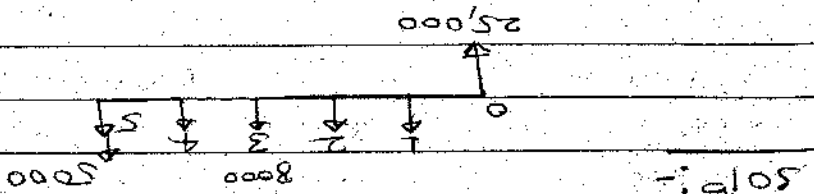
Profit & error method

Q. → Total investment = ₹. 25,000.
 salvage value = ₹. 5000 after 5 years.
 Life time = 5 years

Net Revenue = ₹. 8000 per year.

MARR = 20%.

Is the investment good? use IRR trial & error method



$$PW(1\%) = 0$$

$$5000 (P/F, 1\%, 5) + 8000 (P/A, 1\%, 5) - 25000 = 0$$

$$5000 \frac{(1+1\%)^5 - 1}{1\%} + 8000 * \frac{1 - (1+1\%)^{-5}}{1\%} - 25000 = 0$$

By trial & error

$$\text{At } i = 20\%, \quad PW = 934.3$$

$$\text{At } i = 25\%, \quad PW = -1847.1$$

By interpolation, $i = 21.67\%$ at $PW = 0$.

Putting $i = 21.67\%$ to eqn - (i) $PW = -58.2$

IRR lies between 20% & 21.67% .

Again,

using linear interpolation,

$$i = 21.58\%.$$

Putting, $i = 21.58\%$ to eqn - (i) then,

$$PW(21.58\%) = -1.9$$

IRR lies between 20% & 21.58% .

Again using linear interpolation,

$$i = 21.577\% \left\{ \frac{20 + 21.58}{2} \right\}$$

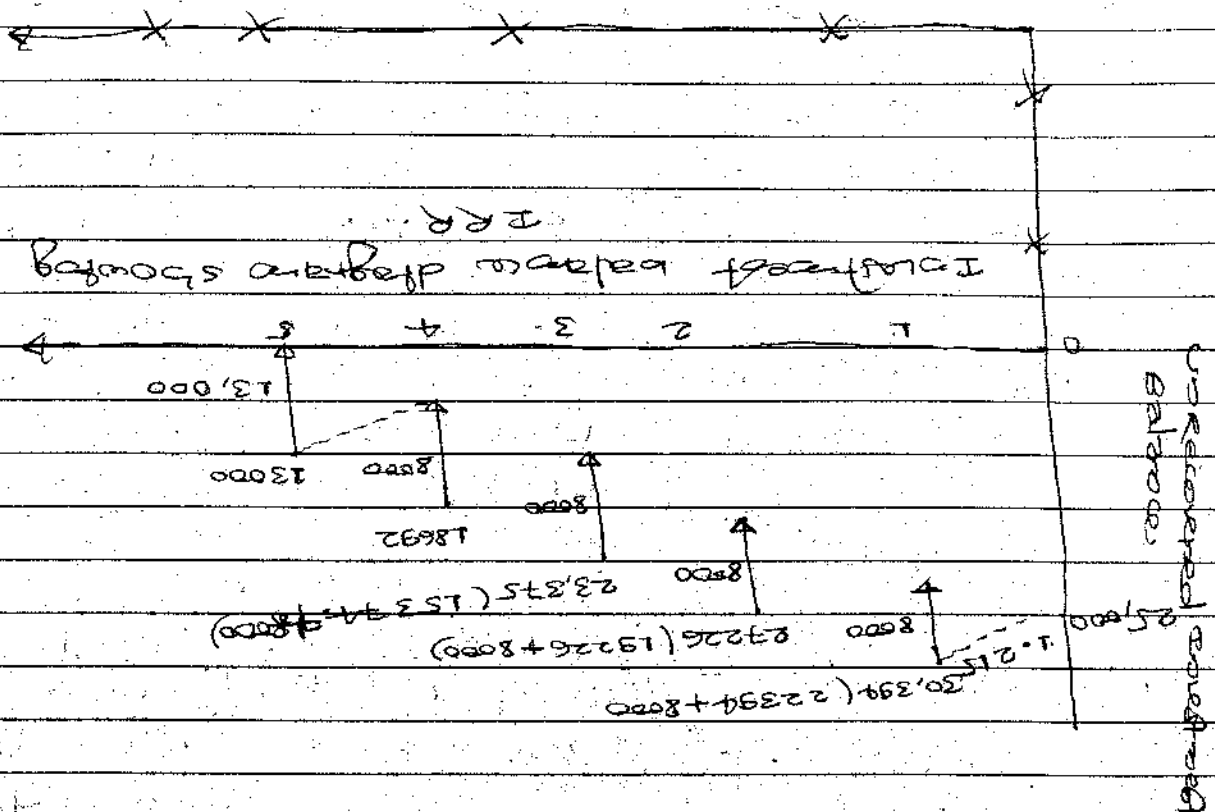
$$PW(21.577\%) = 0$$

Hence, $IRR > MARR (20\%)$, Investment is

acceptable.

US Recovered project balance calculation:

Year	US Repaid balance at beginning of year (Rs.)	Return of un-repaid balance (Rs.)	Recovery (Rs.)	US Repaid balance at end of year (Rs.)
0	-25,000	0	0	-25,000
1	-25,000	$-25,000 \times 0.21577 = -5394.25$	8000	$-25,000 - 5394.25 + 8000 = -22394.25$
2	-22,394.25	-4882	8000	-19226.25
3	-19,226.25	-4198.45	8000	-15374.7
4	-15,374.7	-3317.4	8000	-10692
5	-10692	-2807	13000	0



computer sold method

9/15/84 10:15 AM

$$P(N)(\%) = 0$$

$$8000(P/A)(\%) + 5000(P/E)(\%) - 25000 = 0$$

$$8000 * [(1 + 9)^5 - 1] / [9 * (1 + 9)^5] - 25000 = 0$$

write the eqn to calculator as

$$8 * [(1 + 9)^5 - 1] / [9 * (1 + 9)^5] + 5 / (1 + 9) - 25 = 0$$

press 59977 → calc.

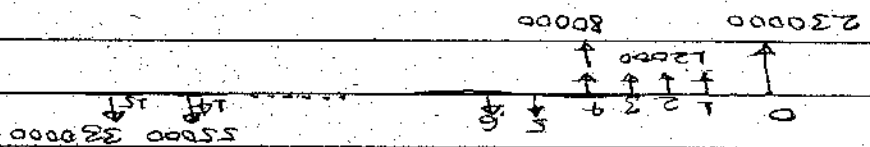
The value to the calculator is the value of

2 R.R. //

ERR Method

External / modified rate of return method

- Q. A consultancy opened now for Rs. 23,000. It will lose Rs. 12,000 each year the first four years. An additional Rs. 8,000 invested in the consultancy during the fourth year will result in a profit of Rs. 55,000 each year from the fifth year to the fifth year. At the end of 15 years the company can be sold for Rs. 33,000. Calculate the external rate of return if external investment rate is 12% per year. Is the investment good? $MARR = 15\%$.



Discount all cash out flow to time zero.

$$23000 + 12000 (P/A, 12\%, 4) + 8000 (P/P, 12\%, 4)$$

$$\Rightarrow 23000 + 12000 * [(1 + 0.12)^4 - 1] / [0.12 * (1 + 0.12)^4] + \frac{8000}{(1 + 0.12)^4}$$

$$\Rightarrow Rs. 31728.8$$

Compound all cash flows to N. i.e. 15 years at 12%

$$55000 (F/A, 12\%, 11) + 33000$$

$$= 55000 * (1 + 0.12)^{11} - 1 / 0.12 + 33000$$

$$= Rs. 1466003$$

Equivalent

$$317288 (F/P, i\%, 15) = 1466003$$

$$i\% = 10.74 \text{ i.e. } ERR = 10.74\% < MARR$$

$\therefore$ Investment is not good.

(Benefit cost ratio method)

B/C Ratio method

a) Find both types of B/C ratio using PM, FWD and

AM method.

Initial investment = Rs. 2000

Revenue/year = Rs. 1000

Expenses/year = Rs. 400

Salvage value = Rs. 400

Useful life = 5 years

MARR = 8%

Soln:-

A) using PM method:-

$$PM(B) = 1000 (P/A, 8\%, 5) = 1000 * 1.085^{-1} = 1000 * 0.6806 = 680.6$$

$$= 68.3992 \cdot 7 = 478.3744$$

$$PM(S) = 400 (P/F, 8\%, 5) = 400 * 1.085^{-5} = 400 * 0.6806 = 272.23$$

$$(1.085^{-1})$$

$$PM(0.8M) = 400 (P/A, 8\%, 5) = 400 * 1.085^{-5} = 400 * 0.6806 = 272.23$$

$$= 68.1756 \cdot 79 = 5384.03$$

Conventional B/C ratio =

$$PM(B)$$

$$PM(I) - PM(S) + PM(0.8M)$$

$$3992.7$$

$$= 2000 - 272.23 + 1756.79$$

$$= 1.146 > 1 \quad OK$$

Modified B/C ratio =

$$PM(B) - PM(0.8M)$$

$$3992.7 - 1756.79$$

$$= 2000 - 272.23$$

$$= 1.294 > 1 \quad OK$$

8) using Actual worth (AW) method.

Corrected E/C ratio = $AW(B)$

$CR + AW(0.8M)$

$$CR = I - 5.$$

$$CR = 2000(A/P, 8\%, 5) - 400(A/P, 8\%, 5)$$

$$= 2000 * 1.085 * 0.08 - 400 * 0.08$$

$$(1.085 - 1)$$

$$= 14.32.7.$$

1000

$$\text{Corrected E/C ratio} = (432.7 + 490) = 1.19671$$

OK //

Modified E/C ratio = $AW(B) - AW(0.4M)$

CR.

$$1000 - 490.$$

$$432.7$$

$$= 1.29471 \text{ OK} //$$

c) using FM method:-

$$FM(B) = 1000(A/P, 8\%, 5)$$

$$= 1000 * (1.085 - 1)$$

0.08

$$= 18.5866.6.$$

$$FM(5) = 18.400.$$

$$FM(0.8M) = 490(A/P, 8\%, 5)$$

$$= 490 * (1.085 - 1)$$

0.08

$$= 18.2581.3$$

(B) (3)

- ↳ Co-terminated method (2.5% & 5% cases)
- ↳ equalized method (PM, AM, & FM)
- ↳ equalized method (CIC method only)

For the first part of the test, the test is repeated for the first part of the test.

Repeatability Assumption: The test is repeated for the first part of the test.

$$= 1.294 > 1.0 \quad \text{OK}$$

$$= 2938.65 - 400$$

$$= 5866.66 - 2581.3$$

$$FM(I) - FM(S)$$

$$\text{Modified B/C ratio} = FM(I) - FM(S) - FM(0.8M)$$

$$= 1.146 > 1.0 \quad \text{OK}$$

$$= 2938.65 - 400 + 2581.3$$

$$= 5866.66$$

$$FM(I) - FM(S) + FM(0.8M)$$

$$\text{Corrected B/C ratio} = FM(I)$$

$$= 28.2938.65$$

$$FM(I) = 2000 (F/P, 8\%, 5) = 2000 * 1.085$$

2015 Fall

Co-terminated Assumption

(study useful life < study period) start at case

3. (b) A large corporation is considering the funding of following mutually exclusive water supply projects.

based on given information determine which alternative

is more desirable to implement by using co-terminated

assumption. (MARR = 10%)

Study period 8 yrs

Alternative	Investment (\$)	Annual Revenue (\$)	Annual Cost (\$)	Useful life	Salvage value (\$)
A	30000	38000	12900	9 yrs	0
B	100000	50000	27660	8 yrs	0

NOTE: - Taking study period as 8 yrs.

Always take study period equal or greater than

useful life of all alternatives, it is given in the question

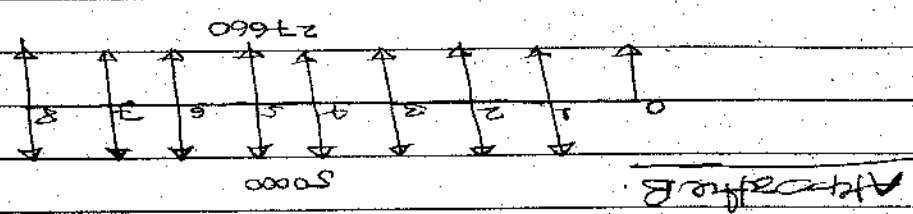


Fig: - cash flow diagram of project B.

So adjustment is required for alternative B because study period is equal to useful life.

Now,

$$\begin{aligned}
 FM(10\%)B &= 10000(F/P, 10\%, 8) + (50000 - 27660) \cdot \left[\frac{(F/P, 10\%, 8)}{(1+0.1)^8 - 1} \right] \\
 &= -10000(1+0.1)^8 + 22390 \cdot 0.1 \\
 &= 98.9118 \cdot 85
 \end{aligned}$$

5/ Select alternative B for two periods.

Here based on co-terminated assumptions and for project B higher (Rs. 4118.85)

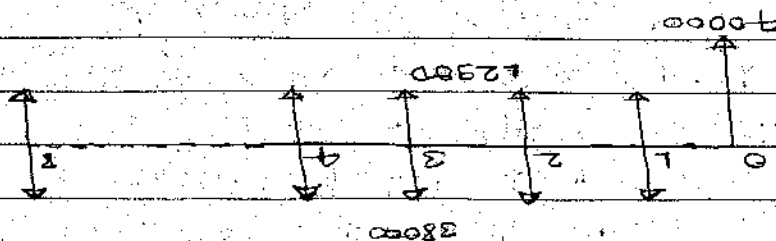
$$= \text{Rs. } 20500.47$$

$$= [-70000 (1 + 0.1)^4 + 25100 * \frac{(1.1)^4 - 1}{0.1}] * (1 + 0.1)^4$$

$$FV(10\%) = [-70000 (F/P, 10\%, 4) + (38000 - 12900) (F/P, 10\%, 4)] * (F/P, 10\%, 4)$$

The cash flow ~~after~~ accumulated at end of useful life is requested for extended periods.

For Alternative A,
Fig:- cash flow of project A



Study period t
(useful life) > (study period)

→ use co-terminated i.e. the proposed market value technique method to develop an estimated market value and determine the best alternative preferred by taking study period of 5 yrs for given infrastructure. (MARR = 10% per year)

Alternative	X	Y
Capital Investment	-35000	-50000
Annual expenses	1500	2500
Annual Revenue	13000	17500
Salvage value	3500	5000
Useful life	5 yrs	8 yrs

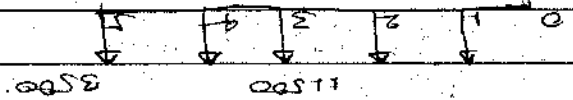
Alternative X :-

Net annual Revenue = Revenue - expenses

$$= 13000 - 1500$$

$$= 11500$$

Fig: - Cash flow diagram of X.



$$PW(10\%)X = -35000(P/F, 10\%, 5) + 11500(P/A, 10\%, 5) + 3500(P/F, 10\%, 5) = -35000(1+0.1)^5 + 11500 \left[\frac{1-1.1^{-5}}{0.1} \right] + 3500$$

$$= 88.17840 \cdot 8$$

For Alternative Y :-

$$\text{Net annual Revenue} = 17500 - 2500$$

$$= 15000$$

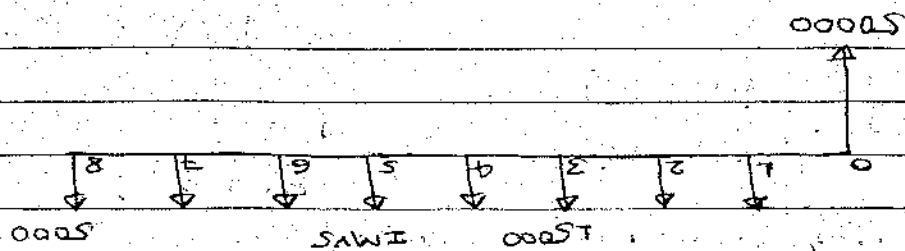
Now, Imputed Market value (IMV) = $22220 + 3756.5$
 $= \text{Rs. } 25976.5$

Also, PM at the end of 5th year of original market value at the end of useful life (8 years).
 i.e. at the end of 5th year of salvage value (Rs. 5000)
 $50, PM (10\%) MV = 5000 (P/F, 10\%, 3)$
 $= 5000 \times (1 + 0.1)^3$
 $= \text{Rs. } 3756.50$

Now, PM at the end of 5th year for remaining 3 years.
 $PM (10\%) CR = 8935 (P/F, 10\%, 3)$
 $= 8935 \times \left[\frac{1.1^3 - 1}{0.1} \right] \times \frac{1.1^3}{1.18}$
 $= \text{Rs. } 22220$

By using imputed market value (IMV) technique,
 Capital Recovery, CR (10%)
 $= 50000 (A/P, 10\%, 8) - 5000 (P/F, 10\%, 8)$
 $= 5000 \times (1.1)^8 \times 0.1 - 5000 \times \frac{1.1^8 - 1}{1.18 - 1}$
 $= 9372.2 - 437.2$
 $= \text{Rs. } 8935$
 $\therefore A = \frac{F \times 1}{(1 + i)^n - 1}$

Fig:- Cash flow diagram of Y.



$$FN(10\%) = -5000(F/P, 10\%, 5) + 1500(F/P, 10\%, 5) + 25976.5$$

$$= -5000 * (1 + 0.1)^5 + 1500 * \left[\frac{1.0}{1 + 0.1^5} \right] + 25976.5$$

$$= 18,370.275$$

Based on the calculated assumptions FN for Alternative Y is higher (18,370.275)

5b)

Alternative Y is best selection.

2017/5

Repeatability Assumption :- Based on PM method

Q.1 The following data have been estimated for two mutually exclusive investment alternatives A & B associated with a small engineering project for which revenues as well as expenses are involved. They have useful lives of four and six years respectively. If the MARR = 10% per year, show which alternative is more desirable by using equivalent worth method use Repeatability Assumption.

Alternative	Capital Investment	Annual Revenue	Annual Expenses	Useful Life (year)	Marketed value
A	Rs -500	1900	-645	4	0
B	Rs -500	2500	-1020	6	0

Soln :-

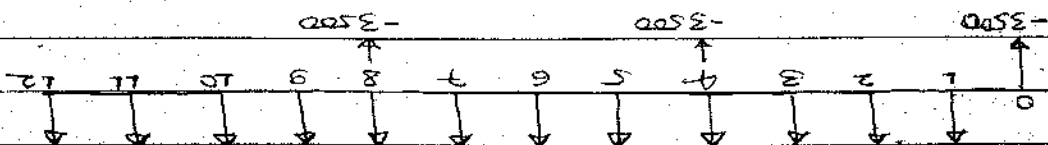
The LCM of useful lives of Alternatives A & B is 12 yrs. Using the Repeatability Assumption and a 12 yrs study period.

per Alternative A :-

$$\left\{ \frac{12}{4} = 3 \text{ i.e. } 4, 8, 12 \text{ yrs} \right\}$$

Net Revenue = Revenue - expenses

$$= 1900 - 645 = \text{Rs. } 1255$$



$$\text{Now, PM (10\%)} A = -3500 - 3500 \left[(P/A, 10\%, 4) + (P/A, 10\%, 8) \right] +$$

$$1255 (P/A, 10\%, 12)$$

$$= -3500 - 3500 \left[(1 + 0.1)^{-4} + (1 + 0.1)^{-8} \right] + 1255 * \frac{1 - 1.1^{-12}}{0.1}$$

$$= \text{Rs. } 1028$$

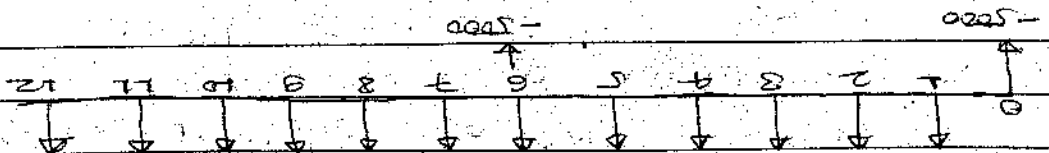
For Alternative B:-

Net Revenue = Revenue - expenses

$$= 2500 - 1020$$

$$= \text{Rs. } 1480$$

$$\left\{ \frac{12}{12} = 2 \text{ i.e. } 6/12 \right\}$$



Note,

$$PW(10\%) B = -5000 + 5000(P/F, 10\%, 6) + 1480(P/F, 10\%, 12)$$

$$= \text{Rs. } 2262$$

Based on PW method we should select

Alternative B because it has larger

value (2262)

NOTE:-

Similarly, we can apply PW & FV method

Capitalized Costs Method (C.C. Method)

a.) A selector is to be made between two structures because revenues do not exist, only negative cash flow accounts (cost) and the market value at the end of useful life are estimated as follows:-

Alternative	Structure M	Structure N
Capital Investment	-12,000	40,000
Market value	0	10,000
Annual expenses	-2000	-1000
useful life	10 yrs	25 yrs

By using the Repetitive Assumption and the method of Analysis. Determine which structure is better if the MARR is 15% per year.

Soln:-

The Annual Worth (AW) over the useful life of each alternative structure at MARR = 15%, is calculated as follows:-

$$AW(15\%)M = -12000(A/P, 15\%, 10) - 2000$$

$$= -12000 * (0.15)^{10} * \frac{0.15}{1 - 0.15^{10}} - 2000$$

$$= -12000 * 0.15 - 2000 = -1800 - 2000 = -3800$$

$AW(15\%)N = -40000(A/P, 15\%, 25) + 10000(A/P, 15\%, 25)$

$$= -40000 * (0.15)^{25} * \frac{0.15}{1 - 0.15^{25}} + 10000 * \frac{0.15}{1 - 0.15^{25}} - 1000$$

$$= -1000 - 7141 = -8141$$

Since, the C.C.s of Structure M & N are as follows:-

$CW(15\%)M = AW(15\%)M = -3800$

$CW(15\%)N = AW(15\%)N = -8141$

So, based on the C.C. of each structure design, Alternative M is selected because it has a lower negative value.

11th 2016 1st IRR Method & Incremental Analysis

e) Two mutually exclusive investment opportunities are there and their prospective cash estimates are given below:-

which project should you select by using IRR (Internal Rate of Return) method? use Repetition Assumption.

Terms	Capital Outflow	Annual Revenue	Annual Cost	Useful Life	Salvage value
A	350000	190000	69500	4	0
B	500000	250000	13830	8	0

MARR = 10% per year.

Sol:-

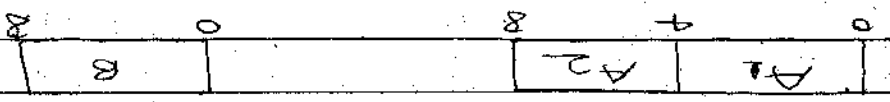
From Repetition Assumption,

LCM of 4 & 8 = 8 yrs.

∴ Study period = 8 yrs. And

both the projects will be analysed over 8 yrs.

Comparison:-



Determination of IRR

$$PWA(\%) = -350000 - 350000(P/F, \%, 4) + (190000 - 69500)(P/F, \%, 8) = 0$$

$$\therefore \% = 16.2\%$$

	C.F. of A		C.F. of B		C.F. of A - B	
	A1	A2	A1+A2		A → B	
0	-350000	-350000	-700000	-500000	-150000	
1	125500	125500	251000	111700	-13800	
2	125500	125500	251000	11	-13800	
3	125500	125500	251000	11	-13800	
4	125500	-350000	-224500	11	336200	
5	125500	125500	251000	11	-13800	
6	125500	125500	251000	11	-13800	
7	125500	125500	251000	11	-13800	
8	125500	125500	251000	11	-13800	

Investment Analysis:-

Here, $IRR_A = 16.2\% > MARR(10\%)$
 $IRR_B = 15.1\% > MARR(10\%)$
 $\therefore 15.1\% = 15.1\%$
 $PNB(8\%) = -500000 + (250000 - 138300)(P/A, 8\%, 8) =$

20175/20157

a.) perform sensitivity analysis using PM method

over a range of $\pm 40\%$ to (a) total investment

(b) Net Annual Revenue (c) salvage value and

(d) useful life.

$$\text{Total Investment} = \text{Rs. } 2,00,000$$

$$\text{Annual Revenue} = \text{Rs. } 50,000$$

$$\text{Annual expenses} = \text{Rs. } 5,000$$

$$\text{Salvage value} = \text{Rs. } 25,000$$

$$\text{Useful life} = 10 \text{ yrs.}$$

$$\text{MARR} = 12\% \text{ per year.}$$

Draw also the sensitivity graph.

Soln:-

Prime eqn:-

$$\text{PW}(12\%) = -2,00,000 + (50,000 - 5,000) (P/A, 12\%, 10)$$

$$+ 25,000 (P/F, 12\%, 10)$$

$$= \text{Rs. } 62,309$$

For $\pm 40\%$ fluctuations to,

(a) Total Investment

$$\text{PW}(12\%) = -2,00,000 \left(1 \pm \frac{40}{100}\right) + (50,000 - 5,000)$$

$$(P/A, 12\%, 10) + 25,000 (P/F, 12\%, 10)$$

calculations has been shown to the table

below.

(b) Net Annual Revenue

$$\text{PW}(12\%) = -2,00,000 (50,000 - 5,000) \left(1 \pm \frac{40}{100}\right)$$

$$(P/A, 12\%, 10) + 25,000 (P/F, 12\%, 10)$$

calculations has been shown to table below.

25000 (LF 90/1.00) (P/F 12%, 10)

Calligraph Sat Sep 10

ଶିକ୍ଷା ମନ୍ତ୍ରାଳୟ (୧)

$$PM(12\%) = -200000 + (50000 - 5000) \left[\frac{P/A, 12\%, 10}{1 + 4\%} \right] + 25000 \left[\frac{P/F, 12\%, 10}{1 + 4\%} \right]$$

- Following are the locations of the major ports of India

Calculation table :-

PM (12%)				
-40%	-20%	0	20%	40%
Investment	102309	62309	22309	-17691
Net Revenue	-39394.6	11457.2	62309	113160.8
Salvage value	59089	60699	62309	63919
Useful life	-2205	38639.5	62309	85165.5
				102384

a) following information has been obtained regarding two projects.

Standard motor	new model motor
10050	12050
130000	156000
20	20
0	0
89.5%	93%
8000	2500
Electricity cost	Rs. 6/kWh

Annual tax & insurance 2% of book value.
 i) At what operating hours are they equivalent?
 ii) If the motor has to be operated 38 hrs a year which one should be selected?
 Take MARR = 10% per year.

Soln:-
 we know that

$$n = \frac{\text{output}}{\text{input}}$$

$$n = \frac{\text{input}}{\text{output}}$$

For standard motor

$$\text{Capital recovery cost} = 130000 (P/P, 10\%, 20) = Rs. 15275$$

$$\text{Input power} = 100 \times 0.746 = 83.35 \text{ kW}$$

$$0.895$$

Assume, both the motors are operated X hrs a year.

$$\begin{aligned} \text{Operating cost for X hours of operation a year} &= 83.35 * X * 6 \\ &= 500.11X \\ \text{Maintenance cost} &= \text{Rs. } 8000 \\ \text{Tax \& Insurance} &= 2\% \text{ of } 130000 = \text{Rs. } 2600 \\ \text{Total Annual cost for} &= 15275 + 500.11X + 8000 + 2600 \end{aligned}$$

$$\Rightarrow 25875 + 500.11X \quad \text{--- (1)}$$

For New Model Motor:-

$$\text{Capital Recovery cost} = 156000 (A/P, 10\%, 20) = \text{Rs. } 18830$$

$$\text{Input power} = 120 * 0.746 = 96.26 \text{ kW}$$

$$\text{Operating cost for X hours of operation a year} = 96.26 * X * 6$$

$$= 577.55X$$

$$\text{Maintenance cost} = \text{Rs. } 2500$$

$$\text{Tax and Insurance} = 2\% \text{ of } 156000$$

$$= \text{Rs. } 3120$$

Total Annual cost for

$$18330 + 577.55X + 2500 + 3120$$

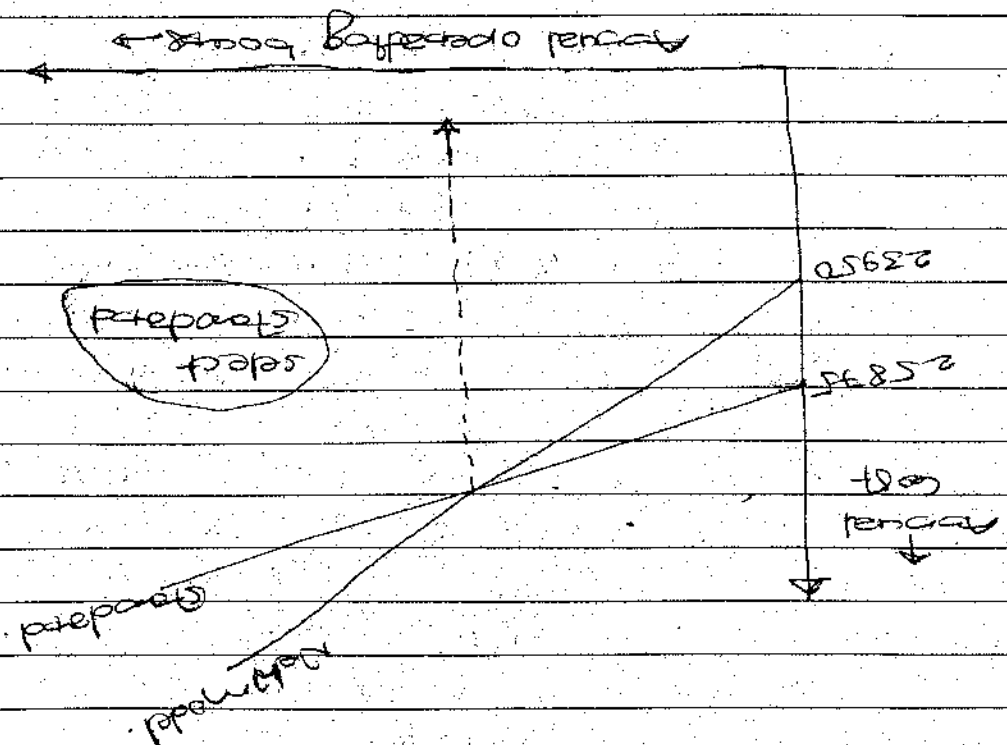
$$\Rightarrow \text{Rs. } 23950 + 577.55X \quad \text{--- (2)}$$

Two motors are said to be equivalent if only if eqn (1) equals to eqn (2)

$$25875 + 500.11X = 23950 + 577.55X$$

$$\therefore X = 24.857 \text{ hrs}$$

If no one is operating at the end of the year
 standard motor would be selected as shown
 by the above.



selection of motor -

∴ greater profit is 24,857 less.

7

$$= \text{Rs. } 570$$

$\therefore$ Manufacturing cost of each machine = $\text{Rs. } 57000 / 100$

$\therefore$ Total cost = prime cost + O/H cost = $53,000 + 4,000 = \text{Rs. } 57,000$

$$= \text{Rs. } 4000$$

$\therefore$ Factory overhead = $200 \times \frac{50,00,000}{2,50,000}$

Total Actual Labor cost = $2,50,000 \text{ hrs}$

Total Actual factory O/H = $\text{Rs. } 50,00,000$

overhead costs

$$\text{Subtotal} = \text{Rs. } 53,000$$

Direct Labor cost = $200 \times 40 = \text{Rs. } 8,000$

Gravel = $100 \times 200 = \text{Rs. } 20,000$

Gut = $300 \times 50 = \text{Rs. } 15,000$

Leather = $50 \times 200 = \text{Rs. } 10,000$

Direct material costs

prime costs:-

50:-

cost of each machine.

and overhead costs and find out the manufacturing

breakdown the costs into components of prime costs

Total Actual direct Labor hours = $2,50,000 \text{ hrs}$

Total Actual factory overhead = $\text{Rs. } 50,00,000$

Labor hours needed = 200 hrs

Gravel = $100 \text{ kg @ Rs. } 200/\text{kg}$

Gut = $300 \text{ m @ Rs. } 50/\text{m}$

Leather = $50 \text{ m @ Rs. } 200/\text{m}$

Labor rate = $\text{Rs. } 40/\text{hr}$

100 machines required.

The following are the cost data for the production of

2015F

On the following information is extracted from the books of a company for a certain year. prepare a cost sheet showing clearly the cost per unit under direct and indirect cost and also the profit loss units.

(IN RS.)

Direct material	24,00,000
Direct labor	5,00,000
Depreciation of factory building	15,000
Grass office expenses	40,000
Depreciation of office building	8,000
Depreciation of staff cars	12,000
Insurance	-
Staff cars	1,500
Office building	1,200
Factory building	1,500
Delivery van maintenance & running expenses	10,000
Salaries (including)	3,00,000
Sales manager	25,000
Factory defect equipment	25,000
Finished goods warehouse expenses	20,000
Electricity (including 4000 hrs. additional office)	40,000
Advertisement	20,000
Guards Ripon Ripon expenses	3,90,000
Sales promotion	5,000
Office administration expenses	50,000
Expenses for participating in industrial exhibition	10,000
Sales	42,00,000

5010 :-

1-2) prime cost

Direct material = Rs. 2400000

Direct Labor = Rs. 500000

Sub-total = Rs. 2900000

2) Overhead cost :-

2-1) production overhead :-

depreciation of factory building = Rs. 15000

factory building insurance = Rs. 1500

Salary (factory staff) = Rs. 25000

Electricity = Rs. 36000

factory repair expenses = Rs. 340000

Sub-total = Rs. 417500

2-2) Administrative overhead :-

depreciation of office building = Rs. 8000

depreciation of staff cars = Rs. 12000

Insurance (staff cars) = Rs. 1500

Insurance (office building) = Rs. 1200

Electricity = Rs. 4000

office Administrative expenses = Rs. 50000

Salaries = Rs. 250000

Sub-total = Rs. 326700

2.3) Selling and Distribution Overhead :-

Branch office expenses = Rs. 40000

Selling and marketing & R.D. = Rs. 10000

Salary (Sales Manager) = Rs. 25000

Advertisement = Rs. 20000

Sales promotion = Rs. 5000

Exhibition = Rs. 10000

Physical goods warehouse expense = Rs. 20000

Sub total = Rs. 130000

Total overhead cost = Rs. 417500 + 326700 + 130000

= Rs. 874200

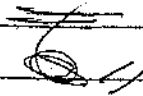
Total cost = prime cost + overhead cost

= Rs. 290000 + Rs. 874200

= Rs. 374200

Revenue from = Rs. 4200000

9



$$\therefore N = 12.74 \text{ yrs}$$

$$N \log 1.03 = \log 3$$

Taking log on both sides

$$1.09N = 3$$

$$\frac{P}{3P} = (1 + 0.09)^N$$

$$3P = P (P/P, 9\%, N)$$

$$P = P (P/P, 9\%, N)$$

we know that

$$\text{Future sum, } P = 3P$$

$$\text{Then, } P (P/P, 9\%, N) = P$$

Let the investment, P will become 3P after N yrs.

sol:-

If the interest rate is 9%, compounded annually?

2. How long does it take for an investment to triple itself?

$$= \text{Rs. } 25773$$

$$= 15000 (F/P, 7\%, 8)$$

$$F = P (F/P, 7\%, N)$$

we know that

$$F = ?$$

$$N = 8 \text{ yrs}$$

9 = 7% per year, compounded annually

$$P = \text{Rs. } 15000$$

sol:-

8 years?

1. Suppose you needed Rs. 15000 at 7% interest rate compounded annually. How much would you have after

3) For an effective rate of 8% compounded annually, find how much can be loaned now if Rs. 25000 will be repaid at the end of 5 years?

soln:-

$$P = ?$$

$$F = \text{Rs. } 25000$$

$$i = 8\% \text{ per year}$$

$$N = 5 \text{ yrs}$$

we know,

$$P = 25000 (P/F, 8\%, 5)$$

$$= \text{Rs. } 17015$$

Ans

4) A man wants to have Rs. 20000 for his studies at the end of 10 yrs. How much money does he have to deposit each year for a saving account that earns 8% every year?

soln:-

$$F = \text{Rs. } 20000$$

~~we know,~~

$$i = 8\% \text{ per year}$$

$$N = 10 \text{ yrs}$$

$$A = ?$$

we have

$$F = 20000 (A/F, 8\%, 10)$$

$$= \text{Rs. } 13800$$

Ans

Ex 4

$$\therefore N = 10.61 \text{ yrs}$$

1.07N = 2.05 (taking log on both sides)

$$30000 = 2000 * \left[\frac{(1+0.07)^N - 1}{0.07} \right]$$

$$P = A * \left[\frac{(1+i)^N - 1}{i} \right]$$

we know that

$$N = ?$$

$$P = 100,000$$

$$i = 7\% \text{ per year}$$

$$A = 10,000$$

Soln:-

monthly saving of Rs 10,000 can be purchased?

Monthly saving of Rs 10,000 can be purchased?

out. If Rs 10,000 is deposited annually at 7% for 10.61 yrs,

into a sinking fund to replace the machine when it wears

6) part of the income that a machine generates is put

Ex 5

$$P = 2000 (P/A, 10\%, 8)$$

we have

$$A = 10,000$$

$$N = 8 \text{ yrs}$$

$$i = 10\% \text{ per year}$$

$$P = ?$$

Soln:-

Monthly saving of Rs 10,000 can be purchased?

Monthly saving of Rs 10,000 can be purchased?

5) How much money should you deposit now to a sinking

7) A person is planning for his retired life and has 10 more years of service. He would like to

deposit 20% of his salary, which is Rs. 10000 at the end of the first year and thereafter he wishes to

deposit the same amount (Rs. 10000) with an annual

increase of Rs. 2000 for the next 9 yrs with an interest

rate of 15%. Find the total amount at the end of the 10th year of the above series.

Soln:-

From the question,

it is a linear, composite gradient series of increasing type. In this series,

$$A_1 = \text{Rs. } 10000$$

$$n = 10 \text{ yrs.}$$

$$i = 15\% \text{ per year.}$$

$$g = \text{Rs. } 2000$$

$$P = ?$$

Converting gradient series into uniform series we get,

$$A_2 = 2000 (A/G, 15\%, 10)$$

$$= \text{Rs. } 6766$$

So, the given gradient series can be replaced by the uniform series having annual cash flow of A_1, A_2

$$A = A_1 + A_2 = \text{Rs. } (10000 + 6766)$$

$$= \text{Rs. } 16766$$

$$P = 16766 (P/A, 15\%, 10)$$

$$= \text{Rs. } 340920$$

9

Gorod.

8) The Annual deposits to the amount of (Rs 1200, Rs 1000, Rs 800, Rs 600, Rs 400) are made into a fund that pays interest at a rate of 9%, compounded annually. Determine the amount to the fund immediately after 5th deposit.

Soln:-

It is a decreasing gradient series, no base,

$$A_1 = \text{Rs. } 1200$$

$$G = \text{Rs. } 200$$

$$N = 5 \text{ year}$$

$$i = 9\% \text{ per year}$$

$$F = ?$$

These series can be replaced by a uniform series having equal payments A as follows.

$$A = A_1 - 200 (A/9, 9\%, 5)$$

$$= 1200 - 200 (A/9, 9\%, 5)$$

$$= \text{Rs. } 834.36$$

Now,

$$F = 834.36 (F/A, 9\%, 5)$$

$$= \text{Rs. } 9993.40$$

//

9) what is the equal payment series for 10 years that is equivalent to a payment series of Rs. 12000 at the end of the first year, decreasing by Rs. 1000 each year over 10 years. Interest rate is 8% compounded annually.

Soln:-

The given series is a decreasing gradient series.

$$A_1 = \text{Rs. } 12000$$

$$G = \text{Rs. } 1000$$

$$N = 10 \text{ yrs.}$$

$$i = 8\% \text{ per year.}$$

The given series can be transformed into a uniform series having an equal payment as follows:-

$$A = A_1 - 1000 (A/G, 8\%, 10)$$

$$= 12000 - 1000 (A/G, 8\%, 10)$$

$$= \text{Rs. } 8128.7$$

8

$$= Rs. 133.10$$

$$= 100 * (1 + 0.1)^5$$

$$F = P * (1 + i)^N$$

OR, we may compute it directly using the formula,

$$\text{Total interest earned} = Rs. 133.10 - 100 = Rs. 33.10$$

Here,

Start of year	Amount left	Interest earned	Amount owed at year end
1	100	$100 * 0.1$	110.00
2	110	$110 * 0.1$	121.00
3	121	$121 * 0.1$	133.10

Compounded interest method:-

$$= Rs. 130$$

$$= 100 (1 + 0.1 * 3)$$

$$F = P (1 + i)^N$$

OR, we may compute it directly using the formula,

$$\text{Future accumulated value} = 100 + 30 = Rs. 130$$

$$= Rs. 30$$

$$= 100 * 3 * 0.1$$

$$\text{Total interest earned, } I = P * N * i$$

Simple interest method

$$F = ?$$

$$i = 10\% \text{ per year}$$

$$N = 3 \text{ yrs}$$

$$P = Rs. 100$$

Sol:-

end of the three years?

1) You lend Rs. 100 for 3 yrs at 10% interest compounded annually. How much would you earn at the end of the three years?

2) A bank charges 1% per month on car loans. What is the APR? What is EAR?

APR? What is EAR?

Soln:-

Annual percentage rate, $r = 1\%$, $12 = 12\%$.
Effective Annual interest rate, $f = \left(1 + \frac{r}{n}\right)^n - 1$

$n =$ Nominal interest rate = 12% per year.

$m =$ No. of compounding periods per year = 12

$\therefore$ Compounded is monthly and there are 12 months in a year.

$$f = \left(1 + \frac{0.12}{12}\right)^{12} - 1$$

$$\therefore f = 12.6825\%$$

3) What will be the maturity amount of Rs. 15000 after 5 years for Nominal rate of 9% per year when compounded 4 years & quarterly.

Soln:-

Nominal rate (r) = 9%

Compounded 4 years, $m = 1$

$$\text{Effective rate, } f = \left(1 + \frac{0.09}{1}\right)^1 - 1 = 9\%$$

Maturity Amount, $F = 15000$ (r/p , 9% , 5)

$$= Rs. 23079.36$$

Compounded 4 quarterly, $m = 4$

$$\text{Effective rate } f = \left(1 + \frac{0.09}{4}\right)^4 - 1$$

$$= 9.308\%$$

$$F = 15000 \left(1 + \frac{0.09308}{1}\right)^5 = Rs. 23907.28$$

$$= 3.22 \text{ years}$$

$$\text{payback period} = \frac{3 + 10000}{95000}$$

End of year	Net cash flows	Cumulative cash flows
0	-85000	-85000
1	15000	-70000
2	25000	-45000
3	35000	-10000
4	45000	35000
5	45000	80000
6	35000	115000

soln:-

End of year	Cash flows
0	-105000 + 20000
1	15000
2	25000
3	35000
4	45000
5	45000
6	35000

cost are as follows:-

1. A company has just bought a new machine at a cost of Rs. 1,05,000 to replace one that had a salvage value of Rs. 20,000. The expected annual after-tax savings are Rs. 25,000. The payback period will exceed the breakeven point.

2) Evaluate the acceptability of the following proposal:
 If managers allocate simple payback period of 5 years and MARR allowable discounted payback period of 6 years. Assume, $MARR = 15\%$ per year.

Cash flow estimated.
 Year. Amount (in thousands)

0	-1500
1	200
2	400
3	450
4	450
5	600
6	800
7	1100

Soln:-

Computation of simple payback period.
 End of year Net cash flow cumulative cash flow

0	-1500	-1500
1	200	-1300
2	400	-900
3	450	-450
4	450	0
5	600	600
6	800	1500
7	1100	2600

Since cumulative cash flow becomes zero at the end of 4 years.

Simple payback period = 4 years.

computation of discounted payback period

Year	Net Cash flow	PI at 15%	Cumulative PI
0	-1500	-1500.00	-1500
1	200	1731.91	-1326.09
2	400	302.46	-1023.63
3	450	295.88	-727.747
4	450	257.29	-470.458
5	600	298.31	-172.152
6	900	389.09	216.9427
7	1100	413.53	630.4734

From the above table,

Cumulative PI is negative at the end of 5 year and +ve at the end of 6 year. Therefore, the discounted payback period lies between 5 & 6 years and is calculated as below:-

$$PI = 5 + \frac{389.09}{172.152} = 5.4 \text{ years}$$

∴ Decision:-

$PI = 5.4 \text{ years} < 5 \text{ yrs}$, the proposal is acceptable And
 $PI = 5.4 \text{ years} < 6 \text{ years}$, the proposal is acceptable

3. Check the feasibility of the following investment by using present worth, future worth and annual worth method. What is the capital recovery amount of this project?

$$\text{First investment} = \text{Rs. } 1,00,000$$

$$\text{Salvage value} = \text{Rs. } 25,000$$

$$\text{Projected period} = 10 \text{ yrs.}$$

$$\text{Annual Revenue} = 20,000$$

$$\text{Annual cost} = 5,000$$

$$MARR = 10\% \text{ per year.}$$

soln:-

present worth method:-

$$PW(10\%) = -1,00,000 + (20,000 - 5,000) (P/A, 10\%, 10) + 25,000$$

$$(P/P, 10\%, 10) = \text{Rs. } 18.07$$

Since $PW(10\%) > 0$, the project is Acceptable.

future worth method:-

$$FW(10\%) = -1,00,000 (F/P, 10\%, 10) + (20,000 - 5,000) (F/A, 10\%, 10)$$

$$+ 25,000 = \text{Rs. } 4691$$

Since $FW(10\%) > 0$, the project is Acceptable.

Annual worth method:-

$$AW(10\%) = -1,00,000 (A/P, 10\%, 10) + (20,000 - 5,000) + 25,000$$

$$(A/P, 10\%, 10) = \text{Rs. } 298$$

Since $AW(10\%) > 0$, the project is Acceptable.

And, capital recovery amount,

$$CR(10\%) = 1,00,000 (A/P, 10\%, 10) - 25,000 (A/P, 10\%, 10) = \text{Rs. } 14,719$$

4) Evaluate the machine X12 on the basis of the PM and FM method. When the MARR is 12% per year, per fixed cost data are as follows:-

Machine X12:-

Investment cost = Rs. 150000

Salvage value = Rs. 35000

Useful life = 15 years

Annual expenses = Rs. 12000

Overhaul cost (Rs.) - end of 5th year = 2900

Overhaul cost (Rs.) - end of 10th year = 60000

50/10:-

Present worth method

$$PW(12\%) = -150000 - 12000(P/A, 12\%, 15) - 29000(P/P, 12\%, 5) - 60000(P/P, 12\%, 10) + 35000(P/F, 12\%, 15)$$

$$= Rs. -258274$$

Since $PW(12\%) < 0$, the project is not acceptable.

Future worth method:-

$$FW(12\%) = -150000(F/P, 12\%, 15) - 12000(F/P, 12\%, 15) - 29000(F/P, 12\%, 5) + 35000$$

$$= Rs.$$

We may also compute the value by direct comparison as we have already calculated PW of the project. The method is explained as below,

$$FW(12\%) = PW(12\%)(F/P, 12\%, 15)$$

$$= -258274(F/P, 12\%, 15)$$

$$= Rs. -1413689$$

Since $PW(12\%) < 0$, the project is not acceptable.

5) A mineral water company is considering constructing a new plant to manufacture its products. The land at which the plant can be sold for £900,000. The building for £300,000, the equipment for £250,000, and £100,000 additional working capital is required. The mineral water is expected to be sold at £750,000 per year for 10 years. At which time the land can be sold for £900,000. The building for £300,000, the equipment for £250,000, and all of the working capital would be recovered. The annual expenses are estimated to be £750,000 and the company requires a mark of 25%. Determine by P10 method whether the company should invest in the new product.

Sol:-

(Amount in £000)

$$\text{Total Profit} = £(300 + 600 + 250 + 100) = £1,250$$

$$\text{Net Annual Revenue} = £(750 - 750) = £0$$

$$\text{Salvage value} = £(900 + 300 + 250) = £1,450$$

$$= £8,800$$

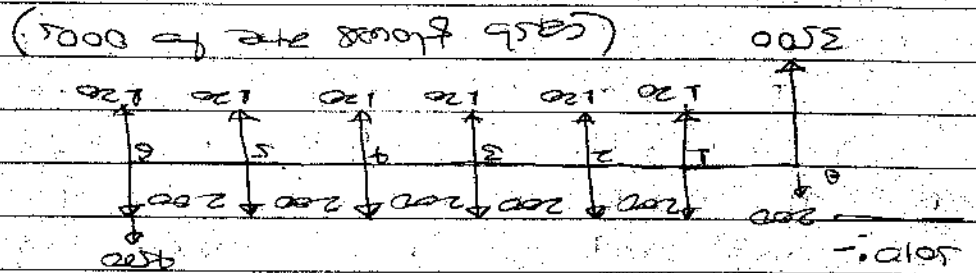
Now,

$$PV(25\%) = -1250 + 2750(P/A, 25\%, 10) + 8000(P/P, 25\%, 10)$$

$$= £8,182.2$$

Since $PV(25\%) < 0$, the company should not invest in it and the project be rejected.

6) Suppose that you purchased a building the years ago for Rs. 350000. Its actual maintenance expenses have been Rs. 12000 per year. At the end of the years (now), you sell the building for Rs. 450000. During the period of ownership, you needed the building for Rs. 20000 per year at the beginning of each year. Use the ANN method to evaluate the investment when your MARR is 12% per year.



$$\begin{aligned}
 ANN(12\%) &= -3500(A/P, 12\%, 5) + (200 - 12) + 4500(A/P, 12\%, 5) \\
 &= 18.183
 \end{aligned}$$

Since $ANN(12\%) < 0$, the project is acceptable.

3) A company is considering the acquisition of a new machine. The estimated cash flows are as follows.

End of year Net Cash flow (£)

0	-100000
1	36000
2	43500
3	48400

Suppose that the machine will be operated a carrying charge of £1500 per year.

to the second year, and zero in the third year.

Compute the Annual equivalent saving of cash per

machine bought if the firm's MARR is 15%.

Soln:-

Let us first compute the Annual worth of

the cash flows. Since the cash flows pattern is uneven

each year, we first compute the PW and convert

it into the AW.

$$PW(15\%) = -100000 + 36000(P/F, 15\%, 1) + 43500(P/F,$$

$$15\%, 2) + 48400(P/F, 15\%, 3)$$

$$= -88157.44$$

$$AW(15\%) = -88157.44(A/P, 15\%, 3) = -£ 6896$$

Let X , £8/yr be the equivalent Annual saving per

machine bought which is to be determined. Equivalently

Annual saving of X can be computed as

$$1500X(P/F, 15\%, 1) + 2500X(P/F, 15\%, 2) + 2000X(P/F,$$

$$15\%, 3) (A/P, 15\%, 3) = 1975.22X$$

From eqn (1) & (2)

$$1975.22X = 6896$$

$$\therefore X = £ 3.49/\text{yr}$$

8. > A Housing company is considering building Apartments at Satdobato, Lalitpur. The company has needed to the following estimates for the project and determined the MARR as 15% per year. Using A/R method, determine the minimum monthly rate to be charged per unit apartment.

No. of apartments = 50

occupancy capacity = 80%

Land cost = Rs. 30,00,000

constructed cost = Rs. 9,50,00,000

Study period = 20 yrs

Annual taxes and insurance = 15% of total investment

Salvage value = 1.2 times the land cost.

SOP:-

(Amounts are in 000000)

The best approach to solve such problems is to equate equivalent A/R of costs to the equivalent A/R of Revenues to be collected.

A/R of costs:-

Capital Recovery costs.

$$\Rightarrow (300 + 950) (A/P, 15\%, 20) - 300 \times 1.2 (A/P, 15\%, 20) \\ \approx \text{Rs. } 196.22$$

Annual taxes, = 15% of $(300 + 950)$

$$= \text{Rs. } 187.5$$

Therefore

$$\text{equivalent A/R cost} = \text{Rs. } 383.722$$

Am of Revenue,

Let x be the monthly net to be charged.

$$\text{Am of Revenue} = x(1.2)(.50)(.80\%) = .480x$$

equating these two terms,

$$.480x = 383.722$$

$$x = 0.799$$

Say, $x = 800$ per month (approximate)

~~28~~

• and demand

dest to pag

5010

$$\%S = 16 \text{ part}$$

$$PM(5\%) = 2.3105 \text{ months}$$

$$\text{Entropy} = 0 - 2 \cdot 3105 = -2 \cdot 3105$$

profit = 15%

$$20.11 \times 10^6 \times 6.5 \times 10^{-7} = (15\%) \text{ PM}$$

$$6565 \cdot T = (6565 \cdot T - 1) - 0 = 6564$$

Practical & Theoretical

Weg zu 'Gefühlswelt' durch 'Lage'

$$\frac{Tx - 2x}{Tx - x} = \frac{4R - 2R}{4R - R}$$

conjugate $\bar{z} = R$ and $x = 0$. $\phi \cdot e \cdot e \cdot e = 0$.

$$R_1 = 10\% \quad R_2 = -2.3105$$

$$x_2 = 1.5\% \quad x_2 = 1.5959$$

Putting these values we get,

$$y - 5 = 0 + 2.3105$$

$$15 - 5 = 1.5959 + 2.3105$$

$$\therefore y = 10.91\%$$

$$y.e. IRR, y = 10.91\%$$

NOTE:-

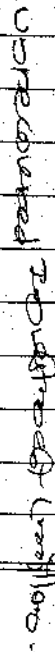
If we solve it by using Excel, we get more accurate value of IRR. & it comes to be 10.18%.

But for our manual calculation 10.91% is acceptable.

Unrecovered Investment calculation

Year	Cash flow	Unrecovered Investment	Start of year	End of year	covered
0	-10	-10.00	-10.00	-10.00	-11.09
1	1.8	-11.09	-9.29	-8.50	-10.30
2	1.8	-10.30	-8.50	-7.63	-9.43
3	1.8	-9.43	-7.63	-6.67	-8.47
4	1.8	-8.47	-6.67	-5.59	-7.39
5	1.8	-7.39	-5.59	-4.40	-6.20
6	1.8	-6.20	-4.40	-3.08	-4.88
7	1.8	-4.88	-3.08	-0.62	-3.42
8	2.8	-3.42			-0.69

• ରେଭିନ୍ସା ଚାକିରୀ



2. Calculate the ERR of the following project if $a = 15\%$ per year.

End of year	Net cash flow (€)
0	-50000
1	16000
2	16000
3	16000
4	16000
5	26000

Soln:-

Step 1:- Discount all cash out flows to time 0 at $a\%$.

€ 50000

Step 2:- Compound all cash inflows to N. at $a\%$.

$$\Rightarrow 16000 (P/A, 15\%, 5) + 10000 = € 117878$$

Step 3:- Make equivalent

$$\Rightarrow 50000 (P/P, i\%, 5) = 117878$$

$$\Rightarrow (P/P, i\%, 5) = 2.36$$

$$\Rightarrow (1+i)^5 = 2.36$$

$$\Rightarrow 1+i = (2.36)^{1/5}$$

$$\Rightarrow 1+i = 1.1871$$

$$\Rightarrow i = 18.71\%$$

$$\therefore ERR = 18.71\%$$

Good